

# Exponent-Sequence and q-Series Frontiers for the Fabius–Rvachev System

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## Provenance

This volume preserves the absorbed theorem and proof bodies, with labels, citation keys, and asset paths mechanically prefixed per part. Part IV is an editorial merge of the two ninth-wave same-topic drafts (shared core stated once, unique layers kept, constants cross-checked); their verification packages are preserved verbatim under `assets/`. Later edits add status-only Lean crosswalk annotations; the historical source hashes below are unchanged. The absorbed drafts are deleted from the working tree; git history is the archive.

- `Fabijs_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex`  
SHA-256 2c068211b3e5f85f410976f89eda3616b3d48c1e1817dabdded0db5e41429a22
- `fabijs_frontier_results/q_richardson_sinc_frontier.tex`  
SHA-256 3b612cf8bbc34c4bfa9b3fbefbee14c618ff2edbbe84bc38d7cbe6baece8fcf4
- `finite_sinc_products_report/finite_sinc_products_report.tex`  
SHA-256 e01c3a2716d1c28ba6982510870775490981056b3f97b2a2b3f79002b2ef11b1 (eighth wave, folded in 2026-08-28)
- `Rvachev_Piecewise_Approximation_Fourier_Images/Rvachev_Piecewise_Approximation_Fourier_Images.tex`  
SHA-256 e70c2dd20b1ad71d44096d9c5c462596e36be202833bdbb5235efd4fba8355fd (ninth wave; merged editorially into Part IV)
- `rvachev_fourier_frontier_report/rvachev_piecewise_fourier_frontier.tex`  
SHA-256 23acde817ac19802ffed9cc19e80a80368d70315f87edf41e9fdc3f54c2a970b (ninth wave; merged editorially into Part IV)

## Part I

# Exponent-Sequence and Newton-Basis Frontiers for the Fabius–Rvachev System

## Abstract

The audited PROVEIT corpus already contains the classical Fabius and Rvachev functions, exact dyadic arithmetic, the full Thue–Morse formula atlas, Gaussian  $q$ -binomial interpolation, all-orders endpoint expansions, lower-branch Lambert- $W$  coordinates, Bell- and Bernoulli-polynomial coefficient calculus, sharp Fourier decay, the integer-zero spectrum of the Rvachev transform, a spectral determinant, rational arithmetic rays, half-integer quadrature identities, logarithmic  $q$ -Richardson acceleration, and a Pascal-weighted hierarchy of compactly supported densities. This report avoids repackaging those results. Instead it identifies a single master object that contains them as special coordinates and exposes several new deductions.

For a nonnegative integer exponent sequence  $a = (a_h)_{h \geq 0}$  with  $A(1/2) < \infty$ , where  $A(q) = \sum_{h \geq 0} a_h q^h$ , define

$$\Phi_a(z) = \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{a_h}, \quad \operatorname{sinc} z = \frac{\sin \pi z}{\pi z}.$$

It is the characteristic function of an explicit compactly supported convolution of uniform variables. We prove a universal dictionary in which the same generating function  $A$  controls: the integer-zero multiplicity  $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$ ; the spectral zeta  $\mathcal{Z}_a(s) = \zeta(s)A(2^{-s})$ ; every even cumulant  $\kappa_{2j} = B_{2j}A(4^{-j})/(2j)$ ; the Fourier-lobe sign; exact digital zero counts; and the order of Prouhet cancellation of each finite sign block. The zero divisor, the spectral zeta quotient, and the complete even-cumulant sequence each determine  $a$  uniquely.

The cumulant samples  $A(4^{-m})$  admit a new exact reconstruction mechanism. Lagrange extrapolation at geometric nodes has Gaussian- $q$ -binomial weights, an exact transformed series, fixed alternating sign, bounded  $\ell^1$  condition number, and certified integer brackets. Thus Richardson extrapolation becomes a stable tomography algorithm for the hidden dyadic exponent sequence.

For polynomial exponent sequences, the Newton expansion  $P(h) = \sum c_r \binom{h}{r}$  gives the finite factorization  $\Phi_P = \prod_r \Phi_{r+1}^{c_r}$  through the audited Pascal–Rvachev hierarchy. Negative Newton coefficients are allowed: whenever  $P(h) \geq 0$  on the nonnegative integers, the resulting alternating quotient is nevertheless entire and positive definite. The even family

$$P_d(h) = \binom{h-1}{d}, \quad d = 0, 2, 4, \dots,$$

is especially striking. It satisfies

$$\Psi_d(z) = \operatorname{sinc}(z)\Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}},$$

has support  $[-1, 1]$ , zero multiplicity  $1 + \binom{\nu_2(n)}{d+1}$ , and converges in total variation to the uniform law. Its transition across the two discontinuities of the limiting box density has a Gaussian boundary layer with an explicit Berry–Esseen rate. For  $d = 2$  this supplies the concrete positive-definite quotient  $\Phi_1\Phi_3/\Phi_2$ .

Finally, a Mellin-reflection identity relates the endpoint Laplace transform directly to the reflected spectral zeta. For polynomial weights it explains the complete hierarchy of logarithmic powers and periodic coefficients, gives a generalized lower-Lambert saddle, and yields the leading inverse-endpoint law. Reasonable conjectures and a detailed research and Lean-formalization program conclude the report. Numerical experiments were independently implemented at high precision; the commented Python source and generated data accompany the paper.

# 1 Audit boundary, source stratification, and novelty discipline

## 1.1 Pinned corpus

The repository changed several times on the audit date, so the source boundary is the exact commit printed on the title page. Its recursive documentation tree contains 67 distinct `.tex` paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (including twelve Fourier-decay dossiers), and seven transcribed or corrected source papers. Every path is listed in [section A](#). Historical revisions were read as mathematical lineages rather than as independent evidence; the archive’s provenance ledger identifies what was superseded, merged, or deliberately frozen.

The living primary exposition separates the bounded distribution function, the compactly supported Rvachev bump, and the signed global extension, and confines itself to claims with matching Lean declarations. The consolidated frontier volume and its current descendants contain the non-formalized but mathematically developed material. In particular, the audit treated the following as prior art rather than as discoveries of this report:

- the integer zero set and multiplicity  $1 + \nu_2(n)$  of the classical Rvachev transform;
- its spectral determinant, spectral zeta, heat trace, local zero jets, and symmetric  $q$ -integer deformation;
- the full signed Thue–Morse formula atlas, Prouhet cancellation, Mahler products, rarefied sums, Mellin–Dirichlet theory, and automatic-sequence machinery;
- exact dyadic values of  $F$ , half-base  $q$ -Pochhammer and Gaussian-binomial formulas, Lagrange interpolation, and several existing Richardson transforms;
- the corrected lower-Lambert endpoint expansion, its nonconstant periodic factor, all-order Bell-polynomial coefficient construction, and the inverse Fabius expansion;
- sharp geometric, arithmetic, metric, and envelope results for the infinite sinc product;
- cyclotomic rational rays, half-integer spectral identities, exact zero jets, and the Pascal hierarchy  $\Phi_r(z) = \prod_h \text{sinc}(z/2^h)^{\binom{h}{r-1}}$ ;
- logarithmic  $q$ -Richardson acceleration of truncated sinc products and phase-locked Richardson extraction of the endpoint periodic term;
- positive Gaussian and Lobatto tail quadrature, exact ordinary-prefix error laws, generalized gamma-convolution duality, and the existing Gaussian rank limit for the Pascal hierarchy.

## 1.2 What is new relative to the pinned tree

Targeted path, phrase, and formula searches found no counterpart of the following theorem package in the audited snapshot:

1. the arbitrary nonnegative exponent-sequence monoid and its single generating-function dictionary for zero spectra, cumulants, digital signs, and endpoint Mellin transforms;
2. exact digital zero-count defects and generalized Prouhet order for an arbitrary exponent parity word, including the first surviving power moment;
3. equivalence and constructive recovery of the exponent sequence from either the zero divisor or all even cumulants;
4. certified Gaussian- $q$ -binomial Richardson tomography of the exponent coefficients;
5. the finite-refinement-rank classification by rational ordinary generating functions;



6. the Newton-basis factorization of every nonnegative integer-valued polynomial weight, including positive-definite alternating quotients of Pascal transforms;
7. the normalized family  $P_d(h) = \binom{h-1}{d}$  for even  $d$ , its sparse zero spectrum, automatic sign selector, exact total-variation law, and Gaussian boundary layer;
8. the general spectral–endpoint Mellin reflection formula  $\mathcal{M}[\log L_a](w) = -\Gamma(w)\zeta(w+1)\mathcal{Z}_a(-w)/\zeta(-w)$ ;
9. the polynomial-weight universality theorem and its generalized Lambert- $W_{-1}$  endpoint coordinate.

A targeted external search located the classical ingredients separately—the original Fabius–Rvachev analysis and arithmetic, lacunary Thue–Morse products, automatic sequences, Berry–Esseen bounds, rational and integer-valued generating functions, Mellin harmonic sums, exponential Tauberian transfer, and Lambert inversion [4, 5, 1, 12, 2, 10, 11, 8, 7, 6]—but not this combined classification. Accordingly, “new” below always means new to the pinned corpus and apparently nonstandard; it is not an assertion of exhaustive worldwide priority.

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**Proof-status convention.** Results explicitly labeled **proved here** are proved in this manuscript. Results labeled **derived from audited corpus theorems** use a theorem already developed in the audited frontier corpus (most often the sharp Pascal-hierarchy Fourier or endpoint asymptotic) and then add a new deduction. Items labeled **conjectural/open** are conjectures or proposed programs.

A fourth, orthogonal label is now carried alongside these three. Wherever a statement of this report—or an exactly reusable core of one—already has a machine-checked counterpart in the repository’s Lean library, that declaration is cited inline in typewriter type—for instance `Fabius.weightedScaleMultiplicity`—together with its module under `Analysis/FabiusFunction/Lean/FabiusFunction/`. Roughly a dozen such citations appear below, alongside the *Post-snapshot Lean crosswalk* paragraphs. In most of them the formal statement is *more general* than the one printed here (arbitrary base instead of 2, arbitrary coefficient ring or module instead of  $\mathbb{R}$ , arbitrary geometric ratio instead of  $q \in (0, 1)$ ). Each citation says which. In particular the finite rational geometric-Lagrange kernel used in the Gaussian- $q$  rows—closed weights, all residual moments and their sign, and the finite exact  $\ell^1$  norm—is formalized at arbitrary ratio  $0 < q < 1$ , so the  $q = 1/4$  instance is covered, and the denominator-free principal specialization with the complete positive-degree geometric-moment identities behind [theorem 5.1](#) is formal as well (see the crosswalk paragraphs there). The post-snapshot module `AnalyticSeriesFilter.lean` now also proves the convergent-series analogue. Its finite filter acts diagonally under Mathlib’s *unconditional* `Summable` hypothesis on each weighted sampled series, splits into an exact finite head and infinite tail, and specializes geometric and geometric-Lagrange rows to the complete Gaussian residual tail, including the shifted-scale form used in (5.6). The hypothesis is sharp at a zero-weight node: the weighted series is then identically zero, and no convergence of the corresponding unweighted sample is required. It does not cover a series convergent only in its natural order at a boundary point, and it proves neither positivity, signs, bounds, uniform convergence, nor a sinc-product instantiation.

An independent reusable layer is now present in `FractionalVolterraCalculus.lean` and `FabiusFractionalVolterra.lean`. Positive affine scale changes on an ordered real interval are formalized. For  $\alpha \leq 0$ , affine covariance is an identity for the totalized Lean interval-integral definition; a classical Riemann–Liouville or integrability interpretation is asserted only for positive order. Gamma-normalized order raising assumes real  $\alpha > 0$ , a continuous Banach-valued primitive, an interval-integrable derivative, and the right derivative `HasDerivWithinAt` on the open interval. Its boundary-free form yields the exact dyadic shift for the signed Fabius extension when  $x \geq 0$ , for the bounded Fabius function when  $0 \leq x \leq 1$ , and the up-Fabius bridge from base  $-1$  when  $x \geq -1$ . Complex orders and powers, Caputo or Riemann–Liouville derivative operators, weighted-monomial or iterated identities, negative- $x$  signed branches, endpoint lattices, hierarchy versions, and inverse/quantile formulas remain frontier. These post-snapshot series and fractional APIs were checked at source checkpoint 22f801337.

These labels concern representation only, not correctness: report-specific probability and sinc-product instantiations still require their own bridges. In particular the sign and bound in (5.7)–(5.8), the infinite condition-number limit, Fourier envelopes for the Pascal hierarchy of rank at least two, Mellin contour displacement, exponential Tauberian transfer, Berry–Esseen for non-identically-distributed summands, total-variation and Wasserstein distances, and the automatic-sequence machinery have no Lean counterpart here.

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## 2 The exponent-sequence convolution monoid

## 2.1 Admissible sequences

Throughout, the normalized entire sinc is

$$\operatorname{sinc} z = \frac{\sin \pi z}{\pi z}, \quad \operatorname{sinc} 0 = 1. \quad (2.1)$$

It is the Fourier transform, under the kernel  $e^{-2\pi izx}$ , of the uniform probability density on  $[-1/2, 1/2]$ .

**Definition 2.1** (Admissible exponent sequence). A sequence  $a = (a_h)_{h \geq 0}$  is *admissible* if

$$a_h \in \mathbb{N}_0, \quad A(q) := \sum_{h \geq 0} a_h q^h, \quad A(1/2) < \infty.$$

Its generalized Rvachev transform is

$$\Phi_a(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{a_h}. \quad (2.2)$$

For each  $h$  and  $1 \leq j \leq a_h$ , let  $U_{h,j}$  be independent and uniform on  $[-1/2, 1/2]$ , and put

$$X_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} U_{h,j}, \quad R_a = \frac{1}{2} A(1/2). \quad (2.3)$$

Absolute convergence is deterministic because the sum of the radii is  $R_a$ .

**Theorem 2.2** (Probability, analyticity, support, and convolution). *For every admissible  $a$ :*

1.  $X_a$  is supported on  $[-R_a, R_a]$  and has characteristic function  $\Phi_a$ ;
2. the product in (2.2) converges locally uniformly on  $\mathbb{C}$ , hence is entire;
3.  $\Phi_{a+b} = \Phi_a \Phi_b$ , and the corresponding probability law is the convolution of the laws for  $a$  and  $b$ ;
4. if  $\sum_h a_h = \infty$ , then  $X_a$  has a compactly supported  $C^\infty$  density; if only finitely many  $a_h$  are nonzero, its law is a finite uniform box spline.

All four assertions are **proved here**. Formally, only the classical case  $a_h \equiv 1$  is represented: `Fabius . rvachevFourier_eq_product` (`FourierProduct.lean`) identifies the Fourier transform of the Rvachev up-function with the sinc product (2.2) at  $a_h \equiv 1$ , in the same normalization (2.1), and `Fabius . ProbabilityRepresentation . rvachevUp_eq_weightedSum_probability_global` (`ProbabilityRepresentation.lean`) is the matching dyadic-uniform-series representation, stated in the Fabius normalization  $\sum_{h \geq 0} 2^{-h-1} V_h$  on  $[0, 1]$ , which is  $(X_a + R_a)/2$  in the notation used here. Nothing about a general admissible  $a$  is formalized.

*Proof.* Independence gives the finite-prefix characteristic products. The variables are bounded and  $\sum_{h,j} |2^{-h} U_{h,j}| \leq R_a$ , so the finite sums converge almost surely and in every  $L^p$ , and their characteristic functions converge to that of  $X_a$ .

Locally,  $\log \operatorname{sinc} w = \mathcal{O}(w^2)$ . Since  $\sum_h a_h 4^{-h} = A(1/4) < \infty$ , the logarithms of the factors converge normally on compact sets avoiding a harmless choice of local branch; standard infinite-product theory then gives an entire limit. Addition of exponent sequences is multiplication of products, which is exactly convolution of independent sums.

If infinitely many individual factors occur, fix  $M$  of them. On the real axis,  $|\operatorname{sinc}(\xi/2^h)| \leq \min(1, 2^h/(\pi|\xi|))$ . The chosen  $M$  factors give  $|\Phi_a(\xi)| \leq C_M(1 + |\xi|)^{-M}$ , while all remaining factors have modulus at most one. Thus  $\Phi_a$  is rapidly decreasing; Fourier inversion gives a  $C^\infty$  density. With finitely many factors one obtains a finite convolution of interval densities, hence a box spline.  $\square$

**Remark 2.3.** The classical Rvachev transform corresponds to  $a_h \equiv 1$ . The audited Pascal hierarchy corresponds to  $a_h = \binom{h}{r-1}$ . The point of [theorem 2.2](#) is not merely that both products exist, but that the collection of all such laws is a commutative convolution monoid whose algebra is the coefficientwise additive monoid of exponent sequences.

### 3 Integer zeros, spectral zeta, digital defects, and lobe signs

#### 3.1 The universal zero-divisor formula

Euler’s product for the normalized sinc gives

$$\operatorname{sinc} z = \prod_{k \geq 1} \left(1 - \frac{z^2}{k^2}\right).$$

An integer  $n \neq 0$  is a zero of the scale- $h$  factor precisely when  $2^h \mid n$ . This immediately yields the following master theorem.

**Theorem 3.1** (Zero divisor and spectral zeta). *Let  $a$  be admissible and define*

$$m_a(n) = \sum_{h=0}^{\nu_2(n)} a_h, \quad n \geq 1. \quad (3.1)$$

Then

$$\Phi_a(z) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{m_a(n)} \quad (3.2)$$

with locally uniform convergence. In particular,  $\operatorname{ord}_{z=\pm n} \Phi_a = m_a(n)$ . For  $\Re s > 1$ ,

$$\mathcal{Z}_a(s) := \sum_{n \geq 1} \frac{m_a(n)}{n^s} = \zeta(s) A(2^{-s}). \quad (3.3)$$

Both statements are **proved here**. The multiplicity function (3.1) is formalized as `Fabius.weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), and in two directions more generally than it is used here: the base 2 is replaced by an arbitrary  $b$ , and the weights  $a_h$  may lie in an arbitrary additive commutative monoid rather than in  $\mathbb{N}_0$ . The canonical product (3.2) is formalized only at  $a_h \equiv 1$ , as `Fabius.rvachevFourierProduct_eq_canonical` (`SincCanonicalProduct.lean`), whose exponent is the classical  $\nu_2(n) + 1$ ; the general- $a$  product and the spectral zeta (3.3) are not formalized.

*Proof.* The scale- $h$  sinc contributes one zero at  $n$  for every representation  $n = 2^h k$  with  $k \geq 1$ , and it occurs with multiplicity  $a_h$ . Summing over  $h \leq \nu_2(n)$  gives (3.1). Moreover

$$\sum_{n \geq 1} \frac{m_a(n)}{n^2} = \sum_{h \geq 0} a_h \sum_{k \geq 1} \frac{1}{(2^h k)^2} = \zeta(2) A(1/4) < \infty,$$

so the genus-zero canonical product converges normally. The same rearrangement for  $\Re s > 1$  gives

$$\sum_{n \geq 1} \frac{1}{n^s} \sum_{2^h \mid n} a_h = \sum_{h \geq 0} a_h 2^{-hs} \zeta(s).$$

□

---

**Spectral–probabilistic dual use of  $A$ .** The spectral side samples  $A$  at  $2^{-s}$ , while the probability side will sample it at  $4^{-j}$ . The entire zero divisor and every even cumulant are therefore two transforms of one coefficient sequence, not merely analogous formulas.

---

### 3.2 Exact counting and a generalized digital defect

Let

$$M_a(N) = \sum_{n=1}^N m_a(n). \quad (3.4)$$

For  $N = \sum_{j \geq 0} b_j(N)2^j$ , define the tail weights

$$\tau_j(a) = \sum_{k \geq 1} a_{j+k} 2^{-k}. \quad (3.5)$$

**Theorem 3.2** (Exact zero count and digit defect). *For every positive integer  $N$ ,*

$$M_a(N) = \sum_{h \geq 0} a_h \left\lfloor \frac{N}{2^h} \right\rfloor, \quad (3.6)$$

$$= NA(1/2) - \rho_a(N), \quad \rho_a(N) = \sum_{h \geq 0} a_h \left\{ \frac{N}{2^h} \right\}, \quad (3.7)$$

$$= NA(1/2) - \sum_{j \geq 0} b_j(N) \tau_j(a). \quad (3.8)$$

This is **proved here**. The floor form (3.6) is formalized as `Fabius.sum_range_weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), over the finite range  $h < N$  that already contains every nonzero term, and again for an arbitrary base  $b \neq 1$  and weights in an arbitrary additive commutative monoid, so the Lean statement is strictly more general than (3.6). The two defect forms (3.7)–(3.8) are not formalized.

*Proof.* Count the multiples of  $2^h$  among  $1, \dots, N$  and sum their contributions  $a_h$ . Subtracting the fractional parts from  $N \sum_h a_h 2^{-h}$  gives (3.7). Finally,

$$\left\{ \frac{N}{2^h} \right\} = \sum_{j=0}^{h-1} b_j(N) 2^{j-h};$$

absolute convergence permits exchanging the sums and yields (3.8).  $\square$

For  $a_h \equiv 1$ , one has  $A(1/2) = 2$  and  $\tau_j = 1$ , recovering the audited identity  $M_a(N) = 2N - s_2(N)$ . For general  $a$ , the ordinary binary weight is replaced by a weighted suffix transform of the exponent sequence.

### 3.3 Lobe signs and generalized Thue–Morse characters

For  $x \in (N, N+1)$ , every canonical-product factor with index  $n \leq N$  is negative and all remaining factors are positive. Hence

$$\text{sgn } \Phi_a(x) = (-1)^{M_a(N)}. \quad (3.9)$$

Modulo two,  $\lfloor N/2^h \rfloor \equiv b_h(N)$ , so the sign is a digit character.

**Theorem 3.3** (Generalized lobe sign). *Put  $\bar{a}_h = a_h \bmod 2$  and*

$$\varepsilon_a(N) = (-1)^{\sum_{h \geq 0} \bar{a}_h b_h(N)}. \quad (3.10)$$

*Then, away from the integer zeros,*

$$\boxed{\text{sgn } \Phi_a(x) = \varepsilon_a(\lfloor x \rfloor)}. \quad (3.11)$$

Moreover  $\varepsilon_a$  is a 2-automatic sequence if and only if the parity word  $(\bar{a}_h)_{h \geq 0}$  is eventually periodic. Both statements are **proved here**. At  $a_h \equiv 1$  the sign law (3.11) is formalized as `Fabiús.rvachevFourierProduct_eq_thueMorse_sign_mul_norm (ThueMorseLobeSign.lean)`, in the equivalent form  $\Phi(x) = (-1)^{s_2(N)} \|\Phi(x)\|$  on the lobe  $(N, N+1)$ . The general parity character (3.10) is not formalized, and neither is the automaticity criterion, for want of any automatic-sequence or 2-kernel theory in the library.

*Proof.* The sign formula follows from (3.9), (3.6), and reduction modulo two. For the automaticity criterion, let  $S$  denote the left shift of the parity word. If  $0 \leq r < 2^k$ , binary concatenation gives

$$\varepsilon_a(2^k n + r) = \varepsilon_a(r) \varepsilon_{S^k a}(n). \quad (3.12)$$

Thus an eventually periodic parity word has only finitely many shifts and therefore gives a finite 2-kernel. Conversely, equality of two shifted sign sequences forces equality of the corresponding shifted parity words, because  $\varepsilon_{S^k a}(2^j) = (-1)^{\bar{a}_{k+j}}$ . A finite 2-kernel therefore implies a finite shift orbit of the parity word, which is equivalent to eventual periodicity. This is the present Walsh-character specialization of the standard finite-kernel characterization of automatic sequences [2].  $\square$

### 3.4 Finite sign products and exact Prouhet order

For  $L \geq 0$ , let

$$O_L(a) = \{0 \leq h < L : a_h \text{ is odd}\}, \quad \omega_L(a) = |O_L(a)|. \quad (3.13)$$

**Theorem 3.4** (Generalized Prouhet block). *The length- $2^L$  sign polynomial factors as*

$$T_{a,L}(z) := \sum_{n=0}^{2^L-1} \varepsilon_a(n) z^n = \prod_{h=0}^{L-1} \left(1 + (-1)^{a_h} z^{2^h}\right). \quad (3.14)$$

*Its zero at  $z = 1$  has exact order  $\omega_L(a)$ . Consequently*

$$\sum_{n < 2^L} \varepsilon_a(n) n^k = 0, \quad 0 \leq k < \omega_L(a), \quad (3.15)$$

*and the first surviving power moment is*

$$\sum_{n < 2^L} \varepsilon_a(n) n^{\omega_L(a)} = (-1)^{\omega_L(a)} \omega_L(a)! 2^{L-\omega_L(a) + \sum_{h \in O_L(a)} h}. \quad (3.16)$$

*This is **proved here**. Its sharp-moment core is formalized, and in greater generality: `Fabiús.sum_powerset_neg_one_pow_pow_card (ThueMorseSparseProuhet.lean)` evaluates*

$$\sum_{T \subseteq S} (-1)^{|T|} \left(x + \sum_{j \in T} w_j\right)^{|S|} = (-1)^{|S|} |S|! \prod_{j \in S} w_j$$

*for an arbitrary finite index set  $S$ , arbitrary step weights  $w_j$  in an arbitrary commutative ring, and every offset  $x$ . Taking  $S = O_L(a)$  and  $w_h = 2^h$  produces exactly the constant  $(-1)^{\omega_L(a)} \omega_L(a)! 2^{\sum_{h \in O_L(a)} h}$  of (3.16), independently of  $x$ , and the companion `Fabiús.sum_powerset_neg_one_pow_eval_of_degree` it is the vanishing (3.15) for every polynomial of degree below  $|S|$ ; `Fabiús.sum_submask_neg_one_pow_pow` is the binary specialization in which  $S$  is the bit support of an integer. What is not formalized is the factorization (3.14) itself, nor the free-even-layer step that sums the offset-independent core over the  $2^{L-\omega_L(a)}$  even-layer offsets.*

*Proof.* Expand the product by choosing the binary digit  $b_h \in \{0, 1\}$  independently. At  $z = 1$ , an odd exponent contributes a factor  $1 - z^{2^h}$  with a simple zero, while an even exponent contributes  $1 + z^{2^h} = 2$ . Therefore the zero order is exactly  $\omega_L$  and its leading coefficient in powers of  $z - 1$  is

$$(-1)^{\omega_L} 2^{L - \omega_L + \sum_{h \in O_L} h}.$$

The derivatives  $T_{a,L}^{(k)}(1)$  are the signed falling-factorial moments. Vanishing for  $k < \omega_L$  implies vanishing of ordinary moments of those degrees. At the first surviving degree, lower falling-factorial terms have already vanished, so multiplying the leading coefficient by  $\omega_L!$  gives (3.16).  $\square$

*Remark 3.5.* The usual Thue–Morse Prouhet theorem is the case  $a_h \equiv 1$ , for which  $\omega_L = L$ . Here the cancellation order is not tied to the block length alone: it is exactly the number of odd spectral layers present before scale  $L$ .

## 4 Bernoulli cumulants, Bell moments, and identifiability

### 4.1 The same generating function at $q = 4^{-j}$

For  $U \sim \text{Unif}[-1/2, 1/2]$ ,

$$\log \mathbb{E} e^{tU} = \log \frac{\sinh(t/2)}{t/2} = \sum_{j \geq 1} \frac{B_{2j}}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.1)$$

**Theorem 4.1** (Cumulant transform). *For an admissible exponent sequence,*

$$\kappa_{2j}(X_a) = \frac{B_{2j}}{2j} A(4^{-j}), \quad \kappa_{2j+1}(X_a) = 0. \quad (4.2)$$

Equivalently,

$$\log \mathbb{E} e^{tX_a} = \sum_{j \geq 1} \frac{B_{2j} A(4^{-j})}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.3)$$

This is [proved here](#).

*Proof.* Cumulants add under independent sums and scale homogeneously. The level- $h$  contribution to the  $2j$ th cumulant is  $a_h 2^{-2jh} B_{2j}/(2j)$ . Summing over  $h$  gives  $A(4^{-j})$ .  $\square$

**Corollary 4.2** (Bell-polynomial moments). *The even moments are*

$$\mathbb{E} X_a^{2n} = \mathcal{B}_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (4.4)$$

or explicitly

$$\mathbb{E} X_a^{2n} = (2n)! \sum_{\substack{j_1, \dots, j_n \geq 0 \\ \sum_{r=1}^n 2j_r = n}} \prod_{r=1}^n \frac{1}{j_r!} \left( \frac{B_{2r} A(4^{-r})}{2r(2r)!} \right)^{j_r}. \quad (4.5)$$

This is a direct, established Bell-polynomial conversion applied to the new master cumulant formula. The conversion itself is formalized: (4.4) is `Fabiús.completeBellPolynomial` (`MomentCumulantAlgebra.lean`), the explicit partition sum (4.5) is `Fabiús.partitionExpSum` (`ExponentialPartition.lean`), and the two are proved to agree by `Fabiús.partitionExpSum_eq_expCoeff` (`ExponentialBell.lean`); all three hold in an arbitrary commutative  $\mathbb{Q}$ -algebra, so the formal statement is considerably more general than this application. Its cumulant input [theorem 4.1](#) is not formalized, so the corollary is not yet formally available as stated.



## 4.2 Zero spectrum and cumulants determine one another

Normalize the even cumulants by

$$c_j(a) = \frac{2j}{B_{2j}} \kappa_{2j}(X_a) = A(4^{-j}). \quad (4.6)$$

**Theorem 4.3** (Four equivalent encodings). *The following data determine one another uniquely:*

1. the exponent sequence  $(a_h)_{h \geq 0}$ ;
2. the integer-zero multiplicity function  $m_a(n)$ ;
3. the quotient  $\mathcal{Z}_a(s)/\zeta(s) = A(2^{-s})$ ;
4. the normalized even-cumulant samples  $(c_j(a))_{j \geq 1}$ .

Constructively,

$$a_v = m_a(2^v) - m_a(2^{v-1}), \quad m_a(2^{-1}) := 0, \quad (4.7)$$

$$a_0 = \lim_{j \rightarrow \infty} c_j, \quad (4.8)$$

$$a_v = \lim_{j \rightarrow \infty} 4^{vj} \left( c_j - \sum_{h=0}^{v-1} a_h 4^{-hj} \right), \quad v \geq 1. \quad (4.9)$$

This is [proved here](#).

*Proof.* The first formula is the first difference of the prefix sum  $m_a(2^v) = \sum_{h \leq v} a_h$ . The spectral-zeta equivalence follows from (3.3). For the cumulants,  $c_j = A(4^{-j})$ , and  $4^{-j} \rightarrow 0$  inside the disk of analyticity of  $A$ . Equality of all samples for two sequences gives an analytic difference with infinitely many zeros accumulating at the interior point 0, hence the generating functions coincide. The explicit limits follow by peeling off one Taylor coefficient at a time; the remaining tail converges absolutely because  $A(1/4) < \infty$ .  $\square$

*Remark 4.4.* The limits in (4.8)–(4.9) are mathematically exact but numerically wasteful. The next section replaces them by finite, sign-certified,  $q$ -binomial extrapolants.

## 5 Certified Gaussian- $q$ -binomial Richardson tomography

The coefficient limits in [theorem 4.3](#) are exact, but they use one cumulant at a time and amplify roundoff by powers of  $4^j$ . The geometric nature of the samples admits a much better reconstruction. The result is not merely an acceleration formula: positivity of the hidden coefficients supplies an alternating sign theorem and hence certified integer brackets.

### 5.1 Lagrange weights on a geometric grid

Fix  $q \in (0, 1)$ . We use

$$(q; q)_r = \prod_{k=1}^r (1 - q^k), \quad \left[ \begin{matrix} r \\ j \end{matrix} \right]_q = \frac{(q; q)_r}{(q; q)_j (q; q)_{r-j}}.$$

For  $r \geq 0$  and  $0 \leq j \leq r$ , let

$$w_{r,j}(q) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-q^k}{q^j - q^k}. \quad (5.1)$$

These are the Lagrange interpolation weights which evaluate, at the origin, a polynomial known at  $1, q, \dots, q^r$ .



**Lemma 5.1** (Closed Gaussian-binomial weights). *One has*

$$w_{r,j}(q) = (-1)^{r-j} \frac{q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}} = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_r} \begin{bmatrix} r \\ j \end{bmatrix}_q. \quad (5.2)$$

Consequently,

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = \delta_{h0} \quad (0 \leq h \leq r). \quad (5.3)$$

For every integer  $h \geq r + 1$  the first nonvanishing value is

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q. \quad (5.4)$$

This is **proved here**. The weights (5.1) are formalized as `Fabiús.geometricLagrangeWeight` (`GeometricLagrange.lean`); the annihilation (5.3) is `Fabiús.sum_geometricLagrangeWeight_mul_pow`, with the normalization at  $h = 0$  recorded as the genuine boundary value  $0^0 = 1$ . The division-free forward and reversed Gaussian numerators are supplied by `GeometricQBinoomialLagrange.lean`, while `Fabiús.sum_geometricLagrangeWeight_mul_pow_of_pos` (`GeometricResidualMoments.lean`) gives (5.4) for every  $h > 0$ ; above-diagonal zero extension contains the range  $1 \leq h \leq r$  as well. These results hold over an arbitrary field whenever  $q^0, \dots, q^r$  are distinct, so positivity and the hypothesis  $q \in (0, 1)$  enter only in later sign arguments. The denominator-free principal specialization  $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$  is proved over commutative semirings in `GeometricCompleteHomogeneous.lean`.

*Proof.* Separating the factors with  $k < j$  and  $k > j$  in (5.1) gives (5.2). The annihilation relations are the exactness of Lagrange interpolation for the monomials  $1, x, \dots, x^r$ .

For completeness, let  $H_h(x_0, \dots, x_r)$  denote the complete homogeneous symmetric polynomial of degree  $h$ . The divided difference of  $x^h$  at  $x_0, \dots, x_r$  is  $H_{h-r}(x_0, \dots, x_r)$ . The interpolation remainder at  $x = 0$ , with  $x_j = q^j$ , therefore gives

$$\sum_{j=0}^r w_{r,j} q^{jh} = (-1)^r q^{0+1+\dots+r} H_{h-r-1}(1, q, \dots, q^r).$$

The principal specialization identity  $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$  yields (5.4); see, for example, the standard finite  $q$ -binomial calculus in [3].  $\square$

**Post-snapshot Lean crosswalk.** The source-level individual quotient is the already compiled theorem `geometricLagrangeWeight_eq_gaussianBinomial_div` in `GeometricQBinoomialLagrange.lean`; it assumes a field and injectivity of  $j \mapsto q^j$  on  $\{0, \dots, r\}$ , and deliberately uses the numerator `gaussianBinomialqr(r-j)`. The symmetric-index notation in (5.2) is the equivalent paper convention, not a newly added generic symmetry theorem. The later source checkpoint `1e761ed77583e9870ceeaeb2c74ac824094f0f3f` adds `completeHomogeneousEval_geometric`, which proves

$$H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$$

over every commutative semiring, without division, distinctness, nonzeroness, or an exclusion of roots of unity. More literally, its right-hand side is the recursive term `gaussianBinomialq(n+r)r`,

which agrees with the displayed quotient in the present field setting. Over a commutative ring, `sum_weight_mul_geometric_pow_of_pos` proves more generally that any row  $a_0, \dots, a_r$  with

$$\sum_{j=0}^r a_j (q^j)^d = 0^d \quad (0 \leq d \leq r)$$

satisfies

$$\sum_{j=0}^r a_j (q^j)^h = (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q \quad (h > 0).$$

The Lagrange specialization is `sum_geometricLagrangeWeight_mul_pow_of_pos`; it gives both (5.3) and (5.4), with mass one at degree zero supplied separately.

The scaled exact-row theorem is genuinely stronger than merely factoring a unit from the nodes. If exactness is assumed directly on  $c, cq, \dots, cq^r$ , then for  $s \geq 0$

$$\sum_{j=0}^r a_j (cq^j)^{r+1+s} = c^{r+1+s} (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} r+s \\ r \end{bmatrix}_q;$$

this is `sum_weight_mul_scaled_geometric_pow_succ_add`, valid even when  $c$  is zero or a zero divisor. For the field-valued Lagrange row and the shifted nodes  $q^{m+j}$ , `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos` gives, for every  $h > 0$ ,

$$\sum_{j=0}^r w_{r,j}(q) (q^{m+j})^h = (-1)^r q^{mh + \binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q.$$

The finite theorems alone make no convergence claim. Their analytic companion `AnalyticSeriesFilter.lean` does supply the exact unconditionally summable Gaussian tail, including the shifted Lagrange row, under summability of each nonzero-weight sample. It supplies neither the one-sided sign nor the numerical remainder bound used below.

## 5.2 Exact transformed series and sign certification

Let

$$\mathcal{R}_{r,m}[A] := \sum_{j=0}^r w_{r,j}(q) A(q^{m+j}), \quad m \geq 1. \quad (5.5)$$

Scaling all interpolation nodes by  $q^m$  leaves the weights unchanged.

**Theorem 5.2** (Exact  $q$ -Richardson tomography). *Suppose  $A(x) = \sum_{h \geq 0} a_h x^h$  converges at  $x = q^m$ . Then*

$$\mathcal{R}_{r,m}[A] = a_0 + (-1)^r q^{\binom{r+1}{2}} \sum_{h \geq r+1} a_h \begin{bmatrix} h-1 \\ r \end{bmatrix}_q q^{mh}. \quad (5.6)$$

If  $a_h \geq 0$ , then

$$(-1)^r (\mathcal{R}_{r,m}[A] - a_0) \geq 0. \quad (5.7)$$

Moreover,

$$0 \leq (-1)^r (\mathcal{R}_{r,m}[A] - a_0) \leq \frac{q^{\binom{r+1}{2}}}{(q; q)_r} \left( A(q^m) - \sum_{h=0}^r a_h q^{mh} \right). \quad (5.8)$$

All three statements are **proved here**. The finite exactness half is formalized: `Fabiús.geometricLagrange_richardson_exact` (`GeometricLagrange.lean`) proves that the weights (5.1) annihilate the first  $r$  geometric error modes of a sample block beginning at an arbitrary index and return the constant term exactly. It is stated for samples in an arbitrary module over an arbitrary field, with no positivity hypothesis,

no separate  $q \neq 0$  condition beyond finite-node injectivity, and no lower bound on the starting index, so it covers more than the real nonnegative-coefficient setting used here. The full transformed tail (5.6) is now the shifted specialization `Fabiús.geometricLagrangeAnalyticSeriesFilter_shifted` (`AnalyticSeriesFilter.lean`) under unconditional summability of the sampled series. Only nonzero Lagrange weights create an unweighted convergence obligation; a zero-weight node is discharged identically. This theorem excludes a conditionally-only boundary series and does not formalize the positivity argument, the sign (5.7), or the bound (5.8).

*Proof.* Insert the Taylor series of  $A$  into (5.5); absolute convergence permits interchanging the two sums. The terms  $1 \leq h \leq r$  vanish by (5.3), while (5.4) gives the remaining terms. The sign follows because Gaussian binomial coefficients are positive for  $0 < q < 1$ . Finally,

$$\left[ \begin{matrix} h-1 \\ r \end{matrix} \right]_q = \frac{\prod_{k=0}^{r-1} (1 - q^{h-1-k})}{(q; q)_r} \leq \frac{1}{(q; q)_r},$$

which proves (5.8).  $\square$

**Corollary 5.3** (Certified integer brackets). *If the coefficients are nonnegative integers, then for any odd  $r_-$  and even  $r_+$ ,*

$$\lceil \mathcal{R}_{r_-,m}[A] \rceil \leq a_0 \leq \lfloor \mathcal{R}_{r_+,m}[A] \rfloor. \quad (5.9)$$

Once the two rounded endpoints coincide,  $a_0$  has been recovered with a proof certificate. For the Fabiús–Rvachev cumulants, the grid parameter is  $q = 1/4$  and the data are  $A(q^m) = 2m\kappa_{2m}/B_{2m}$ .

### 5.3 Conditioning does not deteriorate with order

High-order extrapolation is often unstable because its weights become enormous. The geometric grid behaves unusually well.

**Theorem 5.4** (Exact  $\ell^1$  condition number). *The weight norm is*

$$\sum_{j=0}^r |w_{r,j}(q)| = \frac{(-q; q)_r}{(q; q)_r} = \prod_{k=1}^r \frac{1 + q^k}{1 - q^k}. \quad (5.10)$$

It increases to the finite limit  $(-q; q)_\infty / (q; q)_\infty$ . At the cumulant grid  $q = 1/4$ ,

$$\frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} = 1.96926035366826943194719369697 \dots \quad (5.11)$$

This is [proved here](#). Identity (5.10) is formalized only at the half base  $q = 1/2$ , and is therefore a strict specialization of what is stated here: `Fabiús.sum_abs_discreteLimitWeight` (`FabiúsDiscreteLimitToeplitz.lean`) evaluates the  $\ell^1$  norm of the  $r$ th Toeplitz row as  $(-1/2; 1/2)_r / (1/2; 1/2)_r$ , and `Fabiús.discreteLimitWeight_eq_geometricLagrangeWeight` identifies that row, after the reversal  $j \mapsto r - j$ , with the weights (5.1). General  $q$ , and hence both the cumulant grid  $q = 1/4$  and the limiting value (5.11), is not formalized. The finite  $q$ -binomial theorem used in the proof below is however available in full generality as `Fabiús.finite_qBinomial_theorem` (`FiniteqBinomialCore.lean`), over an arbitrary commutative ring and for every  $q$  and  $z$ , which is exactly the input a general- $q$  proof would need.

*Proof.* Use (5.2), replace  $j$  by  $r - j$ , and apply the finite  $q$ -binomial theorem:

$$\sum_{k=0}^r q^{\binom{k+1}{2}} \left[ \begin{matrix} r \\ k \end{matrix} \right]_q = \prod_{k=1}^r (1 + q^k) = (-q; q)_r.$$

Division by  $(q; q)_r$  gives (5.10). Both infinite products converge for  $|q| < 1$ .  $\square$

**Remark 5.5.** The bounded condition number is a structural advantage over extrapolation on an arithmetic grid. It does not remove the need for accurate high cumulants, but it prevents the interpolation layer itself from introducing an exponentially growing amplification factor.

## 5.4 Recovering the entire exponent sequence

After  $a_0, \dots, a_{v-1}$  have been certified, define the deflated generating function

$$A^{(v)}(x) = x^{-v} \left( A(x) - \sum_{h=0}^{v-1} a_h x^h \right) = \sum_{k \geq 0} a_{v+k} x^k. \quad (5.12)$$

Applying [theorem 5.2](#) to  $A^{(v)}$  recovers  $a_v$  with the same alternating certificate.

**Algorithm 5.6** (Cumulant-to-spectrum tomography). Given certified intervals for the normalized cumulants  $c_m = A(4^{-m})$ :

1. set  $q = 1/4$  and  $v = 0$ ;
2. subtract the already recovered polynomial  $\sum_{h < v} a_h q^{mh}$  from each  $c_m$  and multiply by  $q^{-mv}$ ;
3. evaluate one odd- and one even-order transform ([5.5](#));
4. round outward as in ([5.9](#));
5. increase  $m$  or the orders until the integer interval collapses;
6. increment  $v$  and repeat.

The truncation estimate ([5.8](#)), combined with interval arithmetic, turns the procedure into a fully rigorous recovery algorithm.

---

**Relation to prior Richardson work.** The corpus already contains Gaussian- $q$ -binomial acceleration of finite sinc products, dyadic value sequences, spline approximants, and phase-locked endpoint expansions. The new point here is different: the input is the sequence of *even cumulants*, the unknowns are the integer multiplicities of dyadic uniform layers, and nonnegativity turns the Lagrange remainder into alternating spectral certificates.

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## 6 Shift algebra, refinement systems, and rational classification

### 6.1 The shift is the dilation operator in disguise

Let  $(Sa)_h = a_{h+1}$  and  $\Delta = S - I$ . For signed integer sequences we use the notation  $\Phi_b$  for the analytic germ at the origin obtained by exponentiating  $\sum_h b_h \log \operatorname{sinc}(z/2^h)$ ; negative entries may make its global continuation meromorphic, but every identity below is unambiguous near zero.

**Theorem 6.1** (Master shift-refinement identity). *For every admissible nonnegative sequence,*

$$\boxed{\Phi_a(2z) = \operatorname{sinc}(2z)^{a_0} \Phi_{Sa}(z).} \quad (6.1)$$

More generally, for  $m \geq 1$ ,

$$\Phi_a(2^m z) = \left( \prod_{h=0}^{m-1} \operatorname{sinc}(2^{m-h} z)^{a_h} \right) \Phi_{S^m a}(z). \quad (6.2)$$

Since  $S^m = (I + \Delta)^m$ , one also has the finite-difference factorization

$$\boxed{\Phi_{S^m a}(z) = \prod_{r=0}^m \Phi_{\Delta^r a}(z)^{\binom{m}{r}}.} \quad (6.3)$$

The final identity is initially an identity of analytic germs; all apparent poles cancel whenever the left side is entire. These statements are [proved here](#). The exponent-level content of (6.1)–(6.2)—that passing to  $S^m a$  exposes the first  $m$  layers and shifts the rest—is formalized as `Fabius.weightedScaleMultiplicity_base_pow_mul` (`WeightedScaleMultiplicity.lean`), for an arbitrary base  $b > 1$  and weights in an arbitrary additive commutative monoid. The analytic identities themselves are formalized only at  $a_h \equiv 1$ , as `Fabius.rvachevFourierProduct_two_mul` and `Fabius.rvachevFourierProduct_two_pow_mul` (`SincProductShells.lean`); these are instances of `Fabius.tprod_geom_scale` and `Fabius.tprod_geom_scale_pow` (`GeometricScaleProducts.lean`), which renormalize a scale product  $\prod_n g(z/c^n)$  for an arbitrary factor function  $g$ , an arbitrary nonzero ratio  $c$  and an arbitrary commutative topological monoid—but for one fixed  $g$ , so they do not by themselves reach a variable exponent sequence. The finite-difference factorization (6.3) is not formalized.

*Proof.* The first formula follows by separating the  $h = 0$  factor and reindexing the rest:

$$\prod_{h \geq 0} \operatorname{sinc}(2z/2^h)^{a_h} = \operatorname{sinc}(2z)^{a_0} \prod_{k \geq 0} \operatorname{sinc}(z/2^k)^{a_{k+1}}.$$

Iteration gives (6.2). Finally, exponent sequences add when their transforms multiply, and the binomial identity for the shift gives  $S^m a = \sum_r \binom{m}{r} \Delta^r a$ .  $\square$

*Remark 6.2.* This formula explains why finite differences, Newton interpolation, and refinement equations continually appear together in the Fabius corpus. The coincidence is not merely an analogy:  $S$  acts on the exponent sequence exactly as dilation acts on the sinc product.

## 6.2 When the refinement tower closes

Write

$$\mathcal{V}(a) = \operatorname{span}_{\mathbb{C}}\{S^j a : j \geq 0\}.$$

The following classical sequence-theoretic equivalence acquires a direct analytic meaning in this setting.

**Theorem 6.3** (Rational generating functions are precisely finite refinement systems). *For a complex sequence  $a$  of at most exponential growth, the following are equivalent:*

1.  $\dim \mathcal{V}(a) < \infty$ ;
2.  $a$  satisfies a constant-coefficient linear recurrence from some point onward;
3. its ordinary generating function  $A(q) = \sum_{h \geq 0} a_h q^h$  is rational.

When these conditions hold, the logarithmic transforms  $L_j(z) = \log \Phi_{S^j a}(z)$  span a finite-dimensional space stable under  $z \mapsto 2z$ . Conversely, any finite-dimensional shift-closed exponent model has rational  $A$ . This is [proved here](#) as a synthesis of the standard rational-series theorem with [theorem 6.1](#).

*Proof.* The equivalence of the first three statements is the usual companion-matrix characterization of rational ordinary generating functions [11]. If  $S^d a = \sum_{j < d} \gamma_j S^j a$ , then  $L_d = \sum_{j < d} \gamma_j L_j$  as analytic germs, while

$$L_j(2z) = a_j \log \operatorname{sinc}(2z) + L_{j+1}(z).$$

Thus the vector  $(L_0, \dots, L_{d-1}, \log \operatorname{sinc}(2z))$  obeys a finite refinement system. The converse is the same statement read on the exponent vectors.  $\square$

**Corollary 6.4** (Pole dictionary for rational weights). *If  $A$  is rational, then*

$$\frac{\mathcal{Z}_a(s)}{\zeta(s)} = A(2^{-s}) \tag{6.4}$$

is a finite combination of terms whose poles lie on vertical arithmetic progressions obtained by taking logarithms of the poles of  $A$ . Poles of  $A$  at roots of unity produce log-periodic terms in Mellin asymptotics; poles away from the unit circle produce additional power-law or exponentially modulated scales. The exact transseries classification suggested by this observation is formulated in [theorem 12.3](#).

## 7 Polynomial weights and Newton factorization through the Pascal hierarchy

### 7.1 The audited Pascal–Rvachev transforms

For  $r \geq 1$ , put

$$\Phi_r(z) := \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{\binom{h}{r-1}}. \quad (7.1)$$

The case  $r = 1$  is the classical Rvachev transform. The pinned corpus already develops this Pascal hierarchy: compact support and smoothness, refinement ladders, zero divisors, spectral zeta functions, cumulants, Gaussian rank limits, Fourier envelopes, and Mellin endpoint structure. We use those theorems as a basis and ask which other exponent products can be factored through finitely many  $\Phi_r$ .

### 7.2 Integer-valued polynomials are exactly the finite Newton sector

Let  $P \in \mathbb{Q}[h]$  be integer-valued on  $\mathbb{N}_0$  and nonnegative there. Its Newton expansion is

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r}, \quad c_r = \Delta^r P(0) \in \mathbb{Z}. \quad (7.2)$$

The coefficients  $c_r$  need not be nonnegative even though every  $P(h)$  is.

**Theorem 7.1** (Newton–Rvachev factorization). *Let  $d = \deg P$  and define  $\Phi_P = \Phi_{(P(h))}$ . Then*

$$A_P(q) = \sum_{h \geq 0} P(h) q^h = \sum_{r=0}^d c_r \frac{q^r}{(1-q)^{r+1}}, \quad (7.3)$$

$$\boxed{\Phi_P(z) = \prod_{r=0}^d \Phi_{r+1}(z)^{c_r}}. \quad (7.4)$$

Although the right side may be written as a quotient, it is an entire positive-definite function because the unreduced exponent sequence  $P(h)$  is nonnegative. Its support radius, integer-zero multiplicity, spectral zeta, and even cumulants are

$$R_P = \frac{1}{2} A_P(1/2) = \sum_{r=0}^d c_r, \quad (7.5)$$

$$m_P(n) = \sum_{r=0}^d c_r \binom{\nu_2(n) + 1}{r+1}, \quad (7.6)$$

$$\mathcal{Z}_P(s) = \zeta(s) \sum_{r=0}^d c_r \frac{2^s}{(2^s - 1)^{r+1}}, \quad (7.7)$$

$$\kappa_{2j}(X_P) = \frac{B_{2j}}{2j} \sum_{r=0}^d c_r \frac{4^j}{(4^j - 1)^{r+1}}. \quad (7.8)$$

All formulas are **proved here**. Of these, (7.6) is formalized one Newton component at a time: *Fabius.weightedScaleMultiplicity\_choose* (*WeightedScaleMultiplicity.lean*) proves  $m(n) = \binom{\nu_2(n)+1}{r+1}$  for the single Pascal weight  $h \mapsto \binom{h}{r}$ , for an arbitrary base, its hockey-stick input being *Fabius.inclusivePrefixSum\_choose*. Assembling those components into the signed sum over  $r$ , and every other formula in the theorem, is not formalized.

*Proof.* The generating function  $\sum_{h \geq 0} \binom{h}{r} q^h = q^r / (1 - q)^{r+1}$  gives (7.3). Exponents add under multiplication, proving (7.4). At  $q = 1/2$ , every Newton-basis generating function has value 2, which gives (7.5). The hockey-stick identity  $\sum_{h=0}^v \binom{h}{r} = \binom{v+1}{r+1}$  proves (7.6); (7.7) and (7.8) follow by substituting  $q = 2^{-s}$  and  $q = 4^{-j}$  into (7.3).  $\square$

---

**Positive-definite quotients.** Equation (7.4) gives a systematic source of nonobvious positive-definite quotients of compactly supported entire characteristic functions. Cancellation is certified at the level of dyadic sinc exponents, not guessed from zeros after the fact. This is substantially more flexible than taking ordinary convolutions, which only permits nonnegative exponents in the Pascal basis itself.

---

### 7.3 Leading Fourier and endpoint orders

Let  $c_d = \Delta^d P(0) > 0$  and set

$$p = d + 2, \quad C_P = \frac{c_d}{(d+2)!(\log 2)^{d+1}}. \quad (7.9)$$

Lower Newton components have smaller logarithmic order, so the highest Pascal factor controls the leading scale.

**Theorem 7.2** (Leading Fourier and small-ball laws). *The sharp Pascal-hierarchy envelope theorem in the audited corpus implies*

$$\limsup_{|x| \rightarrow \infty} \frac{\log |\Phi_P(x)|}{(\log |x|)^{d+2}} = -C_P. \quad (7.10)$$

Let

$$Y_P = X_P + R_P \in [0, 2R_P], \quad G_P(x) = \mathbb{P}(Y_P \leq x).$$

The corresponding exponential Tauberian theorem of de Bruijn type [7, 13] gives

$$-\log G_P(x) \sim C_P (\log(1/x))^{d+2} \quad (x \downarrow 0). \quad (7.11)$$

Consequently the inverse endpoint satisfies

$$\boxed{\log \frac{1}{G_P^{-1}(y)} \sim \left( \frac{\log(1/y)}{C_P} \right)^{1/(d+2)} \quad (y \downarrow 0).} \quad (7.12)$$

These statements are **derived from audited corpus theorems** from the established Pascal hierarchy and the standard exponential Tauberian transfer.

*Remark 7.3.* For  $P(h) \equiv 1$ , one has  $d = 0$ ,  $c_0 = 1$ , and  $C_P = 1/(2 \log 2)$ , recovering the quadratic logarithmic scale of the classical Fabius endpoint. Increasing the polynomial degree increases the endpoint flatness from a quadratic to a higher logarithmic power.



## 7.4 The generalized lower-Lambert coordinate

The leading saddle for inverse Laplace analysis solves

$$x = C_P p \lambda^{p-1} e^{-\lambda}. \quad (7.13)$$

It has the explicit lower-branch solution

$$\lambda_P(x) = -(p-1)W_{-1}\left(-\frac{1}{p-1}\left(\frac{x}{C_P p}\right)^{1/(p-1)}\right). \quad (7.14)$$

For  $p = 2$  this is the familiar lower-Lambert coordinate of the classical Fabius analysis; the branch notation and inversion conventions follow [6]. For  $p > 2$  it gives a single closed coordinate in which the polynomial-weight endpoint transseries should be organized; the Mellin calculation in [section 10](#) explains the periodic coefficients that decorate it. Only the  $p = 2$  coordinate is formalized: `Fabius.lowerLambertW` (`LowerLambertW.lean`), a definitional alias of the library’s real lower branch `Real.lambertWLower`, inverts  $w \mapsto we^w$  on  $(-\infty, -1]$ , with functional equation `Fabius.lowerLambertW_mul_exp` and uniqueness `Fabius.lowerLambertW_unique`. The generalized coordinate (7.14) itself has no Lean counterpart.

## 8 An alternating Newton family of positive-definite Pascal quotients

The abstract Newton factorization becomes more compelling when negative coefficients are unavoidable and the quotient simplifies to something not visibly positive definite.

### 8.1 Definition and three exact factorizations

For an even integer  $d \geq 0$ , define the generalized-binomial polynomial sequence

$$P_d(h) = \binom{h-1}{d} \quad (h \in \mathbb{N}_0). \quad (8.1)$$

Here  $\binom{-1}{d} = (-1)^d = 1$ . Thus

$$P_d(0) = 1, \quad P_d(1) = \cdots = P_d(d) = 0, \quad P_d(h) = \binom{h-1}{d} > 0 \quad (h \geq d+1). \quad (8.2)$$

The parity restriction on  $d$  is precisely what makes the  $h = 0$  exponent nonnegative.

**Theorem 8.1** (Alternating Newton family). *Let  $d \geq 0$  be even and put  $\Psi_d = \Phi_{P_d}$ . Then*

$$A_d(q) = 1 + \left(\frac{q}{1-q}\right)^{d+1}, \quad (8.3)$$

$$P_d(h) = \sum_{r=0}^d (-1)^{d-r} \binom{h}{r}, \quad (8.4)$$

$$\Psi_d(z) = \text{sinc}(z) \Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}}. \quad (8.5)$$

In particular,

$$\Psi_2(z) = \frac{\Phi_1(z)\Phi_3(z)}{\Phi_2(z)}. \quad (8.6)$$

Every expression in (8.5) is the characteristic function of a  $C_c^\infty$  probability density supported on  $[-1, 1]$ . These statements are [proved here](#).



*Proof.* The positive-index part is

$$\sum_{h \geq 1} \binom{h-1}{d} q^h = q \sum_{k \geq 0} \binom{k}{d} q^k = \frac{q^{d+1}}{(1-q)^{d+1}},$$

and the exceptional  $h = 0$  term equals 1, proving (8.3). The finite identity (8.4) follows from the backward-shift formula for binomial polynomials, or by comparing coefficients in  $(1+t)^{h-1} = (1+t)^h/(1+t)$  through degree  $d$ . The first product formula separates  $h = 0$  and reindexes  $h = k + 1$ :

$$\Psi_d(z) = \text{sinc}(z) \prod_{k \geq 0} \text{sinc}\left(\frac{z/2}{2^k}\right)^{\binom{k}{d}} = \text{sinc}(z) \Phi_{d+1}(z/2).$$

The second is [theorem 7.1](#). Since the unreduced exponents (8.2) are nonnegative,  $\Psi_d$  is positive definite. Finally,  $A_d(1/2) = 2$ , so the support radius is one.  $\square$

**Corollary 8.2** (Concrete random-series model). *Let  $U \sim \text{Unif}[-1/2, 1/2]$  and let  $X_{d+1}$  have characteristic function  $\Phi_{d+1}$ , independently of  $U$ . Then*

$$Y_d := U + \frac{1}{2}X_{d+1} \tag{8.7}$$

*has characteristic function  $\Psi_d$ . This exhibits the hidden positivity of the alternating Pascal quotient without any appeal to a zero-by-zero cancellation argument.*

## 8.2 Sparse enhancement of the classical zero divisor

**Theorem 8.3** (Exact arithmetic spectrum of  $\Psi_d$ ). *For every positive integer  $n$ ,*

$$\boxed{\text{ord}_{z=n} \Psi_d(z) = 1 + \binom{\nu_2(n)}{d+1}.} \tag{8.8}$$

*Its spectral zeta and even cumulants are*

$$\mathcal{Z}_d(s) = \zeta(s) \left(1 + \frac{1}{(2^s - 1)^{d+1}}\right), \tag{8.9}$$

$$\kappa_{2j}(Y_d) = \frac{B_{2j}}{2j} \left(1 + \frac{1}{(4^j - 1)^{d+1}}\right), \quad \kappa_{2j+1}(Y_d) = 0. \tag{8.10}$$

*In particular,*

$$\text{Var}(Y_d) = \frac{1}{12} \left(1 + 3^{-(d+1)}\right). \tag{8.11}$$

*These are [proved here](#). The hockey-stick step behind (8.8) is formalized as `Fabius.inclusivePrefixSum_choose` (`WeightedScaleMultiplicity.lean`), which proves  $\sum_{h \leq v} \binom{h}{d} = \binom{v+1}{d+1}$  for a general Pascal column; together with `Fabius.weightedScaleMultiplicity` it supplies the whole arithmetic content of (8.8) apart from the exceptional  $h = 0$  term. The spectral zeta (8.9), the cumulants (8.10) and the variance (8.11) are not formalized.*

*Proof.* Use [theorem 3.1](#) and the hockey-stick identity:

$$\sum_{h=0}^v P_d(h) = 1 + \sum_{h=1}^v \binom{h-1}{d} = 1 + \binom{v}{d+1}.$$

The other formulas are obtained by substituting  $q = 2^{-s}$  and  $q = 4^{-j}$  into (8.3); the variance is the  $j = 1$  case because  $B_2/2 = 1/12$ .  $\square$

**Example 8.4** (The first genuinely alternating quotient). For  $d = 2$ ,

$$P_2 = (1, 0, 0, 1, 3, 6, 10, 15, \dots), \quad A_2(q) = 1 + \frac{q^3}{(1-q)^3}.$$

Every odd integer and every integer with  $\nu_2(n) \leq 2$  remains a simple zero. The first extra multiplicity occurs at  $n = 8$ , where the order is 2; at  $n = 16$  it is 5; at  $n = 32$  it is 11. Yet the same entire function is the quotient  $\Phi_1 \Phi_3 / \Phi_2$ .

### 8.3 Automatic lobe signs and sparse Prouhet order

For  $h \geq 1$ , Lucas' theorem gives

$$P_d(h) \equiv 1 \pmod{2} \iff d \text{ is a binary submask of } h-1. \quad (8.12)$$

This parity sequence is periodic, hence the Fourier-lobe sign sequence of  $\Psi_d$  is 2-automatic by [theorem 3.3](#). More precisely, among a complete sufficiently long binary block of values of  $h-1$ , the forced one-bits of  $d$  reduce the density by a factor  $2^{-s_2(d)}$ .

**Corollary 8.5** (Linear Prouhet order with a digital density). *Let*

$$\omega_d(L) = \#\{0 \leq h < L : P_d(h) \text{ is odd}\}.$$

*Then*

$$\omega_d(L) = 2^{-s_2(d)}L + \mathcal{O}_d(1). \quad (8.13)$$

*The length- $2^L$  lobe-sign block attached to  $P_d$  annihilates every power of degree  $< \omega_d(L)$ , and its first surviving power moment is given exactly by (3.16). Thus the Prouhet order remains linear in  $L$ , but its slope is the binary cylinder density selected by  $d$ . The Lucas input (8.12) is formalized as `Fabiús. odd_choose_iff_bitSupport_subset` (`ThueMorseLucasSupport.lean`), in the submask form “ $\binom{n}{k}$  is odd exactly when the binary support of  $k$  is contained in that of  $n$ ”, with `Fabiús. oddBinomialIndices_eq_image_powerset` parametrizing the odd positions of a Pascal row by the powerset of the bit support. The density count (8.13) is not formalized.*

*Proof.* In each block of  $2^k$  consecutive values of  $h-1$ , where  $k$  exceeds the binary length of  $d$ , precisely  $2^{k-s_2(d)}$  numbers contain all one-bits of  $d$ . Partial end blocks account for the bounded error. The Prouhet statement is [theorem 3.4](#).  $\square$

## 9 Gaussian boundary layers and convergence to a box density

The factorization  $Y_d = U + X_{d+1}/2$  makes the large- $d$  limit geometrically transparent. The Pascal variable collapses to zero after normalization, so the new smooth compactly supported density tends to the discontinuous uniform density. The collapse is Gaussian on its natural microscopic scale.

### 9.1 A quantitative central limit theorem for the Pascal component

Let  $X_r$  denote the centered random variable with characteristic function  $\Phi_r$ . It is the sum of  $\binom{h}{r-1}$  independent copies of  $2^{-h}U$ , with  $U \sim \text{Unif}[-1/2, 1/2]$ .

**Theorem 9.1** (Variance and Berry–Esseen rate). *For  $r \geq 1$ ,*

$$\sigma_r^2 := \text{Var}(X_r) = 3^{-(r+1)}, \quad (9.1)$$

$$\sum \mathbb{E}|\text{summand}|^3 = \frac{1}{4 \cdot 7^r}. \quad (9.2)$$

Hence any valid universal Berry–Esseen constant  $C_{\text{BE}}$  gives

$$\sup_{y \in \mathbb{R}} \left| \mathbb{P} \left( \frac{X_r}{\sigma_r} \leq y \right) - \Phi_G(y) \right| \leq C_{\text{BE}} \frac{3\sqrt{3}}{4} \left( \frac{3\sqrt{3}}{7} \right)^r. \quad (9.3)$$

This is **proved here**.

*Proof.* Since  $\text{Var}(U) = 1/12$  and  $\mathbb{E}|U|^3 = 1/32$ ,

$$\begin{aligned} \text{Var}(X_r) &= \frac{1}{12} \sum_{h \geq 0} \binom{h}{r-1} 4^{-h} = \frac{1}{12} \frac{4^{-(r-1)}}{(1 - 1/4)^r} = 3^{-(r+1)}, \\ \sum \mathbb{E}|\text{summand}|^3 &= \frac{1}{32} \sum_{h \geq 0} \binom{h}{r-1} 8^{-h} = \frac{1}{32} \frac{8^{-(r-1)}}{(1 - 1/8)^r} = \frac{1}{4 \cdot 7^r}. \end{aligned}$$

Divide the latter by  $\sigma_r^3$  and apply the Berry–Esseen theorem for independent, not necessarily identically distributed summands [10].  $\square$

*Remark 9.2.* The standardized fourth cumulant is also exact:

$$\frac{\kappa_4(X_r)}{\sigma_r^4} = -2 \left( \frac{3}{5} \right)^{r+1}. \quad (9.4)$$

It decays more slowly than the Berry–Esseen upper bound and predicts the first Edgeworth correction in [theorem 12.1](#).

## 9.2 Boundary scaling of the smooth box approximants

Let  $f_d$  be the density of  $Y_d$ , and put

$$\tau_d = \frac{1}{2} 3^{-(d+2)/2}. \quad (9.5)$$

This is the standard deviation of  $X_{d+1}/2$ .

**Theorem 9.3** (Gaussian boundary layers). *For each fixed bounded interval of  $y$  values,*

$$f_d \left( -\frac{1}{2} + \tau_d y \right) = \Phi_G(y) + \mathcal{O} \left( \left( \frac{3\sqrt{3}}{7} \right)^{d+1} \right), \quad (9.6)$$

$$f_d \left( \frac{1}{2} + \tau_d y \right) = \bar{\Phi}_G(y) + \mathcal{O} \left( \left( \frac{3\sqrt{3}}{7} \right)^{d+1} \right), \quad (9.7)$$

where the implied constant is universal. Away from any shrinking neighborhood of the two endpoints,  $f_d$  converges uniformly to the box density  $\mathbf{1}_{[-1/2, 1/2]}$ . This is **proved here**.

*Proof.* Let  $T_d = X_{d+1}/2$ . Convolution with the unit-height box gives the exact formula

$$f_d(x) = \mathbb{P}(x - 1/2 \leq T_d \leq x + 1/2). \quad (9.8)$$

At  $x = -1/2 + \tau_d y$ , the lower endpoint of the window lies below the support of  $T_d$  for all sufficiently large  $d$ , while the upper endpoint is  $\tau_d y$ . Thus the value is the CDF of  $T_d/\tau_d$  at  $y$ . The right endpoint is the complementary CDF. Apply (9.3). If  $x$  stays a distance much larger than  $\tau_d$  from  $\pm 1/2$ , the standardized threshold tends uniformly to  $\pm\infty$ , giving the final assertion.  $\square$

### 9.3 Exact global distances from the limiting box

**Theorem 9.4** (Exact  $L^1$  identity and transport bounds). *Let  $b(x) = \mathbf{1}_{[-1/2, 1/2]}(x)$ . Then*

$$\|f_d - b\|_{L^1(\mathbb{R})} = \mathbb{E}|X_{d+1}|. \quad (9.9)$$

Consequently,

$$\|f_d - b\|_1 \sim \sqrt{\frac{2}{\pi}} 3^{-(d+2)/2}, \quad (9.10)$$

$$d_{\text{TV}}(\text{Law}(Y_d), \text{Law}(U)) \sim \frac{1}{\sqrt{2\pi}} 3^{-(d+2)/2}, \quad (9.11)$$

$$W_2(\text{Law}(Y_d), \text{Law}(U)) \leq \frac{1}{2} 3^{-(d+2)/2}. \quad (9.12)$$

These statements are *proved here*.

*Proof.* For a fixed shift  $t \in [-1/2, 1/2]$ , the two unit intervals  $[-1/2, 1/2]$  and  $[t - 1/2, t + 1/2]$  overlap in length  $1 - |t|$ . Thus the shifted box puts mass  $|t|$  outside the unshifted box. Averaging over  $T_d = X_{d+1}/2$ , and using equality of the inside deficit and outside excess, gives

$$\|f_d - b\|_1 = 2\mathbb{E}|T_d| = \mathbb{E}|X_{d+1}|.$$

The normalized variables  $X_r/\sigma_r$  converge to a standard Gaussian by [theorem 9.1](#); their second moments are uniformly one, so the absolute first moments converge as well. This proves (9.10), and total variation is half the  $L^1$  distance. Finally couple  $Y_d = U + T_d$  with  $U$  itself and use  $W_2^2 \leq \mathbb{E}T_d^2 = \tau_d^2$ .  $\square$

---

**A smooth-to-discontinuous limit with an exact microscopic profile.** The family  $f_d$  consists entirely of compactly supported  $C^\infty$  densities, yet it tends to a discontinuous box. The global error is exactly an absolute moment of a Pascal–Rvachev variable, while the local endpoint transition is a Gaussian survival function. The same Pascal hierarchy therefore controls both the positive-definite quotient and the singular limit it regularizes.

---

## 10 Mellin reflection, periodic endpoint hierarchies, and Lambert inversion

### 10.1 The endpoint Laplace transform

Shift the random series to its left endpoint:

$$Z_a = X_a + R_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} V_{h,j}, \quad V_{h,j} \sim \text{Unif}[0, 1]. \quad (10.1)$$

Its Laplace transform is

$$L_a(t) = \mathbb{E}e^{-tZ_a} = \prod_{h \geq 0} \left( \frac{1 - e^{-t/2^h}}{t/2^h} \right)^{a_h}. \quad (10.2)$$

Set

$$\lambda(u) = \log \frac{1 - e^{-u}}{u}. \quad (10.3)$$

Then  $\log L_a(t) = \sum_h a_h \lambda(t/2^h)$ .

**Theorem 10.1** (Spectral–endpoint Mellin reflection). *Assume  $a_h$  has polynomial growth. On the strip  $-1 < \Re w < 0$ ,*

$$\int_0^\infty \lambda(u) u^{w-1} du = -\Gamma(w) \zeta(w+1), \quad (10.4)$$

$$\int_0^\infty \log L_a(t) t^{w-1} dt = -\Gamma(w) \zeta(w+1) A(2^w). \quad (10.5)$$

Using the spectral zeta from (3.3), this becomes the reflection formula

$$\mathcal{M}[\log L_a](w) = -\Gamma(w) \zeta(w+1) \frac{\mathcal{Z}_a(-w)}{\zeta(-w)}. \quad (10.6)$$

Thus the endpoint asymptotic and the integer-zero divisor are two Mellin views of the same generating function. The contour-transfer interpretation follows the harmonic-sum Mellin framework of [8]. This is **proved here**. Neither (10.4) nor (10.5) is formalized. The library does prove the sibling real-variable integral `Fabius.bose_mellin_integral_zeta` (`MellinBose.lean`),  $\int_0^\infty x^{a-1} \log(1 - e^{-x}) dx = -\Gamma(a) \zeta(a+1)$  for real  $a > 0$ : the same  $\Gamma \zeta$  right-hand side, but for the bare kernel  $\log(1 - e^{-u})$  and on the half-line to the right of the strip in (10.4), so it is a companion identity and not (10.4) itself. The double pole of  $\Gamma(w) \zeta(w+1)$  at  $w = 0$  that governs the hierarchy below is regularized by `Fabius.tendsto_gammaZetaRegularProduct_finitePart` (`MellinFinitePart.lean`). The subtraction of  $\log u$ , the continuation to  $-1 < \Re w < 0$ , and the complex-variable form all remain to be supplied.

*Proof.* The kernel behaves as  $-u/2 + \mathcal{O}(u^2)$  at zero and as  $-\log u + \mathcal{O}(e^{-u})$  at infinity, so the stated strip is maximal for the elementary integral. Integration by parts followed by the classical Mellin integral for  $(e^u - 1)^{-1}$  gives (10.4). For (10.5), absolute convergence permits summation under the integral and the substitution  $u = t/2^h$  contributes the factor  $2^{hw}$ . Hence the layer sum is  $A(2^w)$ . The last formula follows from  $\mathcal{Z}_a(-w) = \zeta(-w) A(2^w)$ .  $\square$

## 10.2 Pole order is the endpoint logarithmic order

Suppose now that  $a_h = P(h)$  has degree  $d$  and Newton leading coefficient  $c_d > 0$ . Near  $w = 0$ ,  $A_P(2^w)$  has a pole of order  $d+1$ , while  $\Gamma(w) \zeta(w+1)$  has a double pole. At the nonzero dyadic lattice

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (10.7)$$

$A_P(2^w)$  has poles of order  $d+1$  and the Gamma–zeta factor is regular.

**Theorem 10.2** (Polynomial and periodic hierarchy). *As  $t \rightarrow \infty$ , Mellin inversion of (10.5) gives an expansion of the form*

$$\log L_P(t) = -C_P(\log t)^{d+2} + \sum_{j=0}^{d+1} \alpha_j (\log t)^j + \sum_{k \neq 0} t^{-\chi_k} Q_k(\log t) + \mathcal{E}_P(t), \quad (10.8)$$

where  $C_P$  is (7.9), every  $Q_k$  is a polynomial of degree at most  $d$ , and the remainder comes from the exponentially small endpoint correction and from poles farther to the right after contour displacement. The Fourier coefficients of the periodic-polynomial part are explicit residues of  $-\Gamma(w) \zeta(w+1) A_P(2^w)$ .

In particular, degree  $d$  produces:

- a nonperiodic logarithmic polynomial of degree  $d+2$ ;
- log-periodic coefficient polynomials of degree  $d$ ;
- the generalized lower-Lambert saddle (7.14).

This is **proved hereat** the structural level; obtaining a chosen number of explicit coefficients is a finite residue calculation.

*Proof.* The pole at zero has order  $(d + 1) + 2 = d + 3$ , and a Mellin pole of order  $m$  produces a polynomial of degree  $m - 1$  in  $\log t$ . Each nonzero  $\chi_k$  has order  $d + 1$  and contributes  $t^{-\chi_k}$  times a polynomial of degree  $d$ . Since  $t^{-\chi_k} = \exp(-2\pi i k \log_2 t)$ , these terms are periodic in  $\log_2 t$  with polynomial amplitudes. The leading coefficient is inherited from the highest Newton component and agrees with the audited rank- $(d + 1)$  Pascal law, giving  $C_P$ .  $\square$

*Remark 10.3.* For the classical case  $d = 0$ , the periodic terms are scalar Fourier modes. For  $d = 1$  they carry affine functions of  $\log t$ ; in general they are degree- $d$  periodic polynomials. This is a genuine hierarchy rather than a cosmetic perturbation of the rank-one Fabius factor.

### 10.3 Generalized Bernoulli polynomials generate the residues

Define the Nörlund generalized Bernoulli polynomials [9] by

$$\left(\frac{u}{e^u - 1}\right)^\alpha e^{xu} = \sum_{n \geq 0} B_n^{(\alpha)}(x) \frac{u^n}{n!}. \quad (10.9)$$

Every Newton component of  $A_P(e^u)$  has the Laurent expansion

$$\frac{e^{ru}}{(1 - e^u)^{r+1}} = (-1)^{r+1} u^{-r-1} \sum_{n \geq 0} B_n^{(r+1)}(r) \frac{u^n}{n!}. \quad (10.10)$$

**Algorithm 10.4** (Bernoulli–Bell endpoint coefficient engine). For a polynomial weight  $P$  and a desired truncation order:

1. compute its integer Newton coefficients  $c_r$ ;
2. expand  $A_P(e^u)$  using (10.10);
3. multiply by the Laurent series of  $-\Gamma(w)\zeta(w + 1)$  with  $u = w \log 2$ ;
4. take the residue at  $w = 0$  for the nonperiodic logarithmic polynomial and at  $w = \chi_k$  for the periodic coefficient polynomials;
5. solve the saddle recursively in the coordinate (7.14);
6. exponentiate and compose the resulting formal series with complete Bell polynomials.

All ingredients are finite algebra once the desired order and Fourier modes are fixed.

---

**Bernoulli and Bell roles.** Ordinary Bernoulli numbers enter the cumulants through (4.2); generalized Bernoulli polynomials encode the higher-order poles of the rational Newton generating function; complete Bell polynomials convert logarithmic cumulants and saddle corrections into moments, amplitudes, and inverse-series coefficients. The three appearances are different faces of one coefficient calculus rather than unrelated special-function decorations.

---

### 10.4 Inverse endpoint consequence

Combining (10.8) with saddle inversion yields an all-orders formal expansion for  $G_P$  and its inverse in the generalized Lambert coordinate. The leading term was proved in theorem 7.2; the exact periodic-polynomial transseries requires the uniform contour estimates already carried out in the classical corpus to be lifted to higher-order poles. We isolate that analytic completion as theorem 12.4 rather than silently promoting a formal residue calculation to a theorem.

## 11 Numerical experiments and reproducibility audit

The numerical work was designed to falsify algebraic mistakes, not to replace proofs. The accompanying script `numerical_experiments.py` uses 100-digit arithmetic for the  $q$ -Richardson identities and deterministic FFT inversion for the boundary-layer curves. No Monte Carlo sampling is used. The archive contains the script, generated CSV tables, verification log, and both figures.

### 11.1 Exact identity checks

The script independently verifies:

1. the closed Lagrange weights (5.2);
2. annihilation of  $1, q^h, \dots, q^{rh}$  and the exact survivor (5.4);
3. the generating function (8.3) for  $d = 0, 2, 4, 6$  at several high-precision arguments;
4. the multiplicity formula (8.8);
5. the complete transformed series (5.6);
6. the condition-number product (5.10).

Every check passed at tolerances between  $10^{-65}$  and  $10^{-80}$ .

### 11.2 Tomography of the first hidden exponent

For the new  $d = 2$  family,

$$A_2(x) = 1 + \left( \frac{x}{1-x} \right)^3, \quad a_0 = 1, \quad q = 1/4.$$

Table 1 records  $\mathcal{R}_{r,1}[A_2] - a_0$ . The exact alternating sign is visible, and six interpolation steps reduce an initial  $3.7 \times 10^{-2}$  error to  $3.6 \times 10^{-16}$  without a large weight norm.

Table 1: Certified  $q$ -Richardson errors for  $A_2$  at  $m = 1, q = 1/4$ .

| Order $r$ | $\mathcal{R}_{r,1}[A_2] - 1$    | Predicted side |
|-----------|---------------------------------|----------------|
| 0         | $3.7037037037 \times 10^{-2}$   | upper          |
| 1         | $-1.1950617284 \times 10^{-2}$  | lower          |
| 2         | $6.9704584427 \times 10^{-4}$   | upper          |
| 3         | $-6.0938671972 \times 10^{-6}$  | lower          |
| 4         | $1.0586844464 \times 10^{-8}$   | upper          |
| 5         | $-4.0454865708 \times 10^{-12}$ | lower          |
| 6         | $3.5620070678 \times 10^{-16}$  | upper          |

The limiting interpolation condition number was recomputed as

$$1.96926035366826943194719369697 \dots,$$

in agreement with (5.11). This is small enough that the observed convergence reflects the exact tail annihilation rather than a cancellation between huge floating-point weights.

### 11.3 The sparse zero spectrum for $d = 2$

The generated file `zero_multiplicities_d2.csv` compares the prefix sum of the original exponents with  $1 + \binom{\nu_2(n)}{3}$ . The first exceptional values are

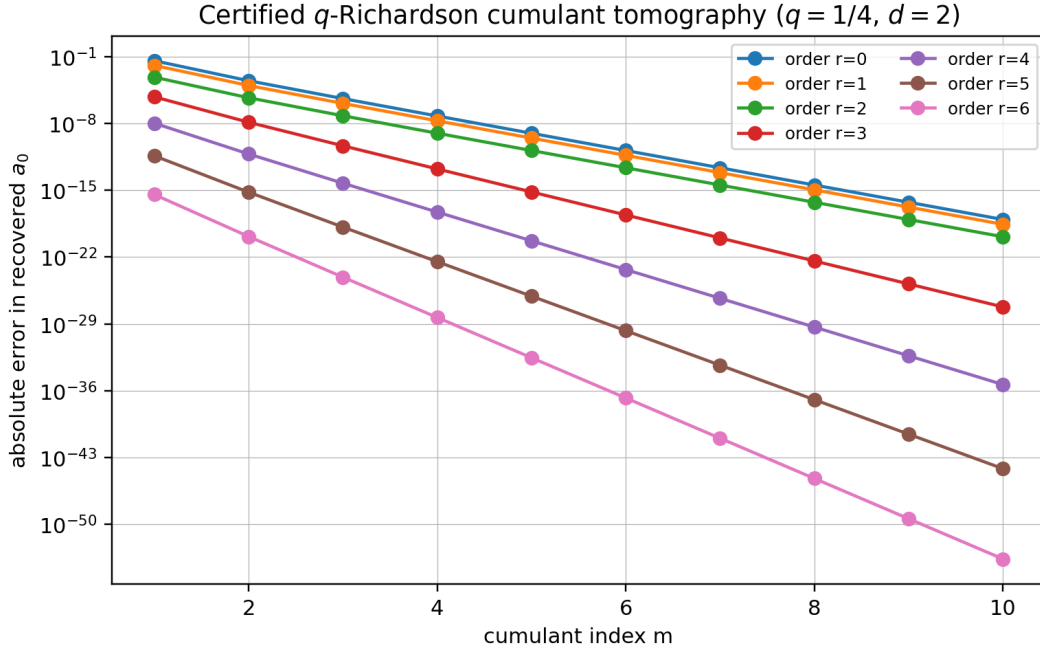


Figure 1: Absolute errors in the recovery of  $a_0$  from normalized cumulants  $A_2(4^{-m})$ . The CSV file retains the signs; the semilogarithmic plot shows the many orders of magnitude gained by increasing either the cumulant index  $m$  or the interpolation order  $r$ .

| $n$ | $\nu_2(n)$ | zero order |
|-----|------------|------------|
| 8   | 3          | 2          |
| 16  | 4          | 5          |
| 24  | 3          | 2          |
| 32  | 5          | 11         |
| 40  | 3          | 2          |
| 48  | 4          | 5          |
| 64  | 6          | 21         |

The spectrum is therefore classical at almost every integer but develops rapidly increasing multiplicity on a sparse tower of highly divisible integers.

#### 11.4 Deterministic boundary-layer inversion

For ranks  $r = 3, 5, 7, 9$  (equivalently  $d = 2, 4, 6, 8$ ), the script evaluates the exact truncated characteristic product of  $X_r$ , inverts it on a nonaliasing FFT interval, normalizes the resulting density, and compares its standardized CDF with the Gaussian. The largest observed CDF errors on  $[-4, 4]$  are shown in table 2.

Table 2: Deterministic FFT check of the standardized Pascal CDF.

| Rank $r$ | family degree $d = r - 1$ | maximum CDF error on $[-4, 4]$ |
|----------|---------------------------|--------------------------------|
| 3        | 2                         | $7.6043 \times 10^{-3}$        |
| 5        | 4                         | $2.4624 \times 10^{-3}$        |
| 7        | 6                         | $9.3598 \times 10^{-4}$        |
| 9        | 8                         | $6.4752 \times 10^{-4}$        |



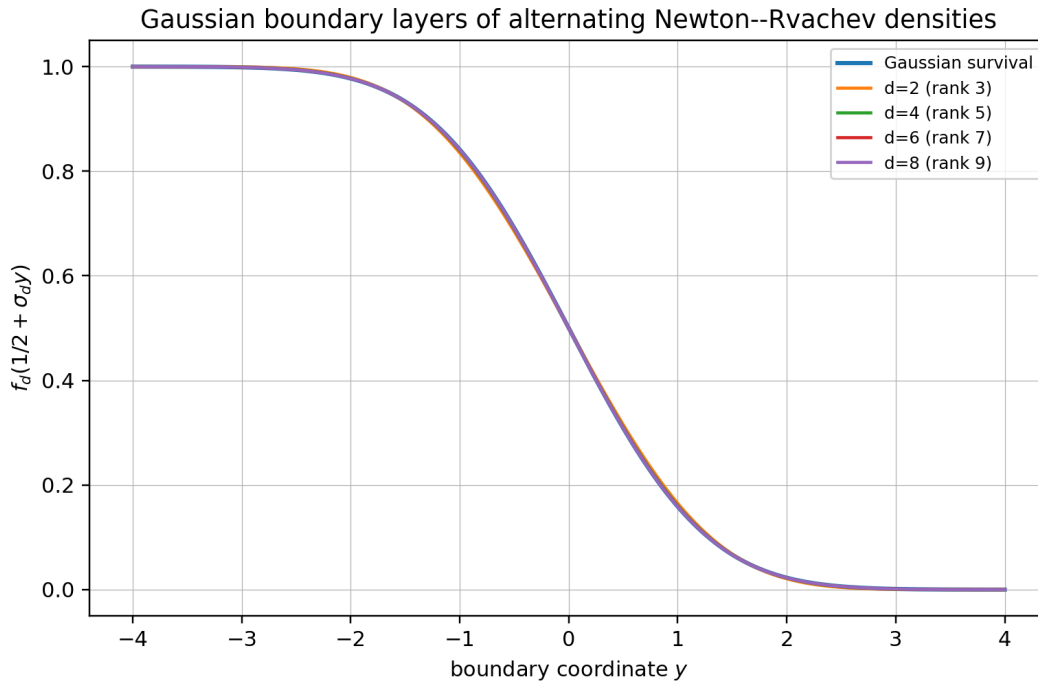


Figure 2: Right-edge boundary profiles  $f_d(1/2 + \tau_d y)$  compared with the Gaussian survival function  $1 - \Phi_G(y)$ . The curves are obtained by deterministic Fourier inversion of the Pascal–Rvachev characteristic products.

The errors decrease consistently with [theorem 9.1](#). The theorem’s universal Berry–Esseen bound is intentionally conservative; the exact standardized fourth cumulant (9.4) gives a better explanation of the observed first correction.

### 11.5 Reproduction command

From the extracted archive, the full experiment can be rerun with

```
python numerical_experiments.py --output-dir reproduced-data
```

The script checks its own exact identities with assertions before writing any figure. A failure therefore stops the run rather than silently producing visually plausible output.

## 12 Conjectures and frontier problems

The following statements are deliberately separated from the proved theorem package. Each is supported by the exact structure above, but each still requires a substantive analytic, optimization, or arithmetic argument.

**Conjecture 12.1** (Uniform Edgeworth boundary expansion). *Let  $r = d + 1$  and  $\tau_d$  be (9.5). For every fixed  $J$  and every fixed number of  $y$ -derivatives, the two boundary profiles admit a uniform expansion*

$$f_d\left(\frac{1}{2} + \tau_d y\right) = \bar{\Phi}_G(y) + \varphi(y) \sum_{j=1}^J E_j(y; r) + o\left((3/5)^{J(r+1)}\right), \quad (12.1)$$

where  $\varphi$  is the Gaussian density and each  $E_j$  is an explicit Hermite polynomial whose coefficients are Bell polynomials in the standardized cumulants. The first correction is governed by  $\kappa_4/\sigma^4 = -2(3/5)^{r+1}$ .

**Why it is plausible.** The triangular array has bounded summands, an entire characteristic function, and standardized cumulants that decay exponentially with rank. A Fourier proof should be considerably cleaner than a generic Edgeworth theorem because the exact product provides uniform control in both the central and large-frequency regions.

**Conjecture 12.2** (Minimax property of alternating  $q$ -tomography). *Fix  $q, m, r$  and the cone*

$$\mathcal{C}_M = \left\{ A(x) = \sum_{h \geq 0} a_h x^h : a_h \geq 0, A(q^m) \leq M \right\}.$$

*Among all linear estimators using  $A(q^m), \dots, A(q^{m+r})$  and exact on polynomials of degree at most  $r$ , the Lagrange estimator (5.5) minimizes the worst one-sided error on  $\mathcal{C}_M$ . Moreover, consecutive odd/even transforms form the narrowest possible universal bracket under these constraints.*

**Why it is plausible.** The remainder kernel in (5.6) has one sign on the coefficient cone and is the unique kernel annihilating the first  $r$  monomials. That uniqueness is already formal: `Fabius.eq_lagrangeEvalWeight_of_moments` (`GeometricLagrange.lean`) shows that on any finite family of distinct field-valued nodes the Lagrange row is the only row with the prescribed moments, with no nonzeroness or positivity assumption on the competing weights. A proof of the conjecture should be expressible as a finite-dimensional linear-programming duality or a Chebyshev-system extremal argument.

**Conjecture 12.3** (Rational refinement transseries). *Let  $A$  be rational, nonnegative coefficientwise, and analytic at the origin. Then the complete large- $t$  transseries of  $\log L_a(t)$  is determined functorially by the poles of  $A$ : each pole  $\rho$  of order  $m$  produces vertical Mellin lattices*

$$w = \frac{\log \rho + 2\pi i k}{\log 2}, \quad k \in \mathbb{Z},$$

*with degree- $(m-1)$  logarithmic amplitudes. Poles on the unit circle yield bounded log-periodic phases; poles of other moduli yield additional power scales. No further frequencies occur.*

**Research route.** Start from (10.5); establish a common meromorphic continuation and vertical bounds for the rational factor, Gamma, and zeta; then shift the Mellin contour across a finite set of pole lattices. The main technical issue is uniform summability of the residue series when several pole lattices have the same real part.

**Conjecture 12.4** (All-orders generalized inverse-Fabius expansion). *For every nonnegative integer-valued polynomial  $P$  of degree  $d$ , the endpoint CDF  $G_P$  and its inverse admit all-orders expansions in the generalized lower-Lambert coordinate (7.14). Their coefficients are finite differential polynomials in the degree- $d$  periodic amplitudes of (10.8); equivalently, they are algorithmically generated from Nörlund Bernoulli polynomials, Gamma-zeta jets, and complete Bell polynomials. The expansion is uniform under bounded shifts of the Lambert phase.*

**Why it matters.** This would extend the corpus's inverse Fabius theory from the rank-one quadratic logarithmic saddle to every polynomial Newton sector and would make the Lambert–Bernoulli–Bell connection exact rather than merely formal.

**Open problem 12.5** (The Newton positivity cone). *Classify the integer vectors  $(c_0, \dots, c_d)$  for which*

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r} \geq 0 \quad (h \in \mathbb{N}_0).$$

Describe the extreme rays, the indecomposable characteristic functions  $\prod_r \Phi_{r+1}^{c_r}$ , and the minimal Pascal numerator and denominator after all entire cancellations. Determine whether the alternating vectors  $((-1)^d, (-1)^{d-1}, \dots, 1)$  for even  $d$  generate distinguished extreme rays under a natural support normalization.

**Open problem 12.6** (Arithmetic of generalized half-integer spectra). For a rational generating function  $A$  with integer coefficients, classify the fields, traces, and norms of

$$\Phi_a\left(k + \frac{1}{2}\right).$$

The audited half-integer report handles the classical exponent sequence. The master zero/cumulant dictionary suggests that rational  $A$  should produce finite-dimensional cyclotomic trace recurrences, but the orders and positivity constraints will depend on the pole data of  $A$ .

**Open problem 12.7** (Optimal smooth approximation of the box within the Newton family). At fixed support and fixed highest Newton degree, choose a nonnegative integer-valued polynomial  $P$  to minimize one of

$$\|f_P - b\|_1, \quad W_2(\text{Law}(X_P), \text{Law}(U)), \quad \sup_{|\xi| \leq T} |\Phi_P(\xi) - \text{sinc}(\xi)|.$$

The family  $P_d(h) = \binom{h-1}{d}$  is natural because it appends one rescaled Pascal law to a box, but it is not yet known to be extremal under any of these criteria.

## 13 Formalization and computational research program

The theorem package is unusually suitable for staged Lean formalization: its algebraic core is finite, and the analytic results can be added only after the exact identities are stable. A dependency-aware order is as follows.

### 13.1 Phase I: finite algebra and digital combinatorics

1. Define exponent prefixes and prove  $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$  for finite products.
2. Prove the finite zero-count identity  $M_a(N) = \sum_h a_h \lfloor N/2^h \rfloor$  and the binary defect (3.8).
3. Define the generalized sign character and prove (3.12); formalize the automaticity equivalence for eventually periodic parity words.
4. Prove the finite sign-product factorization (3.14), its exact order at one, and the first surviving moment (3.16).
5. The geometric Lagrange weights, Gaussian-binomial closed form, exact principal specialization, survivor moments, finite residual sign, and exact finite norm are now formalized. The exact transformed series is also formalized under unconditional summability in `AnalyticSeriesFilter.lean`; the report's positivity, sign-certified bound, uniform estimates, and sinc-product application remain open.

These statements require only finite sums, polynomials, valuations, and existing  $q$ -binomial infrastructure. Several of them are in fact already discharged in the library, in greater generality than is needed here: step 1 and the floor identity of step 2 by `Fabijs.weightedScaleMultiplicity` and `Fabijs.sum_range_weightedScaleMultiplicity`; the sharp moment of step 4 by `Fabijs.sum_powerset_neg_one_pow_pow_card`, together with its vanishing companion `Fabijs.sum_powerset_neg_one_pow_eval_of_degree_lt`; and the annihilation and first surviving moment of step 5 by `Fabijs.sum_geometricLagrangeWeight_mul_pow` and

`Fabijs.sum_geometricLagrangeWeight_mul_pow_succ`, with `Fabijs.geometricLagrange_richardson_exact` already giving module-valued Richardson exactness. What genuinely remains in Phase I is the binary defect (3.8), the automaticity equivalence of step 3, the sign-product factorization (3.14) and its free-even-layer count, and the report-specific ordered sign and bound built on the now-formal general- $q$  survivor. The closed Gaussian-binomial form and every positive survivor moment are no longer Phase I obligations.

### 13.2 Phase II: probability and entire products

1. Construct  $X_a$  from finite uniform sums and pass to the bounded infinite series using the summability hypothesis  $A(1/2) < \infty$ .
2. Prove local uniform convergence of  $\Phi_a$  and the convolution-monoid law.
3. Derive cumulants from the uniform Bernoulli expansion and moments from the existing Bell-polynomial library.
4. Formalize identifiability first from zero multiplicities, then from analytic samples  $A(4^{-j})$ .
5. Build polynomial exponent sequences from Newton expansions and prove the entire cancellation behind (7.4).

### 13.3 Phase III: asymptotics and inverse functions

1. Reuse the corpus’s Mellin kernel theorem to prove (10.5) for polynomial weights.
2. Formalize pole orders and the leading coefficient  $C_P$  before attempting the full periodic residue expansion.
3. Transfer the existing Pascal-hierarchy Fourier envelope to a general polynomial by the Newton factorization and lower-order comparison.
4. Prove the leading small-ball and inverse laws. The library contains no de Bruijn–Voss exponential Tauberian transfer, so this step must begin by isolating that theorem as a reusable library component.
5. Only then lift the classical all-orders saddle and inverse-Lambert machinery to the higher-order pole hierarchy.

### 13.4 Certified computation

The tomography algorithm is a natural application for exact rationals and interval arithmetic. At  $q = 1/4$  all Lagrange weights are rational, as are the normalized cumulants of integer exponent sequences. A verified executable can therefore recover coefficients with no floating-point assumptions. For asymptotic experiments, Arb-style ball arithmetic would certify Mellin residues, Lambert branches, and boundary-layer errors. The supplied Python script is a reproducible exploratory implementation, not a substitute for that final certification layer.

## 14 Conclusion

The central result of this report is a change of viewpoint. Instead of treating the classical Rvachev transform, the Pascal hierarchy, its zero spectrum, its cumulants, and its endpoint asymptotics as separate constructions, one begins with the exponent generating function

$$A(q) = \sum_{h \geq 0} a_h q^h.$$

The same  $A$  then appears at four geometrically related arguments:

$$\begin{aligned} A(1/2) &: \text{support radius,} \\ A(2^{-s}) &: \text{spectral zeta and zero divisor,} \\ A(4^{-j}) &: \text{even cumulants,} \\ A(2^w) &: \text{endpoint Mellin transform.} \end{aligned} \tag{14.1}$$

Its coefficient parity is the Fourier-lobe character, its prefix sums are zero multiplicities, its rationality is finite refinement rank, and its Newton coordinates are Pascal–Rvachev exponents.

Three consequences seem especially durable.

First, the spectral and probabilistic data are identifiable: the zero divisor or the full even-cumulant sequence determines the hidden dyadic convolution law. The Gaussian- $q$ -binomial Richardson transform makes that statement constructive and supplies certified alternating bounds.

Second, nonnegative integer-valued polynomials form a rich positive-definite quotient sector. Negative Pascal exponents are not pathologies when they recombine into a nonnegative scale sequence. The family  $P_d(h) = \binom{h-1}{d}$  gives the explicit quotient  $\Phi_1\Phi_3/\Phi_2$  at  $d = 2$ , sparse enhanced zero divisors, automatic lobe signs, and a smooth approximation to the box with an exact Gaussian edge profile.

Third, Mellin reflection makes the endpoint and spectrum literally the same meromorphic object viewed at  $s$  and  $-s$ . Polynomial degree becomes pole order; pole order becomes the power of the endpoint logarithm; the dyadic pole lattice becomes the periodic coefficient hierarchy; and the leading saddle is solved by a generalized lower branch of Lambert  $W$ . Nörlund Bernoulli polynomials and complete Bell polynomials then supply the coefficient algebra required for an all-orders inverse theory.

These deductions are new relative to the pinned 67-source corpus, but they are not presented as the end of the subject. Their main value is that they turn a collection of striking special identities into a classification problem with exact algebraic coordinates, stable reconstruction algorithms, and a clear route to formal verification.

## A Complete recursive corpus inventory

The following path ledger is the authoritative source list used for the nonduplication audit. Every path is relative to `Analysis/FabijsFunction/docs/`. The pinned commit is `32d6d36c51d803289e6d6a0dc0c37753766eba47`. The count is 67. Frozen revisions were read for distinctive claims and provenance, but a repeated theorem in several revisions was counted once when determining novelty.

| No. | Relative path                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <code>FabijsFunction_Mathematical_Glossary/FabijsFunction_Mathematical_Glossary.tex</code>                                         |
| 2   | <code>Fabijs_Function_and_Rvachev_Up/Fabijs_Function_and_Rvachev_Up.tex</code>                                                     |
| 3   | <code>Non_Elementarity_of_the_Fabijs_Function/Non_Elementarity_of_the_Fabijs_Function.tex</code>                                   |
| 4   | <code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>                                                                   |
| 5   | <code>non-formalized-research-frontiers/non-formalized-research-frontiers.tex</code>                                               |
| 6   | <code>archive/early-notes/FabijsAsymptotic/FabijsAsymptotic.tex</code>                                                             |
| 7   | <code>archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex</code> |
| 8   | <code>archive/glossary/FabijsFunction_Mathematical_Glossary-2/FabijsFunction_Mathematical_Glossary.tex</code>                      |
| 9   | <code>archive/glossary/FabijsFunction_Mathematical_Glossary/FabijsFunction_Mathematical_Glossary.tex</code>                        |
| 10  | <code>archive/lean-walkthrough/Fabijs_Function_Lean_Walkthrough/Fabijs_Function_Lean_Walkthrough.tex</code>                        |

*Continued on the next page*

*Corpus inventory (continued)*

| No. | Relative path                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 11  | archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex                                                                 |
| 12  | archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex                                               |
| 13  | archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex                                               |
| 14  | archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex                                                 |
| 15  | archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex                                                                 |
| 16  | archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex                                                                   |
| 17  | archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex                                               |
| 18  | archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex                                                |
| 19  | archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex                                                  |
| 20  | archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex                                                                 |
| 21  | archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex                                                                 |
| 22  | archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex                                               |
| 23  | archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex |
| 24  | archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex                                   |
| 25  | archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex                                                       |
| 26  | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex                                         |
| 27  | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex                                       |
| 28  | archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex                                               |
| 29  | archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex                                       |
| 30  | archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex                                     |
| 31  | archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex                                                           |
| 32  | archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex                             |
| 33  | archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex                                                           |
| 34  | archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex                                           |
| 35  | archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex                               |
| 36  | archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex                                                         |
| 37  | archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex                                           |

*Continued on the next page*

*Corpus inventory (continued)*

| No. | Relative path                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38  | non-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex                                                                          |
| 39  | non-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex                                                              |
| 40  | non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex                                                                                        |
| 41  | non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex                                                                                          |
| 42  | non-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex                                                                                              |
| 43  | non-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex                                                                  |
| 44  | non-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex                                                              |
| 45  | non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex                                                                                                      |
| 46  | non-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex                                                                                                |
| 47  | non-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex                                                                                                 |
| 48  | non-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex                                                               |
| 49  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex                                                                           |
| 50  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex                                                                        |
| 51  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex                                         |
| 52  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex                                                   |
| 53  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex                                         |
| 54  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex |
| 55  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex                                                                           |
| 56  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex                                                                           |
| 57  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex                                                                     |
| 58  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex                                                           |
| 59  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex                                                                     |
| 60  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex                                                                     |
| 61  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex                                                                                                                                      |
| 62  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex                                                                                                                                                |
| 63  | papers/arXiv-1502.06738v2/Lacunary_products.tex                                                                                                                                                     |
| 64  | papers/arXiv-1702.05442v1/09-Function.tex                                                                                                                                                           |

*Continued on the next page*

*Corpus inventory (continued)*

| No. | Relative path                                                |
|-----|--------------------------------------------------------------|
| 65  | <code>papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex</code> |
| 66  | <code>papers/arXiv-2211.13570v1/document.tex</code>          |
| 67  | <code>papers/ubiq15/ubiq15-corrected.tex</code>              |

## B Result ledger and dependency map

Table 4 records the logical status and principal inputs of the report’s main claims. It is intended to make later formalization and peer checking easier: a reader can separate finite algebra from analytic continuation, and new deductions from imported frontier theorems.

Table 4: Result and proof-dependency ledger.

| ID | Statement                                                                      | Status  | Main inputs                                         |
|----|--------------------------------------------------------------------------------|---------|-----------------------------------------------------|
| A1 | Exponent-sequence probability law, support, entire product, convolution monoid | proved  | uniform characteristic function; normal convergence |
| A2 | Zero multiplicity $m_a(n)$ and zeta $\zeta(s)A(2^{-s})$                        | proved  | Euler product; Tonelli/rearrangement                |
| A3 | Digital zero defect and generalized lobe character                             | proved  | binary floors; parity                               |
| A4 | Automaticity iff the parity word is eventually periodic                        | proved  | exact 2-kernel/shift identity                       |
| A5 | General Prouhet order and first surviving power moment                         | proved  | finite product; falling factorials                  |
| B1 | Cumulant transform $B_{2j}A(4^{-j})/(2j)$                                      | proved  | uniform cumulants; additivity                       |
| B2 | Identifiability from zeros, zeta, or all even cumulants                        | proved  | prefix differences; identity theorem                |
| B3 | Gaussian- $q$ -binomial tomography and one-sided remainder                     | proved  | Lagrange interpolation; principal specialization    |
| B4 | Bounded exact condition number $(-q; q)_r/(q; q)_r$                            | proved  | finite $q$ -binomial theorem                        |
| C1 | Shift/refinement identity and rational finite-rank classification              | proved  | reindexing; rational-series theorem                 |
| C2 | Newton factorization of nonnegative polynomial weights                         | proved  | integer-valued polynomial calculus                  |
| C3 | Leading Fourier, small-ball, and inverse laws                                  | derived | audited Pascal hierarchy; de Bruijn transfer        |
| D1 | Alternating family $P_d(h) = \binom{h-1}{d}$ and positive quotient             | proved  | Newton identity; random-series factorization        |
| D2 | Sparse spectrum, exact cumulants, automatic Prouhet selector                   | proved  | hockey-stick; Lucas theorem                         |
| D3 | Gaussian edge layer, exact $L^1$ law, TV and $W_2$ bounds                      | proved  | Berry–Esseen; interval overlap                      |
| E1 | Mellin reflection between endpoint Laplace and spectral zeta                   | proved  | Mellin kernel; geometric scaling                    |



| ID | Statement                                  | Status                 | Main inputs                                   |
|----|--------------------------------------------|------------------------|-----------------------------------------------|
| E2 | Polynomial/log-periodic pole hierarchy     | proved<br>structurally | Mellin residues; rational<br>pole orders      |
| E3 | All-orders generalized inverse transseries | conjectural            | uniform higher-pole<br>saddle analysis needed |

**Lean crosswalk for B3–B4.** The finite weight, all-moment, residual-sign, and finite-norm identities in B3–B4 are formalized in `GeometricQBinomialLagrange.lean`, `GeometricLagrangeQBinomial.lean`, and `GeometricLagrangeQMoments.lean`. The exact unconditionally summable transformed series, including the shifted Gaussian tail, is formalized in `AnalyticSeriesFilter.lean`; zero-weight nodes require no convergence of their unweighted samples. The conditionally convergent boundary, the ordered one-sided sign and bound, the actual exponent-generating-function instantiation, uniform estimates, and the infinite condition-number limit remain results of this report rather than claims about the current Lean corpus.

## C Reproducibility manifest

The companion archive contains:

| Path                                                        | Purpose                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------|
| <code>Exponent_Sequence_Newton_Rvachev_Frontiers.tex</code> | Complete manuscript source.                                 |
| <code>Exponent_Sequence_Newton_Rvachev_Frontiers.pdf</code> | Rendered report.                                            |
| <code>numerical_experiments.py</code>                       | Commented high-precision and FFT verification program.      |
| <code>figures/richardson_tomography.png</code>              | Figure used in <a href="#">figure 1</a> .                   |
| <code>figures/boundary_layers.png</code>                    | Figure used in <a href="#">figure 2</a> .                   |
| <code>data/richardson_tomography.csv</code>                 | Full signed extrapolation errors.                           |
| <code>data/boundary_layer_errors.csv</code>                 | Deterministic standardized-CDF errors.                      |
| <code>data/zero_multiplicities_d2.csv</code>                | Direct and closed-form zero orders for the $d = 2$ family.  |
| <code>data/verification_log.txt</code>                      | Precision settings, assertion status, and headline values.  |
| <code>README.md</code>                                      | Build and reproduction commands.                            |
| <code>requirements.txt</code>                               | Exact Python package versions used for the numerical audit. |
| <code>SHA256SUMS</code>                                     | Checksums for the report, code, figures, and data.          |

The numerical script fixes the precision and grid sizes explicitly. Its formulas use `numpy.sinc(x) = sin( $\pi x$ )/( $\pi x$ )`, matching the normalization in (2.1). The FFT interval is wider than the compact support, so periodic copies do not overlap. All exact checks run before figure generation.

## References

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- [14] V. Reshetnikov, *ProveIt: formal and expository development of the Fabius function*, GitHub repository, pinned in this report at commit 32d6d36c51d803289e6d6a0dc0c37753766eba47, 27 August 2026.

## Part II

# q-Binomial Richardson Acceleration of Geometric Sinc Products

### Abstract

The Fourier transform of Rvachev’s compactly supported up-function is the geometric sinc product

$$P(\xi) = \prod_{\nu \geq 1} \text{sinc}(\xi/2^\nu).$$

The repository corpus develops its large-frequency decay, its connection with the Fabius function and the signed Thue–Morse sequence, dyadic exact values, small-argument and inverse asymptotics, and several Richardson-type limits. This report studies a different asymptotic variable: the truncation level in the finite product  $P_n(\xi) = \prod_{\nu=1}^n \text{sinc}(\xi/2^\nu)$ .

For a general geometric base  $b > 1$ , with  $q = b^{-2}$ , we prove, in the disc  $|q^n \xi^2| < \pi^2 b^2$ , the total exact analytic factorization

$$P_{b,n}(\xi) = P_b(\xi) \exp \left( \sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn} \right) = P_b(\xi) \sum_{r \geq 0} \alpha_r(b) \xi^{2r} q^{rn}.$$

Off the zero set of  $P_b$ , division gives the corresponding quotient form. The cumulants  $c_k(b)$  are explicit Bernoulli-number combinations and the coefficients  $\alpha_r(b)$  are complete Bell polynomials. Lagrange evaluation at the geometric nodes  $1, q, \dots, q^m$  then gives a universal extrapolant

$$\mathcal{R}_{m,n} P_b = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}, \quad w_j^{(m)} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}.$$

On the same disc and off the zero set of  $P_b$ , its complete error series has the closed  $q$ -binomial form

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) \xi^{2r} q^{rn}.$$

Consequences include an exact sign law, non-asymptotic divided-difference bounds, a rigorous a posteriori error certificate, and a uniformly bounded stability constant  $\prod_{r=1}^m (1 + q^r)/(1 - q^r)$ . For  $q = 1/4$  this constant stays below 1.970, even as the order tends to infinity.

After inverse Fourier transformation, the same filter combines consecutive finite box splines. It matches the first  $2m$  moments of the limiting up-density *exactly at every truncation level*, and converges in every  $C^s$ -norm at rate  $q^{(m+1)n}$ . The first error profile is an explicit even derivative of the up-function. Translating to the Fabius function gives the same accelerated approximation, while inversion on compact interior intervals yields an explicit first correction for the inverse Fabius function. These statements are proved here and checked symbolically and numerically by the accompanying Python program. They are described as *repository-new*: the claim is that they do not occur in the audited `*.tex` corpus under the stated repository path, not that a priority search over the entire literature has been completed.

**Status of results.** Every result labelled theorem, proposition, lemma, or corollary below has a proof in this report. The numerical tables are reproducible from `verify_frontier_results.py`. The denominator-free finite  $q$ -binomial theorem is formalized in `FiniteQBinomialCore.lean`, while `GeometricLagrange.lean` proves mass one, cancellation of the first geometric modes, the first surviving moment, and module-valued Richardson exactness. The already compiled `GeometricQBinomialLagrange.lean` identifies the individual weights with their Gaussian-binomial quotients under injectivity of the finite geometric node set. At post-snapshot source checkpoint `1e761ed77583e9870cceaeb2c74ac824094f0f3f`, `GeometricCompleteHomogeneous.lean` and `GeometricResidualMoments.lean` additionally formalize the denominator-free principal specialization and the complete positive-degree Gaussian  $q$ -moment formula, including common scales and shifted blocks. At the reconciled source checkpoint `f77ff7c296b99050107948f5aa2a9488aec01d05`, the rational layer is implemented in

```
GeometricQBinomialLagrange.lean
GeometricRichardson.lean
GeometricLagrangeWeights.lean
GeometricLagrangeQBinomial.lean
GeometricLagrangeQMoments.lean.
```

These modules prove the finite rational  $0 < q < 1$  weight formulas, the  $W_m(q^r)$  and  $q$ -Pochhammer moment presentations, all residual moments and their sign, and the exact finite  $\ell^1$  norm, including dedicated  $q = 1/4$  corollaries. The generic transformed-error series is now formalized in `AnalyticSeriesFilter.lean` under unconditional summability, with nonzero-weight hypotheses only at samples that actually contribute. Identifying its coefficients with the sinc-product tail, and proving the report's signs, remainder and bracketing bounds, uniform spline limits, infinite condition-number limit, and inverse-function consequences, remain unformalized.

# 1 Audit scope and the non-duplication boundary

## 1.1 How the corpus was used

The source audit covered every LaTeX family in the repository documentation tree [1], including the current expository volumes, the formalization walkthrough, the non-elementarity article, the mathematical glossary, the consolidated frontier volume, the Thue–Morse formula atlas, all waves of the Fourier-decay discussion, the paper-source subdirectories, and the historical archive. The archive index was used to track renamed and byte-identical historical variants, so that an old statement was not mistaken for a new one merely because it appeared under a superseded filename. A source map is given in [section B](#).

The audit revealed four especially important boundaries.

- B1.** The main exposition and its supplements already contain the normalized infinite sinc product, the box-convolution construction of up, its relation to the Fabius function, dyadic identities, q-Pochhammer and q-binomial formulae, and the signed Thue–Morse derivative comb.
- B2.** The asymptotic documents already contain a corrected Lambert– $W$  endpoint scale, a periodic dyadic phase, and all-order small-argument expansions for the Fabius function and its inverse. Those results concern  $x \downarrow 0$ , not finite-product truncation  $n \rightarrow \infty$ .
- B3.** The Fourier-decay waves study  $|P(\xi)|$  as  $|\xi| \rightarrow \infty$ , including dyadic oscillation and exceptional frequencies. They do not develop the exact analytic expansion of  $P_{b,n}/P_b$  in the small geometric variable  $q^n$ , nor the extrapolation filter derived below.
- B4.** The frontier volume already has q-Richardson ideas for discrete Fabius spline rows and degree-dependent approximants. We therefore do not present a generic Richardson recipe as new. The new object here is the *Fourier-tail truncation sequence*  $P_{b,n}$ , for which the error coefficients, all q-moments, stability constant, moment matching, and inverse-function consequence can all be written exactly.

The external search performed for this report also found general literature on Richardson extrapolation and on powers of sinc expressed through Bell polynomials, but no source matching the combined construction in [sections 3 to 5](#). This is supporting evidence only, not an exhaustive priority claim.

## 1.2 What is new in this report

The principal contributions are the following.

- N1.** A one-variable analytic tail function  $\Phi_b$  whose Taylor cumulants are rational Bernoulli combinations and whose Taylor coefficients are complete Bell polynomials.
- N2.** A q-Lagrange operator  $W_m(T_q) = \prod_{r=1}^m (T_q - q^r)/(1 - q^r)$  that annihilates the first  $m$  truncation modes exactly.
- N3.** An all-order q-binomial error identity, not merely a leading-order Richardson estimate.
- N4.** A closed stability constant with a finite infinite-order limit, and a successive-level error certificate requiring no knowledge of the infinite product.
- N5.** Exact matching of the first  $2m$  moments of the limiting Rvachev density by a signed combination of finite box splines, for every  $n$ .
- N6.** A  $C^\infty$  superconvergence theorem with an explicit first derivative profile, followed by an interior inverse-Fabius perturbation formula.
- N7.** A kernel-independent extension showing that the same q-filter accelerates any geometric product whose elementary factor has an even logarithmic Taylor series.

## 2 Geometric sinc products and finite Rvachev splines

Fix  $b > 1$  and set

$$q = b^{-2} \in (0, 1), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Define

$$P_b(\xi) = \prod_{\nu=1}^{\infty} \text{sinc}(\xi/b^\nu), \quad P_{b,n}(\xi) = \prod_{\nu=1}^n \text{sinc}(\xi/b^\nu). \quad (2.1)$$

The convergence is locally uniform in  $\mathbb{C}$ , since  $1 - \text{sinc}(\xi/b^\nu) = O(b^{-2\nu})$  on compact sets.

Let

$$h_{b,\nu}(x) = \frac{b^\nu}{2} \mathbf{1}_{[-b^{-\nu}, b^{-\nu}]}(x), \quad u_{b,n} = h_{b,1} * \cdots * h_{b,n}. \quad (2.2)$$

With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\xi) e^{i\xi x} d\xi,$$

we have  $\widehat{h}_{b,\nu}(\xi) = \text{sinc}(\xi/b^\nu)$ , hence

$$\widehat{u_{b,n}} = P_{b,n}. \quad (2.3)$$

The infinite convolution  $u_b = *_{\nu \geq 1} h_{b,\nu}$  is a probability density supported on  $[-1/(b-1), 1/(b-1)]$ , and  $\widehat{u}_b = P_b$ . The product decays faster than every power, so  $u_b \in C_c^\infty(\mathbb{R})$ . At  $b = 2$ ,  $u_2$  is the normalized Rvachev up-function.

### 2.1 The finite Thue–Morse derivative comb

The bridge to the signed Thue–Morse sequence is already part of the repository background, but the exact normalization is useful for what follows. Write  $t_k = (-1)^{s_2(k)}$ , where  $s_2(k)$  is the binary digit sum.

**Proposition 2.1** (distributional Thue–Morse comb). *For  $n \geq 1$ ,*

$$D^n u_{2,n} = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} t_k \delta_{-1+(2k+1)2^{-n}}. \quad (2.4)$$

*Proof.* For  $a > 0$ ,

$$D \left( \frac{1}{2a} \mathbf{1}_{[-a,a]} \right) = \frac{1}{2a} (\delta_{-a} - \delta_a).$$

Differentiate every factor in (2.2). Choosing the right endpoint in the  $j$ -th factor contributes one minus sign. If the choices are encoded by the binary digits of  $k$ , the coefficient is  $(-1)^{s_2(k)}$ , while the location is

$$-\sum_{j=1}^n 2^{-j} + 2 \sum_{j=1}^n \varepsilon_j 2^{-j} = -1 + 2^{-n} + (2^{1-n})k.$$

Finally  $\prod_{j=1}^n (2 \cdot 2^{-j})^{-1} = 2^{n(n-1)/2}$ . □

Thus every finite product in (2.1) is simultaneously a Fourier approximation and an integrated Thue–Morse object. The acceleration below acts on both descriptions at once.

### 3 The exact Bernoulli–Bell tail expansion

#### 3.1 A universal tail function

The key observation is that the two variables  $n$  and  $\xi$  enter the omitted tail only through  $q^n \xi^2$ .

**Definition 3.1.** For  $z$  near zero, define

$$\Phi_b(z) = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}(b^{-\ell} \sqrt{z})}. \quad (3.1)$$

The expression is independent of the sign of  $\sqrt{z}$ , and hence defines an ordinary analytic function of  $z$ .

**Theorem 3.2** (exact tail factorization). *The function  $\Phi_b$  is analytic in  $|z| < \pi^2 b^2$ , meromorphic in  $\mathbb{C}$ , and has its nearest singularity to the origin at  $z = \pi^2 b^2$ . For every  $n \geq 0$ ,*

$$P_{b,n}(\xi) = P_b(\xi) \Phi_b(q^n \xi^2). \quad (3.2)$$

*The identity is entire in  $\xi$ ; at common zeros it is interpreted by analytic continuation. Moreover,*

$$\Phi_b(z) = \frac{\Phi_b(qz)}{\text{sinc}(b^{-1} \sqrt{z})}. \quad (3.3)$$

*Proof.* Cancel the first  $n$  factors in  $P_b/P_{b,n}$  and write  $\nu = n + \ell$ :

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}(\xi/b^{n+\ell})} = \Phi_b(q^n \xi^2).$$

The zeros of  $\text{sinc}(b^{-\ell} \sqrt{z})$  occur at  $z = \pi^2 k^2 b^{2\ell}$ ,  $k \in \mathbb{Z} \setminus \{0\}$ . The nearest is  $\pi^2 b^2$ . Splitting off the  $\ell = 1$  factor and reindexing proves (3.3).  $\square$

#### 3.2 Bernoulli cumulants

The classical expansion

$$\log \text{sinc } w = - \sum_{k=1}^{\infty} \frac{\zeta(2k)}{k \pi^{2k}} w^{2k}, \quad |w| < \pi, \quad (3.4)$$

turns the geometric tail into an explicitly summable cumulant series.

**Theorem 3.3** (Bernoulli cumulants). *For  $|z| < \pi^2 b^2$ ,*

$$\Phi_b(z) = \exp \left( \sum_{k=1}^{\infty} c_k(b) z^k \right), \quad (3.5)$$

where

$$c_k(b) = \frac{\zeta(2k)}{k \pi^{2k} (b^{2k} - 1)} \quad (3.6)$$

$$= \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(b^{2k} - 1)}. \quad (3.7)$$

*In particular  $c_k(b) > 0$  for every  $k \geq 1$ .*

*Proof.* Insert (3.4) into (3.1), interchange absolutely convergent sums, and use

$$\sum_{\ell=1}^{\infty} b^{-2k\ell} = \frac{1}{b^{2k} - 1}.$$

The Bernoulli form follows from Euler's formula for  $\zeta(2k)$ . Positivity is clear from (3.6).  $\square$

Combining theorems 3.2 and 3.3 gives the exact truncation expansion

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \exp\left(\sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn}\right). \quad (3.8)$$

This is an equality, not a merely formal asymptotic series, whenever  $|\xi| < \pi b^{n+1}$ .

### 3.3 Bell coefficients and their recurrence

Write

$$\Phi_b(z) = \sum_{r=0}^{\infty} \alpha_r(b) z^r, \quad \alpha_0(b) = 1. \quad (3.9)$$

**Proposition 3.4** (complete Bell-polynomial formula). *Let  $Y_r$  denote the complete exponential Bell polynomial. Then*

$$\alpha_r(b) = \frac{1}{r!} Y_r(1!c_1(b), 2!c_2(b), \dots, r!c_r(b)), \quad (3.10)$$

and equivalently

$$r\alpha_r(b) = \sum_{k=1}^r k c_k(b) \alpha_{r-k}(b). \quad (3.11)$$

Every  $\alpha_r(b)$  is strictly positive.

*Proof.* Equation (3.10) is the defining exponential generating identity for complete Bell polynomials. Differentiating (3.5) and comparing coefficients gives (3.11). Positivity follows inductively from  $c_k(b) > 0$ .  $\square$

For the Rvachev base  $b = 2$ , the first cumulants are

$$\begin{aligned} c_1 &= \frac{1}{18}, & c_2 &= \frac{1}{2700}, & c_3 &= \frac{1}{178605}, \\ c_4 &= \frac{1}{9639000}, & c_5 &= \frac{1}{478533825}, & c_6 &= \frac{691}{15688261338750}. \end{aligned}$$

The corresponding Bell coefficients are

$$\alpha_0 = 1, \quad \alpha_1 = \frac{1}{18}, \quad \alpha_2 = \frac{31}{16200}, \quad (3.12)$$

$$\alpha_3 = \frac{2347}{42865200}, \quad \alpha_4 = \frac{1904369}{1311675120000}, \quad \alpha_5 = \frac{3309760579}{88561680752160000}. \quad (3.13)$$

The exact rationals are useful because no numerical fitting is needed at any order.

## 4 q-Lagrange Richardson extrapolation

### 4.1 The filter polynomial

Let  $T_q$  be the dilation operator  $(T_q f)(z) = f(qz)$ . Define

$$W_m(t) = \prod_{r=1}^m \frac{t - q^r}{1 - q^r}, \quad W_0(t) = 1. \quad (4.1)$$

Thus  $W_m(1) = 1$  and  $W_m(q^r) = 0$  for  $1 \leq r \leq m$ . Expanding  $W_m(t) = \sum_{j=0}^m w_j^{(m)} t^j$  gives the following exact weights.

**Theorem 4.1** (geometric Lagrange weights). *For  $0 \leq j \leq m$ ,*

$$w_j^{(m)} = \prod_{\substack{0 \leq \ell \leq m \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}, \quad (4.2)$$

where  $(q; q)_r = \prod_{s=1}^r (1 - q^s)$ . These are precisely the Lagrange weights that recover the value at zero from data at  $1, q, \dots, q^m$ .

*Proof.* The first expression is the Lagrange cardinal basis evaluated at zero. Separating factors with  $\ell < j$  and  $\ell > j$ , then collecting powers of  $q$ , gives the second expression. Alternatively, the q-binomial theorem expands (4.1) directly.  $\square$

**Lean boundary for the individual weights.** The source-level closed quotient predates the new all-residual tranche. Over a field, under  $\text{InjOn}(j \mapsto q^j, \{0, \dots, m\})$ , the declaration

`geometricLagrangeWeight_eq_gaussianBinomial_div`

in `GeometricQBinomialLagrange.lean` gives the numerator with the deliberate reverse index `gaussianBinomialqm(m-j)`. The displayed symmetric-index notation in (4.2) is the equivalent paper convention, not a claim that this checkpoint added a generic Gaussian-symmetry theorem. The same module also contains the following cleared-denominator identity:

`finiteQPochhammerIn_mul_geometricLagrangeWeight_eq.`

In the present real range  $0 < q < 1$ , the required finite-node injectivity is automatic. These pointwise weight results should not be attributed to the later principal-specialization module.

**Definition 4.2** (accelerated finite product). For  $m, n \geq 0$ , define

$$\mathcal{R}_{m,n} P_b(\xi) = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}(\xi). \quad (4.3)$$

With  $z = q^n \xi^2$ , the tail factorization gives the compact operator identity

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} = (W_m(T_q) \Phi_b)(z). \quad (4.4)$$

The right side is well defined as an analytic continuation even at common zeros.



## 4.2 A Romberg-style triangular algorithm

The factorization of  $W_m$  gives a recursion that avoids an explicit weighted sum.

**Algorithm 4.3** (q-Richardson table). Start with  $A_{0,n} = P_{b,n}(\xi)$ . For  $m \geq 1$ , form

$$A_{m,n} = \frac{A_{m-1,n+1} - q^m A_{m-1,n}}{1 - q^m}. \quad (4.5)$$

Then  $A_{m,n} = \mathcal{R}_{m,n} P_b(\xi)$ .

*Proof.* The polynomial recursion

$$W_m(t) = \frac{t - q^m}{1 - q^m} W_{m-1}(t)$$

becomes (4.5) after applying it to the shift  $n \mapsto n + 1$ . □

For  $b = 2$ ,  $q = 1/4$ , the first filters are

$$\mathcal{R}_{1,n} P = \frac{4P_{n+1} - P_n}{3}, \quad (4.6)$$

$$\mathcal{R}_{2,n} P = \frac{P_n - 20P_{n+1} + 64P_{n+2}}{45}, \quad (4.7)$$

$$\mathcal{R}_{3,n} P = \frac{-P_n + 84P_{n+1} - 1344P_{n+2} + 4096P_{n+3}}{2835}. \quad (4.8)$$

The coefficients are exact rationals; the included program generates arbitrary orders.

## 4.3 The exact q-moment identity

**Lemma 4.4** (all geometric moments). For every integer  $r \geq 0$ , let  $M_r^{(m)} = \sum_{j=0}^m w_j^{(m)} q^{jr}$ . Then

$$M_r^{(m)} = W_m(q^r). \quad (4.9)$$

Consequently,

$$M_0^{(m)} = 1, \quad M_r^{(m)} = 0 \quad (1 \leq r \leq m), \quad (4.10)$$

and for  $r \geq m + 1$ ,

$$M_r^{(m)} = q^{mr} \frac{(q^{1-r}; q)_m}{(q; q)_m} \quad (4.11)$$

$$= (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q. \quad (4.12)$$

*Proof.* Equation (4.9) follows from the coefficient definition of  $W_m$ . The roots give (4.10). For  $r > m$ , evaluate the product in (4.1):

$$W_m(q^r) = \prod_{\ell=1}^m \frac{q^r - q^\ell}{1 - q^\ell} = (-1)^m q^{\binom{m+1}{2}} \prod_{\ell=1}^m \frac{1 - q^{r-\ell}}{1 - q^\ell},$$

which is (4.12); (4.11) is the equivalent q-Pochhammer form. □

**Post-snapshot Lean status of the moment formula.** The principal specialization

$$h_s(1, q, \dots, q^m) = \begin{bmatrix} s+m \\ m \end{bmatrix}_q$$

is `completeHomogeneousEval_geometric`; it holds over every commutative semiring, with no division or restriction on  $q$ . Its Lean right-hand side is the recursive term `gaussianBinomialq(s+m)m`; in the present real range this is the quotient denoted by  $\begin{bmatrix} s+m \\ m \end{bmatrix}_q$ . More generally, over a commutative ring, `sum_weight_mul_geometric_pow_of_pos` proves that any row  $a_0, \dots, a_m$  satisfying

$$\sum_{j=0}^m a_j (q^j)^d = 0^d \quad (0 \leq d \leq m)$$

has, for every  $r > 0$ ,

$$\sum_{j=0}^m a_j (q^j)^r = (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q.$$

Here and in the two formulas below, the literal Lean coefficient is the recursive `gaussianBinomial`; the quotient notation is valid in this report's real range  $0 < q < 1$ . The above-diagonal vanishing of the recursive Gaussian coefficient includes all cancellations in (4.10); degree zero remains the separate mass-one theorem. For the Lagrange row, finite-node injectivity supplies the low moments and `sum_geometricLagrangeWeight_mul_pow_of_pos` is exactly the displayed formula.

The same checkpoint also records the scale rather than cancelling it. If a row is exact directly on  $c, cq, \dots, cq^m$ , meaning

$$\sum_{j=0}^m a_j (cq^j)^d = 0^d \quad (0 \leq d \leq m),$$

then `sum_weight_mul_scaled_geometric_pow_succ_add` gives, for  $s \geq 0$ ,

$$\sum_{j=0}^m a_j (cq^j)^{m+1+s} = c^{m+1+s} (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} m+s \\ m \end{bmatrix}_q.$$

This hypothesis is imposed after scaling and remains meaningful when  $c$  is zero or a zero divisor. In the field-valued shifted block used here, `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos` specializes it to

$$\sum_{j=0}^m w_j^{(m)} (q^{n+j})^r = (-1)^m q^{nr + \binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \quad (r > 0).$$

These declarations are finite algebraic statements. The separate module `AnalyticSeriesFilter` .lean now establishes the corresponding finite-sum/ infinite-series interchange and exact Gaussian tail under unconditional summability of the contributing samples; a zero filter weight creates no unweighted summability obligation. It does not identify the coefficient sequence below with an actual sinc-product quotient, cover conditionally-only boundary convergence, or establish signs, bounds, asymptotics, or uniform convergence. The finite declarations in this paragraph were checked at source checkpoint 1e761ed77583e9870ceeaeb2c74ac824094f0f3f.

#### 4.4 The complete error series

**Theorem 4.5** (exact  $q$ -binomial error). *Let  $z = q^n \xi^2$ , and suppose  $|z| < \pi^2 b^2$ . If  $P_b(\xi) \neq 0$ , then*

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r=m+1}^{\infty} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) z^r. \quad (4.13)$$

Without the nonvanishing hypothesis, the total statement is the multiplied identity

$$\mathcal{R}_{m,n}P_b(\xi) - P_b(\xi) = P_b(\xi)(-1)^m q^{\binom{m+1}{2}} \sum_{r=m+1}^{\infty} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) z^r. \quad (4.14)$$

At a common zero of  $P_b$  and all the finite products, both sides of (4.14) vanish. In particular, for fixed  $m$  and fixed  $\xi \neq 0$  with  $P_b(\xi) \neq 0$ , as  $n \rightarrow \infty$ ,

$$\frac{\mathcal{R}_{m,n}P_b(\xi)}{P_b(\xi)} - 1 \sim (-1)^m q^{\binom{m+1}{2}} \alpha_{m+1}(b) \xi^{2m+2} q^{(m+1)n}. \quad (4.15)$$

*Proof.* Apply the total tail identity (3.2) separately to each  $P_{b,n+j}(\xi)$ , multiply by  $w_j^{(m)}$ , and sum. This gives

$$\mathcal{R}_{m,n}P_b(\xi) = P_b(\xi) \sum_{j=0}^m w_j^{(m)} \Phi_b(q^j z)$$

without division, including at zeros. Insert  $\Phi_b(z) = \sum_{r \geq 0} \alpha_r z^r$ ; absolute convergence in the stated disc permits the finite sum to pass through the series. The coefficient of  $z^r$  is  $\alpha_r W_m(q^r)$ , so [theorem 4.4](#) gives (4.14). Division by  $P_b(\xi)$  gives (4.13) exactly on its stated locus.

Finally, analyticity at zero makes the residual series its first nonzero term plus  $O(z^{m+2})$ . Since  $z = q^n \xi^2$ , while  $\alpha_{m+1}(b) > 0$  by [theorem 3.4](#) and  $\xi \neq 0$ , that term is nonzero and yields (4.15).  $\square$

This identity is substantially stronger than the statement that Richardson extrapolation raises the order. Every residual mode and its exact  $q$ -binomial multiplier are known.

#### 4.5 Divided differences and non-asymptotic sign bounds

Let  $x_j = q^j z$ . The quantity  $\sum_j w_j^{(m)} \Phi_b(x_j)$  is the value at zero of the degree- $m$  Lagrange interpolant through the nodes  $x_0, \dots, x_m$ . The standard interpolation remainder therefore yields a second exact form.

**Theorem 4.6** (divided-difference form). For  $0 \leq z < \pi^2 b^2$ ,

$$(W_m(T_q)\Phi_b)(z) - 1 = (-1)^m q^{\binom{m+1}{2}} z^{m+1} \Phi_b[0, q^m z, q^{m-1} z, \dots, z], \quad (4.16)$$

where the bracket denotes a divided difference. Since  $\Phi_b$  is absolutely monotone on this interval,

$$q^{\binom{m+1}{2}} \alpha_{m+1}(b) z^{m+1} \leq |(W_m(T_q)\Phi_b)(z) - 1| \quad (4.17)$$

$$\leq \frac{q^{\binom{m+1}{2}} z^{m+1}}{(m+1)!} \Phi_b^{(m+1)}(z). \quad (4.18)$$

The sign is exactly  $(-1)^m$ .

*Proof.* The Lagrange remainder at zero is

$$p_m(0) - \Phi_b(0) = (-1)^m \left( \prod_{j=0}^m x_j \right) \Phi_b[0, x_0, \dots, x_m],$$

and  $\prod_j x_j = q^{\binom{m+1}{2}} z^{m+1}$ . Because every Taylor coefficient  $\alpha_r$  is positive, all derivatives of  $\Phi_b$  are positive and increasing on the stated interval. The Hermite–Genocchi formula bounds the divided difference between  $\Phi_b^{(m+1)}(0)/(m+1)! = \alpha_{m+1}$  and  $\Phi_b^{(m+1)}(z)/(m+1)!$ .  $\square$

**Corollary 4.7** (relative bracketing). For real  $\xi$  with  $|\xi| < \pi b^{n+1}$  and  $P_b(\xi) \neq 0$ ,

$$\operatorname{sgn} \left( \frac{\mathcal{R}_{m,n}P_b(\xi)}{P_b(\xi)} - 1 \right) = (-1)^m. \quad (4.19)$$

Thus even and odd extrapolation orders bracket the exact product in relative terms. If  $P_b(\xi) > 0$ , the bracketing is also an ordinary numerical ordering.

#### 4.6 Uniform stability of arbitrarily high order

Richardson formulae are often numerically unattractive because their coefficient norms grow rapidly. The geometric nodes here behave unusually well.

**Theorem 4.8** (exact Lebesgue constant). *The amplification factor of the filter is*

$$\Lambda_m(q) = \sum_{j=0}^m |w_j^{(m)}| = \prod_{r=1}^m \frac{1+q^r}{1-q^r} = \frac{(-q; q)_m}{(q; q)_m}. \quad (4.20)$$

For fixed  $q \in (0, 1)$ ,

$$\Lambda_\infty(q) = \prod_{r=1}^{\infty} \frac{1+q^r}{1-q^r} < \infty. \quad (4.21)$$

For  $q = 1/4$ ,

$$\Lambda_\infty(1/4) = 1.969260353668269431947193696967 \dots \quad (4.22)$$

*Proof.* The signs in (4.2) alternate, so

$$\sum_j |w_j^{(m)}| = (-1)^m W_m(-1) = \prod_{r=1}^m \frac{1+q^r}{1-q^r}.$$

The logarithm of the infinite product is  $\sum_r [\log(1+q^r) - \log(1-q^r)] = O(\sum_r q^r)$ , hence it converges.  $\square$

Consequently, absolute input errors bounded by  $\varepsilon$  produce an output error no larger than  $\Lambda_\infty(q)\varepsilon$ , uniformly in the extrapolation order. This is a decisive difference from equally spaced high-order extrapolation.

#### 4.7 A rigorous a posteriori certificate

The positive-coefficient error series also compares consecutive truncation levels.

**Theorem 4.9** (successive-level certificate). *Fix  $m$  and a real  $\xi$ . Assume  $|\xi| < \pi b^{n+1}$  and  $P_b(\xi) \neq 0$ . Let  $R_n = \mathcal{R}_{m,n}P_b(\xi)$ . Then*

$$|R_{n+1} - R_n| \leq |R_n - P_b(\xi)| \leq \frac{|R_{n+1} - R_n|}{1 - q^{m+1}}. \quad (4.23)$$

Furthermore,

$$P_b(\xi) - R_n \sim \frac{R_{n+1} - R_n}{1 - q^{m+1}}. \quad (4.24)$$

*Proof.* By (4.13), the relative error is  $(-1)^m A_n$ , where

$$A_n = q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r \xi^{2r} q^{rn} > 0.$$

Every term is multiplied by  $q^r$  when  $n$  increases by one. Hence  $0 < A_{n+1} \leq q^{m+1} A_n$ . Therefore

$$(1 - q^{m+1})A_n \leq A_n - A_{n+1} \leq A_n.$$

Multiplying by  $|P_b(\xi)|$  proves (4.23). The leading term has ratio  $q^{m+1}$ , which proves (4.24).  $\square$

This certificate is practical: it estimates the unknown infinite product from two consecutive accelerated values and uses only the known number  $1 - q^{m+1}$ .

## 5 Physical-space consequences for Rvachev splines

Define the signed extrapolated spline

$$U_{b;m,n}(x) = \sum_{j=0}^m w_j^{(m)} u_{b,n+j}(x). \quad (5.1)$$

It has total mass one and support in  $[-1/(b-1), 1/(b-1)]$ . It need not be nonnegative, but

$$\|U_{b;m,n}\|_{L^1} \leq \Lambda_m(q) < \Lambda_\infty(q). \quad (5.2)$$

Its Fourier transform is  $\mathcal{R}_{m,n}P_b$ .

### 5.1 All-order inverse-differential expansion

**Theorem 5.1** (all-order spline expansion). *For every pair of nonnegative integers  $N, s$ ,*

$$u_{b,n} = \sum_{r=0}^N (-1)^r \alpha_r(b) q^{rn} u_b^{(2r)} + O_{C^s} \left( q^{(N+1)n} \right) \quad (n \rightarrow \infty). \quad (5.3)$$

*Equivalently, as an all-order asymptotic operator identity,*

$$u_{b,n} \sim \exp \left( \sum_{k \geq 1} (-1)^k c_k(b) q^{kn} D^{2k} \right) u_b. \quad (5.4)$$

*Proof.* Fourier transformation turns  $(-1)^r D^{2r} u_b$  into  $\xi^{2r} P_b(\xi)$ , so the displayed finite sum is exactly the first  $N+1$  terms of (3.8).

For completeness, split the inverse Fourier integral at  $|\xi| = b^n$ . On the low-frequency region,  $z = q^n \xi^2 \leq 1$ , and analyticity of  $\Phi_b$  gives a uniform Taylor remainder bounded by  $C q^{(N+1)n} |\xi|^{2N+2} |P_b(\xi)|$ . This remains integrable after multiplication by  $|\xi|^s$ , because  $P_b$  is rapidly decreasing.

On  $|\xi| \geq b^n$ , the first  $n$  factors satisfy

$$|P_{b,n+j}(\xi)| \leq b^{n(n+1)/2} |\xi|^{-n}, \quad j \geq 0,$$

while the remaining factors have modulus at most one. The corresponding weighted Fourier tail is  $O(b^{-n^2/2+O(n)})$ , hence smaller than every fixed geometric rate. The same bound applies to  $P_b$ . Fourier inversion proves the assertion in  $C^s$ .  $\square$

### 5.2 Superconvergence of the extrapolated spline

**Theorem 5.2** (q-binomial  $C^\infty$  expansion). *For fixed  $m, s$  and every  $N \geq m+1$ ,*

$$\begin{aligned} U_{b;m,n} - u_b &= q^{\binom{m+1}{2}} \sum_{r=m+1}^N (-1)^{m+r} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) q^{rn} u_b^{(2r)} \\ &\quad + O_{C^s} \left( q^{(N+1)n} \right). \end{aligned} \quad (5.5)$$

*In particular,*

$$\|U_{b;m,n} - u_b\|_{C^s} \leq C_{b,m,s} q^{(m+1)n}, \quad (5.6)$$

*and the normalized error has the explicit limit*

$$q^{-(m+1)n} (U_{b;m,n} - u_b) \longrightarrow -q^{\binom{m+1}{2}} \alpha_{m+1}(b) u_b^{(2m+2)} \quad \text{in } C^s(\mathbb{R}). \quad (5.7)$$

*Proof.* Apply inverse Fourier transformation to (4.13), using  $\mathcal{F}^{-1}[\xi^{2r} P_b] = (-1)^r u_b^{(2r)}$ . The remainder estimate is the same low/high-frequency argument as in theorem 5.1. At  $r = m + 1$ , the sign is  $(-1)^{m+m+1} = -1$ , which gives (5.7).  $\square$

For  $b = 2$ , each additional extrapolation level changes the basic convergence factor from  $4^{-n}$  to  $4^{-(m+1)n}$ , while the filter norm remains below 1.970.

### 5.3 Exact moment matching

The cancellation is stronger than fast convergence: a finite set of moments is already exact.

**Theorem 5.3** (exact moment matching). *For every  $m, n \geq 0$  and every integer  $0 \leq \ell \leq 2m$ ,*

$$\int_{\mathbb{R}} x^\ell U_{b;m,n}(x) dx = \int_{\mathbb{R}} x^\ell u_b(x) dx. \quad (5.8)$$

*All odd moments vanish by symmetry. The first missed even moment is known exactly:*

$$\int_{\mathbb{R}} x^{2m+2} (U_{b;m,n}(x) - u_b(x)) dx = -(2m+2)! q^{\binom{m+1}{2}} \alpha_{m+1}(b) q^{(m+1)n}. \quad (5.9)$$

*Proof.* By (4.13),  $\widehat{U}_{b;m,n} - \widehat{u}_b$  has a zero of order at least  $2m+2$  at  $\xi = 0$ . Since

$$\widehat{f}^{(\ell)}(0) = (-i)^\ell \int x^\ell f(x) dx,$$

all moments through degree  $2m+1$  agree. Symmetry handles every odd degree. The coefficient of  $\xi^{2m+2}$  in the Fourier error is  $(-1)^m q^{\binom{m+1}{2}} \alpha_{m+1} q^{(m+1)n}$ . Converting the derivative at zero back to the  $(2m+2)$ -nd moment contributes  $(-1)^{m+1}$ , leaving the negative sign in (5.9).  $\square$

*Remark 5.4* (quadrature interpretation). For any polynomial  $p$  of degree at most  $2m$ ,

$$\int p(x) U_{b;m,n}(x) dx = \int p(x) u_b(x) dx.$$

Thus  $U_{b;m,n}$  is a signed, compactly supported, uniformly stable density surrogate with exact polynomial expectations through degree  $2m$ .

## 6 Fabius and inverse-Fabius consequences

Under the normalization used throughout the repository, the Fabius function on  $[0, 1]$  is the left half of the translated up-function:

$$F(x) = u_2(x-1), \quad 0 \leq x \leq 1. \quad (6.1)$$

Define the accelerated finite-spline approximation

$$F_{m,n}(x) = U_{2;m,n}(x-1), \quad q = \frac{1}{4}. \quad (6.2)$$

**Corollary 6.1** (accelerated Fabius approximation). *For every  $m, s \geq 0$ ,*

$$\|F_{m,n} - F\|_{C^s([0,1])} = O\left(4^{-(m+1)n}\right), \quad (6.3)$$

and

$$4^{(m+1)n} (F_{m,n} - F) \longrightarrow -4^{-\binom{m+1}{2}} \alpha_{m+1}(2) F^{(2m+2)} \quad \text{in } C^s([0, 1]). \quad (6.4)$$

The global statement is possible because all derivatives of the compactly supported up-function vanish at the endpoints. Inversion is more delicate:  $F'(x)$  tends to zero at the ends, and this is precisely where the repository's Lambert– $W$  asymptotics become relevant.

**Theorem 6.2** (interior inverse-Fabius superconvergence). *Let  $K = [a, b] \Subset (0, 1)$ , and put  $J = F(K)$ . Let  $G = F^{-1}$  on  $J$ . For sufficiently large  $n$ , the restriction of  $F_{m,n}$  to a neighborhood of  $K$  is strictly increasing and has an inverse branch  $G_{m,n}$  on  $J$ . Then*

$$\|G_{m,n} - G\|_{C^s(J)} = O\left(4^{-(m+1)n}\right) \quad (6.5)$$

for every fixed  $s$ , and uniformly for  $y \in J$ ,

$$4^{(m+1)n}(G_{m,n}(y) - G(y)) \longrightarrow 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) \frac{F^{(2m+2)}(G(y))}{F'(G(y))}. \quad (6.6)$$

*Proof.* On a slightly larger compact interval  $K' \Subset (0, 1)$ , continuity and strict monotonicity give  $F' \geq c > 0$ . The  $C^1$  convergence in [theorem 6.1](#) implies  $F'_{m,n} \geq c/2$  for large  $n$ , so the inverse branch exists. Standard differentiation of inverse functions gives (6.5).

For the leading term, write  $F_{m,n} = F + \varepsilon_n$ , set  $x = G(y)$ , and solve

$$0 = F(x + \delta_n) - F(x) + \varepsilon_n(x + \delta_n) = F'(x)\delta_n + \varepsilon_n(x) + o(4^{-(m+1)n}).$$

Equation (6.4) gives  $\varepsilon_n(x) \sim -4^{-\binom{m+1}{2}} \alpha_{m+1} 4^{-(m+1)n} F^{(2m+2)}(x)$ , which proves (6.6).  $\square$

## 6.1 Relation to Lambert– $W$ endpoint asymptotics

The inverse theorem is intentionally restricted to compact interior intervals. Near  $y = 0$ , the factor  $1/F'(G(y))$  becomes large and fixed-order inversion is no longer uniform. The repository's endpoint theory expresses the dominant scale of  $G(y)$  using a Lambert– $W$  variable together with a periodic dyadic phase. The two results therefore occupy complementary limits:

| regime                                         | small parameter | governing mechanism               |
|------------------------------------------------|-----------------|-----------------------------------|
| $y \downarrow 0$                               | $1/\log(1/y)$   | Lambert– $W$ and dyadic phase     |
| $n \rightarrow \infty, y \in J \Subset (0, 1)$ | $4^{-n}$        | Bernoulli–Bell tail and q-filter. |

A genuinely new two-parameter problem is to choose  $n = n(y)$  so that (6.6) remains uniform while  $y \downarrow 0$ . The exact factor  $F^{(2m+2)}(G(y))/F'(G(y))$ , combined with the repository's all-order endpoint expansion, reduces that problem to a concrete matching calculation rather than a vague stability question; see [section 10](#).

## 7 Numerical verification

All computations in this section used 90-decimal-digit arithmetic. The infinite product was continued until a geometric bound for the omitted logarithmic tail fell below  $10^{-70}$ . The explicit weighted formula and the triangular table (4.5) were evaluated independently and agreed to the working precision. Symbolic checks verified (4.12) for a range of orders as an identity in the symbol  $q$ .

### 7.1 Absolute errors

For  $b = 2$ , the reference values are

$$\begin{aligned} P(1) &= 0.9456037838988178752649922474364995 \dots, \\ P(3) &= 0.5857191765350684829125292624510928 \dots, \\ P(10) &= -0.03192757801277845828274435480432725 \dots \end{aligned}$$

The observed reduction on increasing  $n$  is approximately  $4^{-(m+1)}$ , exactly as predicted.

Table 5: Absolute errors at  $\xi = 1$ .

| $n$ | $m = 0$                 | $m = 1$                  | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|--------------------------|--------------------------|--------------------------|
| 3   | $8.2128 \times 10^{-4}$ | $1.1050 \times 10^{-7}$  | $3.0877 \times 10^{-12}$ | $1.9989 \times 10^{-17}$ |
| 4   | $2.0524 \times 10^{-4}$ | $6.9036 \times 10^{-9}$  | $4.8225 \times 10^{-14}$ | $7.8050 \times 10^{-20}$ |
| 5   | $5.1304 \times 10^{-5}$ | $4.3143 \times 10^{-10}$ | $7.5345 \times 10^{-16}$ | $3.0485 \times 10^{-22}$ |
| 6   | $1.2826 \times 10^{-5}$ | $2.6964 \times 10^{-11}$ | $1.1772 \times 10^{-17}$ | $1.1908 \times 10^{-24}$ |

Table 6: Absolute errors at  $\xi = 3$ .

| $n$ | $m = 0$                 | $m = 1$                 | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|-------------------------|--------------------------|--------------------------|
| 3   | $4.5982 \times 10^{-3}$ | $5.5691 \times 10^{-6}$ | $1.4003 \times 10^{-9}$  | $8.1582 \times 10^{-14}$ |
| 4   | $1.1454 \times 10^{-3}$ | $3.4676 \times 10^{-7}$ | $2.1800 \times 10^{-11}$ | $3.1753 \times 10^{-16}$ |
| 5   | $2.8608 \times 10^{-4}$ | $2.1652 \times 10^{-8}$ | $3.4031 \times 10^{-13}$ | $1.2392 \times 10^{-18}$ |
| 6   | $7.1504 \times 10^{-5}$ | $1.3529 \times 10^{-9}$ | $5.3162 \times 10^{-15}$ | $4.8397 \times 10^{-21}$ |

## 7.2 Leading constants

At  $\xi = 3$ , divide the signed error by  $q^{(m+1)n}$ . The predicted limits from (4.15) are

| $m$ | predicted limit                       | computed at $n = 11$                  |
|-----|---------------------------------------|---------------------------------------|
| 0   | 0.29285958826753424146                | 0.29285960991271731281                |
| 1   | -0.022696618090733903713              | -0.022696619832602340475              |
| 2   | 0.000365296265765254499               | 0.000365296293045216575               |
| 3   | $-1.36214818263598388 \times 10^{-6}$ | $-1.36214828256045162 \times 10^{-6}$ |

The alternating signs and limiting constants are resolved without fitting any free parameter.

## 7.3 A posteriori estimates

For example, at  $\xi = 3$ ,  $n = 4$ ,  $m = 3$ , the true absolute error is  $3.17531383672 \times 10^{-16}$ . The computable difference between successive accelerated values is  $3.16292144780 \times 10^{-16}$ , and [theorem 4.9](#) gives the upper bound  $3.17532506132 \times 10^{-16}$ . The certificate therefore encloses the true error in an interval of relative width about  $4 \times 10^{-6}$ , using no reference value of  $P(3)$ .

## 8 A universal geometric-product theorem

The q-filter is not tied to the sine function. Its universality clarifies which parts of the construction are special to Rvachev’s function.

**Theorem 8.1** (universal analytic-factor acceleration). *Let  $\varphi$  be an even analytic function near zero with  $\varphi(0) = 1$ , nonzero in a neighborhood of zero, and*

$$\log \varphi(w) = - \sum_{k=1}^{\infty} a_k w^{2k}. \quad (8.1)$$

For  $b > 1$ , define

$$Q_b(\xi) = \prod_{\nu \geq 1} \varphi(\xi/b^\nu), \quad Q_{b,n}(\xi) = \prod_{\nu=1}^n \varphi(\xi/b^\nu).$$



Table 7: Absolute errors at  $\xi = 10$ , beyond the first sign change of the infinite product but still inside the tail disk for the displayed  $n$ .

| $n$ | $m = 0$                 | $m = 1$                 | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|-------------------------|--------------------------|--------------------------|
| 3   | $2.9276 \times 10^{-3}$ | $3.9468 \times 10^{-5}$ | $1.1012 \times 10^{-7}$  | $7.1217 \times 10^{-11}$ |
| 4   | $7.0230 \times 10^{-4}$ | $2.3635 \times 10^{-6}$ | $1.6504 \times 10^{-9}$  | $2.6705 \times 10^{-13}$ |
| 5   | $1.7380 \times 10^{-4}$ | $1.4617 \times 10^{-7}$ | $2.5525 \times 10^{-11}$ | $1.0327 \times 10^{-15}$ |
| 6   | $4.3341 \times 10^{-5}$ | $9.1120 \times 10^{-9}$ | $3.9782 \times 10^{-13}$ | $4.0240 \times 10^{-18}$ |

Then locally at  $q^n \xi^2 = 0$ ,

$$\frac{Q_{b,n}(\xi)}{Q_b(\xi)} = \exp \left( \sum_{k \geq 1} \frac{a_k}{b^{2k} - 1} \xi^{2k} q^{kn} \right). \quad (8.2)$$

Let its Taylor coefficients in  $z = q^n \xi^2$  be  $\beta_r$ . The same weights (4.2) satisfy

$$\frac{\sum_{j=0}^m w_j^{(m)} Q_{b,n+j}(\xi)}{Q_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \beta_r z^r. \quad (8.3)$$

If all  $a_k \geq 0$ , then all  $\beta_r \geq 0$ , and the sign law, divided- difference bounds, and a posteriori certificate remain valid.

*Proof.* The proof of [theorem 3.3](#) uses only the geometric sum  $\sum_{\ell \geq 1} b^{-2k\ell} = 1/(b^{2k} - 1)$ . The q-filter calculation then uses only the Taylor expansion and [theorem 4.4](#).  $\square$

Thus the sinc-specific information is concentrated in the Bernoulli values of  $a_k = \zeta(2k)/(k\pi^{2k})$ . The q-binomial extrapolation, its stability, and its exact mode cancellation belong to the geometry of the scales rather than to the elementary factor.

## 9 Lean formalization roadmap

The analytic proof naturally separates into formalization-sized modules.

1. **Finite q-algebra (formalized).** The finite  $q$ -Pochhammer and Gaussian-binomial core is in `FiniteQBinomialCore.lean`; the closed weights are in `GeometricQBinomialLagrange.lean`; and the principal specialization, all positive moments, scaled exact-row formula, and shifted Lagrange specialization are in `GeometricCompleteHomogeneous.lean` and `GeometricResidualMoments.lean`. The report-facing rational geometric modules cited above, at checkpoint `f77ff7c296b99050107948f5aa2a9488aec01d05`, additionally identify the weight polynomial with  $W_m$ , prove the  $q$ -Pochhammer moment presentation, the sign of every residual moment, the exact finite norm, and dedicated quarter-base corollaries. In particular the cancellations  $1 \leq r \leq m$  and (4.12) have exact Lean counterparts.
2. **Formal and unconditionally summable series (partially formalized).** The coefficientwise geometric filter and its Gaussian residual are in `GeometricPowerSeriesFilter.lean`; `AnalyticSeriesFilter.lean` proves the matching exact tail for an abstract unconditionally summable coefficient series. What remains here is to derive the Bell recurrence (3.11) from  $\Phi = \exp(\sum c_k z^k)$  and connect that concrete series to the sinc-product quotient.
3. **Analytic sinc logarithm.** Establish (3.4) in a disk, sum the geometric scales, and identify the analytic power series with the actual product. Existing repository lemmas about the sinc product should reduce the amount of new infrastructure.

4. **Interpolation and positivity (partially formalized).** The finite Lagrange weights, exact residual sign, and abstract unconditionally summable Gaussian tail are proved. The divided-difference representation for the actual sinc transform, its absolute-monotonicity argument, quantitative bounds, and uniform control remain frontier work.
5. **Fourier/measure layer.** Define the uniform box measures and their finite convolutions. Moment matching follows from derivatives of the characteristic function at zero and may be formalized before the full  $C^\infty$  theorem.
6. **Asymptotic layer.** Prove low-frequency Taylor bounds and the high-frequency superexponential estimate. This is the technically heaviest step, but it is independent of q-binomial algebra.
7. **Inverse-function layer.** Apply a quantitative inverse function theorem on compact intervals where  $F'$  has a positive lower bound, then derive (6.6) from the first-order perturbation identity.

The accompanying Python file supplies exact test vectors for the algebraic modules and high-precision regression tests for the analytic statements.

## 10 Further frontier problems

### 10.1 A matched Lambert–Richardson endpoint theory

Let  $G = F^{-1}$  and let  $G_{m,n}$  be the accelerated inverse branch. Determine the largest endpoint region  $0 < y \leq y_n$  on which

$$G_{m,n}(y) - G(y) \sim 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) 4^{-(m+1)n} \frac{F^{(2m+2)}(G(y))}{F'(G(y))}$$

remains uniform. Substitution of the repository’s Lambert– $W$  expansion for  $G(y)$  should identify a sharp transition curve  $y_n$ , together with a dyadically periodic modulation. This would connect two asymptotic theories that are presently separate.

### 10.2 Optimal order in finite precision

Although  $\Lambda_m(1/4) < 1.970$ , the finite products themselves may suffer relative loss near zeros. A complete finite-precision theory should choose  $m$  and  $n$  adaptively from: (i) the certificate (4.23); (ii) the distance to the shared zero set; and (iii) the working precision. Because the filter norm is uniformly bounded, one expects the optimal order to be governed mainly by product evaluation rather than extrapolation instability.

### 10.3 Positivity restoration

The signed density  $U_{b;m,n}$  has exact low moments but need not be positive. Is there a nonlinear, positivity-preserving correction that retains  $q^{(m+1)n}$  convergence and the moment constraints? One possibility is a small entropy projection onto the convex set of nonnegative densities with the same moments. The exact first missed moment (5.9) provides a natural scale for such a correction.

### 10.4 Dyadic rational evaluation from accelerated splines

At a dyadic point, each finite spline  $u_{2,n+j}$  is exactly computable from integrated Thue–Morse data and q-binomial formulae already present in the repository. Substituting those values into (4.3) should yield explicit rational or q-rational approximants to  $F(x)$  with a certified  $4^{-(m+1)n}$  remainder. This would produce a new bridge between exact dyadic arithmetic and the analytic sinc tail.

## 10.5 Complex domains and pole-aware extrapolation

The nearest tail pole at  $z = \pi^2 b^2$  controls the Taylor expansion. For complex  $\xi$ , one can conformally map the slit  $z$ -plane before applying a geometric interpolation filter. A pole-aware variant may retain exact  $q$ -mode cancellation while extending useful acceleration beyond the first omitted sinc zero.

## 10.6 Other refinable and compactly supported functions

Apply [theorem 8.1](#) to elementary factors arising from compactly supported refinable distributions, Bernoulli convolutions, or symmetric B-splines. The main classification question is: which elementary factors have nonnegative logarithmic cumulants, and therefore inherit the exact sign law and successive-level certificate?

## 11 Conclusion

The truncation of the Rvachev sinc product carries more structure than a generic convergent product. Its omitted tail is an analytic function of the single geometric variable  $z = 4^{-n}\xi^2$ . Bernoulli numbers describe the logarithm of that function, Bell polynomials describe the function itself, and Lagrange interpolation on the nodes  $1, q, \dots, q^m$  produces a  $q$ -binomial Richardson filter with a complete, sign-definite error series.

The most notable consequences are exact rather than heuristic: the  $q$ -moments of the weights are known in closed form; the filter norm stays bounded at arbitrarily high order; consecutive accelerated levels certify the error; and the associated signed box splines match the first  $2m$  moments of the limiting up-density for every  $n$ . Fourier inversion converts the same algebra into  $C^\infty$ -superconvergence with an explicit derivative profile. Translation and local inversion then give a new interior approximation theory for the Fabius function and its inverse, complementary to the repository's Lambert- $W$  endpoint asymptotics.

These results create concrete next steps: connect the compiled formal and unconditionally summable Gaussian filter to the concrete sinc-product/Bell series, prove its ordered and uniform analytic error estimates, build a dyadic exact-evaluation implementation, and develop a matched two-parameter theory linking truncation acceleration to the periodic Lambert phase near the endpoint.

## A Exact coefficients and filters at $b = 2$

The next two Bell coefficients, useful for regression tests, are

$$\alpha_6 = \frac{1884491755011469}{1980124051433319792000000},$$

$$\alpha_7 = \frac{151616054172597829}{6278781742186853835936000000}.$$

The order-four filter is

$$\mathcal{R}_{4,n}P = \frac{1}{722925}P_n - \frac{4}{8505}P_{n+1} + \frac{64}{2025}P_{n+2} - \frac{4096}{8505}P_{n+3} + \frac{1048576}{722925}P_{n+4}.$$

The order-five weights are

$$\left( -\frac{1}{739552275}, \frac{4}{2168775}, -\frac{64}{127575}, \frac{4096}{127575}, -\frac{1048576}{2168775}, \frac{1073741824}{739552275} \right).$$

Despite the large integer numerators, the sum of absolute values is below 1.968 at order five.

## B Corpus audit map

This appendix records the source families used to establish the non-duplication boundary. Historical variants were read for content rather than treated as independent discoveries.

### Current top-level expositions

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`
- `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`
- `non-formalized-research-frontiers/non-formalized-research-frontiers.tex`

### Current frontier drafts

- `drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex`
- LaTeX source in `Rvachev_Up_Fourier_Decay-2`
- LaTeX source in `Rvachev_Up_Fourier_Decay-3`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Comparative_Audit`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Gentle_Guide`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Second_Wave_Audit`
- LaTeX source in `asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions`
- LaTeX source in `rvachev_up_fourier_decay-1`
- LaTeX sources in `rvachev_up_fourier_decay-4` through `rvachev_up_fourier_decay-8`

### Paper-source directories

- `AIF_2017__67_2_637_0.tex` and its corrected source
- `Lacunary_products.tex` (arXiv:1502.06738v2)
- `09-Function.tex` (arXiv:1702.05442v1)
- `157-Arithmetic-v3.tex` (arXiv:1702.06487v3)
- `document.tex` (arXiv:2211.13570v1)
- `ubiq15-corrected.tex`

### Historical archive families

- `Fabius_Asymptotic.tex`
- `K-fold summation over signed Thue-Morse.tex`
- `Non_Elementarity_of_the_Fabius_Function.tex`
- `Small_Argument_Asymptotics.tex`
- `fabius-inverse-dyadic-closed-form.tex`
- Historical primary exposition variants under `Fabius_Function_and_Rvachev_Up`, `Fabius_and_Rvachev_Up-2`, `Fabius_and_Rvachev_up`, and `Fabius_function_and_up_function`
- `Fabius_Function_Missing_Parts` variants
- `Fabius_q_Special_Functions` and `fabius-q-special-functions`

- `Fabius_Function_Lean_Walkthrough` and `fabius_lean_walkthrough` variants
- `FabiusFunction_Mathematical_Glossary` variants
- `Fabius_Dyadic_Asymptotic_Bridge`
- `Fabius_Dyadic_Formulae_and_Alternative_Representations`
- `Fabius_Dyadic_Formulae_to_Asymptotics`
- `Fabius_Dyadic_q_Connections`
- `Fabius_Thue_Morse_Convergence_Rate`
- `Fabius_Thue_Morse_Convergence_Rates`
- `Primary_Exposition_Gap_Register`
- `Repeated_Integration_and_Rvachev_Up`
- `Rvachev_Up_from_Repeated_Integration`
- Earlier consolidated non-formalized-research-frontiers source
- `Thue_Morse_Formula_Atlas-1`, `Thue_Morse_Formula_Atlas-2`, and `Thue_Morse_Formula_Atlas-3`

The current consolidated frontier source explicitly records the absorption of several former notebooks and their checksums. That provenance was used when checking whether a proposed statement was genuinely absent or merely moved.

## C Reproducibility notes

The archive accompanying this report contains:

- `q_richardson_sinc_frontier.tex` — the present source;
- `q_richardson_sinc_frontier.pdf` — the rendered report;
- `verify_frontier_results.py` — exact symbolic and high-precision numerical checks;
- `numerical_results.txt` and `numerical_results.csv` — machine-readable output;
- `README.txt` — compilation and reproduction commands.

The script checks the  $q$ -moment identity symbolically for several arbitrary orders, generates the exact Bernoulli and Bell coefficients, verifies agreement between the explicit and triangular filters, reproduces tables 5 to 7, checks the leading constants, and asserts the two-sided certificate (4.23) at representative points.

## References

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## Part III

# Finite Dyadic Sinc Products and Piecewise-Polynomial Approximants to Rvachev’s Up-Function

### Abstract

Let

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

under the Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i x \xi} dx$ . This report studies the inverse transforms  $p_n$  of the first  $n$  factors. They are nonstationary dyadic box splines: compactly supported, nonnegative, even, log-concave, piecewise polynomial of degree  $n - 1$ , and  $C^{n-2}$ . Their knots unexpectedly form a uniform dyadic grid, while the jumps of the top derivative carry the signed Thue–Morse sequence. An explicit truncated-power formula, exact cell formulas, recursions, Fourier-zero multiplicities, partition-of-unity identities, cumulants, and sharp derivative geometry are obtained.

The main convergence result is an exact law, not merely an upper bound. For every  $r \geq 0$  and  $n \geq r + 3$ ,

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The extremum occurs at an explicit dyadic point. The associated distribution functions satisfy the exact Kolmogorov law  $d_K = 1/(9 \cdot 4^n)$ . At all orders,  $p_n$  admits a uniform differential expansion whose coefficients are complete Bell polynomials in Bernoulli-number data from  $1/\Phi$ ; geometric Richardson combinations then give signed  $4^{-Mn}$  extrapolation with uniformly bounded coefficient variation.

A second, positive acceleration theory is developed from a new uniform-scale-mixture representation of up. The exact tail is a mixture of uniform boxes governed by a probability measure  $\nu$  on  $[0, 1]$ . Positive Gauss, Radau, and Lobatto quadrature for  $\nu$  turns the continuum tail into finite positive mixtures of boxes. A one-signed Peano-kernel argument and the dyadic derivative plateaux give exact  $C^r$  errors for the whole hierarchy. In particular, a single variance-matched extra box yields an exact  $16^{-n}$  law, resolving a positive variance-matching direction left open in the repository’s frontier volume; right-Radau rules preserve the exact support  $[-1, 1]$  and attain arbitrarily high algebraic orders. Reproducible FFT and exact-rational experiments verify the formulas to numerical precision.

## Contents

# 1 Scope, source audit, and novelty boundary

## 1.1 Documents examined

The source audit covered the living TeX documents under `Analysis/FabiusFunction/docs`, especially the primary synthesis *The Fabius Function and Rvachev’s Up-Function*, the mathematical glossary, the Lean walkthrough, and the consolidated volume *Non-formalized Research Frontiers*. The recursive archive index was also examined to identify the historical TeX families: early notes, standalone studies, primary-exposition drafts, supplements,  $q$ -calculus articles, Lean walkthroughs, glossaries, and per-topic frontier dossiers. The archive explicitly records frozen first-introduced versions and prunes byte-identical copies already present in the live tree, so the living consolidated sources are the appropriate mathematical baseline for avoiding duplication [1, 2, 3].

The existing corpus already develops the probability law of up, the infinite sinc product, exact dyadic arithmetic, Thue–Morse identities, derivative tilings, moments,  $q$ -binomial structure, endpoint Lambert– $W$  asymptotics, and a repeated-integration spline  $f_n$ . In particular, the frontier volume proves a sharp error law for  $f_n$  and explicitly contrasts it with the ordinary prefix density  $p_n$ , whose sharp law is not studied there. It then asks for a positive compactly supported tail replacement with matching variance, predicting a possible  $16^{-n}$  rate but asserting no such scheme [2].

## 1.2 What is new here

The principal additions of this report are:

1. the exact uniform-grid truncated-power representation of the ordinary inverse transform  $p_n$ , with Thue–Morse jump weights;
2. a sharp derivative-plateau theorem for  $p_n$ ;
3. the exact  $C^r$  prefix error law and exact Kolmogorov distance;
4. a uniform all-orders expansion of  $p_n$  in even derivatives of up, with Bell–Bernoulli coefficient formulas;
5. stable  $q = 1/4$  Richardson weights and their closed  $q$ -binomial form;
6. a uniform-scale-mixture representation of the exact tail;
7. a general positive quadrature theorem giving exact error constants for Gauss, Radau, and Lobatto tail surrogates;
8. explicit positive fourth- and sixth-order schemes, including exact-support accelerants.

Every theorem below is accompanied by a mathematical proof. These results have not been added to the repository’s Lean development and should therefore be read as a research contribution awaiting independent review and formalization. Numerical experiments are used as verification, never as a substitute for proof.

# 2 Fourier normalization and the probabilistic head–tail model

## 2.1 The infinite product

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i x \xi} d\xi. \quad (2.1)$$

For  $a > 0$ , define the centered uniform density

$$u_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x). \quad (2.2)$$



Then

$$\widehat{u}_a(\xi) = \text{sinc}(2\pi a\xi). \quad (2.3)$$

Rvachev's normalized up-function is the even  $C^\infty$  probability density on  $[-1, 1]$  whose transform is

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right). \quad (2.4)$$

This normalization agrees with the compactly supported function studied by Rvachev and Arias de Reyna [7, 8].

## 2.2 Partial products and random sums

For  $n \geq 1$ , set

$$\Phi_n(\xi) = \prod_{j=0}^{n-1} \text{sinc}\left(\frac{\pi\xi}{2^j}\right), \quad p_n = \mathcal{F}^{-1}\Phi_n. \quad (2.5)$$

By equation (2.3),

$$p_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}}. \quad (2.6)$$

Let  $U_0, U_1, \dots$  be independent  $\text{Unif}[-1, 1]$  variables and put

$$S_n = \sum_{j=0}^{n-1} 2^{-j-1} U_j, \quad X = \sum_{j=0}^{\infty} 2^{-j-1} U_j. \quad (2.7)$$

Then  $p_n$  is the density of  $S_n$  and  $\text{up}$  is the density of  $X$ . Introduce

$$\delta_n = 2^{-n}, \quad A_n = 1 - 2^{-n}. \quad (2.8)$$

The omitted tail is a scaled independent copy of the whole sum:

$$X \stackrel{d}{=} S_n + \delta_n X', \quad X' \stackrel{d}{=} X, \quad X' \text{ independent of } S_n. \quad (2.9)$$

Consequently,

$$\Phi(\xi) = \Phi_n(\xi) \Phi(\delta_n \xi), \quad \text{up} = p_n * \rho_{\delta_n}, \quad \rho_{\delta}(x) = \delta^{-1} \text{up}(x/\delta). \quad (2.10)$$

The identity equation (2.10) is the engine behind both the exact error law and all-order acceleration.

## 3 Exact inverse transforms: dyadic box splines with Thue–Morse knots

### 3.1 Uniform knot formula

Write

$$\tau_k = (-1)^{s_2(k)}, \quad (3.1)$$

where  $s_2(k)$  is the binary digit sum. For  $m = 0$ , the convention  $y_+^0 = \mathbf{1}_{(0, \infty)}(y)$  is used away from knots.

**Theorem 3.1** (Exact truncated-power formula). *For every  $n \geq 1$ ,*

$$p_n(x) = \frac{2^{n(n-1)/2}}{(n-1)!} \sum_{k=0}^{2^n-1} \tau_k \left( x + A_n - \frac{k}{2^{n-1}} \right)_+^{n-1}. \quad (3.2)$$



The knots are the uniformly spaced points

$$t_{n,k} = -A_n + \frac{k}{2^{n-1}}, \quad 0 \leq k < 2^n. \quad (3.3)$$

In distributions,

$$D^n p_n = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} \tau_k \delta_{t_{n,k}}. \quad (3.4)$$

*Proof.* For one box,

$$Du_a = \frac{1}{2a}(\delta_{-a} - \delta_a).$$

Convolving the  $n$  derivatives in [equation \(2.6\)](#) gives a signed atomic measure. Choose a binary vector  $b = (b_0, \dots, b_{n-1}) \in \{0, 1\}^n$ ; its atom lies at

$$-A_n + \sum_{j=0}^{n-1} b_j 2^{-j} = -A_n + \frac{k}{2^{n-1}}, \quad k = \sum_{j=0}^{n-1} b_j 2^{n-1-j},$$

and its sign is  $(-1)^{\sum b_j} = \tau_k$ . The normalization is

$$\prod_{j=0}^{n-1} \frac{1}{2 \cdot 2^{-j-1}} = \prod_{j=0}^{n-1} 2^j = 2^{n(n-1)/2}.$$

This proves [equation \(3.4\)](#). Integrating  $n$  times from  $-\infty$ , using compact support, gives [equation \(3.2\)](#).  $\square$

The equal spacing is special to the dyadic weights. The  $2^n$  subset sums of  $1, 1/2, \dots, 2^{1-n}$  fill a complete integer grid after multiplication by  $2^{n-1}$ , while their parity signs are exactly Thue–Morse. Thus a nonstationary convolution of boxes of different widths becomes a uniform-knot spline with highly structured signed top derivative.

### 3.2 Cell formula and exact arithmetic

Set

$$y = 2^{n-1}(x + A_n). \quad (3.5)$$

Inside the support, [equation \(3.2\)](#) becomes

$$p_n(x) = \frac{2^{-(n-1)(n-2)/2}}{(n-1)!} \sum_{0 \leq k \leq \lfloor y \rfloor} \tau_k (y - k)^{n-1}. \quad (3.6)$$

This has several immediate consequences.

**Corollary 3.2** (Piecewise-polynomial and arithmetic structure). *The function  $p_n$  is supported on  $[-A_n, A_n]$ , is polynomial of degree at most  $n-1$  on each of the  $2^n-1$  open knot cells, and belongs to  $C^{n-2}(\mathbb{R})$  for  $n \geq 2$ . At rational arguments it has rational values; at dyadic arguments all computations may be performed with integers after one global power-of-two scaling.*

On the cell  $(t_{n,J}, t_{n,J+1})$ ,

$$p_n^{(n-1)}(x) = 2^{n(n-1)/2} \sum_{k=0}^J \tau_k. \quad (3.7)$$

The elementary Thue–Morse prefix identity

$$\sum_{k=0}^J \tau_k = \begin{cases} \tau_{J/2}, & J \text{ even,} \\ 0, & J \text{ odd} \end{cases} \quad (3.8)$$

therefore shows that every second top-derivative cell is identically zero, while the remaining cells carry a compressed Thue–Morse pattern.

### 3.3 Convolution and scaling recursions

Appending the final box in [equation \(2.6\)](#) gives

$$p_n(x) = 2^{n-1} \int_{x-2^{-n}}^{x+2^{-n}} p_{n-1}(t) dt, \quad (3.9)$$

$$p'_n(x) = 2^{n-1} (p_{n-1}(x + 2^{-n}) - p_{n-1}(x - 2^{-n})). \quad (3.10)$$

The product shell  $\Phi_n(\xi) = \text{sinc}(\pi\xi)\Phi_{n-1}(\xi/2)$  yields the equivalent dilation recursion

$$p_n(x) = \int_{2x-1}^{2x+1} p_{n-1}(s) ds, \quad (3.11)$$

$$p'_n(x) = 2(p_{n-1}(2x+1) - p_{n-1}(2x-1)). \quad (3.12)$$

These formulas permit exact construction of all cell polynomials by repeated primitive and shift operations. They avoid the severe cancellation that can occur in the global truncated-power sum for large  $n$ .

## 4 Structural properties of the finite-product splines

### 4.1 Shape, plateau, and partition of unity

**Proposition 4.1** (Basic geometry). *For every  $n \geq 1$ :*

- (i)  $p_n \geq 0$ ,  $p_n$  is even, and  $\int p_n = 1$ ;
- (ii)  $\text{supp } p_n = [-A_n, A_n]$ ;
- (iii)  $p_n$  is log-concave and therefore unimodal;
- (iv)  $p_n(x) = 1$  for  $|x| \leq \delta_n$ ;
- (v) for  $n \geq 2$ ,

$$\sum_{m \in \mathbb{Z}} p_n(x - m) = 1, \quad p_n(x) + p_n(1 - x) = 1 \quad (0 \leq x \leq 1). \quad (4.1)$$

*Proof.* The first three assertions follow from convolution of centered uniform densities; log-concavity is preserved by convolution. For the plateau, split off the largest box  $u_{1/2}$ . The remaining convolution is supported on  $[-1/2 + \delta_n, 1/2 - \delta_n]$ . If  $|x| \leq \delta_n$ , the interval  $[x - 1/2, x + 1/2]$  contains that support, so convolution with  $u_{1/2}$  integrates the entire remaining density and equals 1.

At every nonzero integer  $k$ , the factor  $\text{sinc}(\pi\xi)$  gives  $\Phi_n(k) = 0$ , while  $\Phi_n(0) = 1$ . Poisson summation therefore gives the first identity in [equation \(4.1\)](#). On  $[0, 1]$ , the support radius is less than 1, so only the translates  $m = 0, 1$  survive; evenness converts  $p_n(x - 1)$  to  $p_n(1 - x)$ .  $\square$

### 4.2 Fourier zeros

Let  $v_2(k)$  denote the exponent of 2 in a nonzero integer  $k$ .

**Proposition 4.2** (Exact zero multiplicities). *For  $k \in \mathbb{Z} \setminus \{0\}$ ,*

$$\text{ord}_{\xi=k} \Phi_n = 1 + \min\{v_2(k), n - 1\}. \quad (4.2)$$

*For fixed  $k$ , this stabilizes to  $1 + v_2(k)$ , the multiplicity of the corresponding zero of  $\Phi$ .*

This 2-adic spectral stratification is the Fourier counterpart of the Thue–Morse jump pattern. It also explains why the splines reproduce constants by integer translation, while higher-order cancellation appears selectively on dyadic frequency sublattices.

### 4.3 Cumulants and the variance defect

Let  $B_{2m}$  be the Bernoulli numbers with  $B_2 = 1/6$  and  $B_4 = -1/30$ . The even cumulants of  $X$  are

$$\kappa_{2m}(X) = \frac{B_{2m}}{2m(1 - 2^{-2m})}, \quad \kappa_{2m+1}(X) = 0. \quad (4.3)$$

Indeed, a uniform  $U \sim \text{Unif}[-1, 1]$  has  $\kappa_{2m}(U) = 2^{2m} B_{2m}/(2m)$ , and the geometric weights sum exactly. From [equation \(2.9\)](#),

$$\kappa_{2m}(S_n) = (1 - \delta_n^{2m})\kappa_{2m}(X). \quad (4.4)$$

In particular,

$$\mathbb{E}X^2 = \frac{1}{9}, \quad \mathbb{E}X^4 = \frac{19}{675}, \quad \mathbb{E}X^6 = \frac{583}{59535}, \quad (4.5)$$

and

$$\text{Var } X - \text{Var } S_n = \frac{\delta_n^2}{9}. \quad (4.6)$$

The raw prefix's leading  $4^{-n}$  error is exactly this missing variance transported through the second derivative.

### 4.4 Sharp derivative plateaux

Define

$$c_m = 1 - 2^{-m} \quad (m \geq 0), \quad c_0 = 0. \quad (4.7)$$

The living primary exposition proves the corresponding sharp derivative norm for up through its exact dyadic derivative tiling [1]. The finite splines possess a stronger local plateau.

**Theorem 4.3** (Finite-prefix derivative geometry). *Fix  $m \geq 0$  and  $n \geq m + 1$ . Interpreting the top derivative at the limiting case  $n = m + 1$  as its canonical almost-everywhere step representative, one has*

$$\|p_n^{(m)}\|_{L^\infty} = 2^{\binom{m+1}{2}}. \quad (4.8)$$

Moreover,

$$p_n^{(m)}(x) = (-1)^m 2^{\binom{m+1}{2}} \quad \text{for almost every } x \in [c_m - \delta_n, c_m + \delta_n]. \quad (4.9)$$

For  $n \geq m + 2$  the derivative is classical and the equality holds pointwise on the closed interval. By evenness, the reflected plateau has the corresponding parity sign.

*Proof.* Factor

$$p_n = p_{m+1} * q_{m+1,n},$$

where  $q_{m+1,n}$  is the convolution of the remaining boxes and is a probability density supported on

$$[-2^{-m-1} + \delta_n, 2^{-m-1} - \delta_n].$$

By [equations \(3.7\) and \(3.8\)](#), the step function  $p_{m+1}^{(m)}$  takes only the values 0 and  $\pm 2^{\binom{m+1}{2}}$ . Convolution with a probability measure cannot increase its supremum norm, proving the upper bound.

The point  $c_m$  is the midpoint of the rightmost nonzero cell of  $p_{m+1}^{(m)}$ , whose half-width is  $2^{-m-1}$ . Translating the support of  $q_{m+1,n}$  across any  $x \in [c_m - \delta_n, c_m + \delta_n]$  remains inside that cell. Hence the convolution averages a constant and gives [equation \(4.9\)](#), which also proves sharpness.  $\square$

## 5 Sharp convergence of the ordinary partial products

### 5.1 Smooth-test and transport estimates

The coupling [equation \(2.9\)](#) immediately gives, for  $1 \leq p < \infty$ ,

$$W_p(\mathcal{L}(S_n), \mathcal{L}(X)) \leq \delta_n (\mathbb{E}|X|^p)^{1/p} \leq \delta_n. \quad (5.1)$$

The mean-zero tail improves smooth tests by one full power. If  $\varphi \in C^2(\mathbb{R})$  has bounded second derivative, then

$$|\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)| \leq \frac{\delta_n^2}{18} \|\varphi''\|_\infty. \quad (5.2)$$

For convex  $\varphi$ , the left side has the nonnegative sign  $\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)$  by conditional Jensen. The constant is sharp for  $\varphi(x) = x^2$ .

### 5.2 Exact $C^r$ error law

**Theorem 5.1** (Sharp ordinary-prefix law). *For every integer  $r \geq 0$  and every  $n \geq r + 3$ ,*

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.3)$$

The extremum is attained at

$$x = c_{r+2} = 1 - 2^{-(r+2)}, \quad (5.4)$$

and the signed value is

$$p_n^{(r)}(c_{r+2}) - \text{up}^{(r)}(c_{r+2}) = (-1)^{r+1} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.5)$$

If  $\|f\|_{C^r} = \max_{0 \leq j \leq r} \|f^{(j)}\|_\infty$ , then the same right-hand side is exactly  $\|p_n - \text{up}\|_{C^r}$ .

*Proof.* Let  $g = p_n^{(r)}$ . From [equation \(2.10\)](#),

$$\text{up}^{(r)}(x) = \mathbb{E}g(x - \delta_n X).$$

The function  $g$  has a bounded second weak derivative when  $n \geq r + 3$ . Taylor's formula with integral remainder gives

$$g(x - h) = g(x) - hg'(x) + h^2 \int_0^1 (1-t)g''(x - th) dt.$$

Set  $h = \delta_n X$  and take expectations. Since  $\mathbb{E}X = 0$  and  $\mathbb{E}X^2 = 1/9$ ,

$$\begin{aligned} \left| \text{up}^{(r)}(x) - p_n^{(r)}(x) \right| &\leq \frac{\delta_n^2}{18} \left\| p_n^{(r+2)} \right\|_\infty \\ &= \frac{\delta_n^2}{18} 2^{\binom{r+3}{2}}, \end{aligned}$$

where [theorem 4.3](#) was used in the second line. This is the claimed upper bound.

At  $x = c_{r+2}$ , the entire interval  $[x - \delta_n, x + \delta_n]$  lies in the plateau from [equation \(4.9\)](#) for  $p_n^{(r+2)}$ . The Taylor remainder therefore averages a constant, and every inequality above is an equality. Its sign is obtained from the plateau sign  $(-1)^{r+2}$ . Finally, the constants in [equation \(5.3\)](#) increase strictly with the derivative order, proving the  $C^r$  statement.  $\square$

For  $r = 0$ ,

$$\|p_n - \text{up}\|_\infty = \frac{4}{9 \cdot 4^n} \quad (n \geq 3). \quad (5.6)$$

This is exactly one half of the repeated-integration error  $8/(9 \cdot 4^n)$  in the frontier volume: omitting the tail has variance defect  $-1/9$ , whereas replacing it by a full-radius uniform box has variance defect  $+2/9$ .

### 5.3 Exact Kolmogorov distance

Let

$$P_n(x) = \int_{-\infty}^x p_n(t) dt, \quad P(x) = \int_{-\infty}^x \text{up}(t) dt. \quad (5.7)$$

**Corollary 5.2** (Exact CDF convergence). *For every  $n \geq 2$ ,*

$$\|P_n - P\|_{\infty} = \frac{1}{9 \cdot 4^n}. \quad (5.8)$$

The maximum is attained at  $x = 1/2$ , where  $P_n(x) - P(x) = 1/(9 \cdot 4^n)$ .

*Proof.* Apply the proof of [theorem 5.1](#) to  $g = P_n$ . Now  $g'' = p'_n$ , whose supremum is 2 by [theorem 4.3](#). The derivative  $p'_n$  is identically  $-2$  on  $[1/2 - \delta_n, 1/2 + \delta_n]$ , so equality occurs at  $1/2$ .  $\square$

### 5.4 $L^p$ consequences and endpoint flatness

For  $1 \leq p < \infty$  and  $n \geq r + 3$ ,

$$\|p_n^{(r)} - \text{up}^{(r)}\|_{L^p} \leq 2^{1/p} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.9)$$

The all-orders expansion in [section 6](#) sharpens this to

$$4^n \|p_n^{(r)} - \text{up}^{(r)}\|_{L^p} \rightarrow \frac{1}{18} \|\text{up}^{(r+2)}\|_{L^p}, \quad 1 \leq p \leq \infty. \quad (5.10)$$

Although  $p_n$  vanishes on the thin endpoint layers  $A_n < |x| \leq 1$ , this causes no  $O(\delta_n)$  uniform loss. Since every derivative of  $\text{up}$  vanishes at  $\pm 1$ , for every  $M$  and fixed  $r$ ,

$$\sup_{A_n \leq |x| \leq 1} |\text{up}^{(r)}(x)| = O_{r,M}(\delta_n^M). \quad (5.11)$$

The endpoint discrepancy is beyond every algebraic order; the  $4^{-n}$  error is an interior moment effect.

## 6 All-orders differential expansion

### 6.1 The reciprocal product and Bell–Bernoulli coefficients

Near the origin, define

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{j=1}^{\infty} \alpha_j (2\pi z)^{2j}\right) = \sum_{m=0}^{\infty} a_m (2\pi z)^{2m}. \quad (6.1)$$

The logarithm of the sine product gives

$$\alpha_j = \frac{|B_{2j}|}{2j(2j)!(1 - 2^{-2j})}. \quad (6.2)$$

Thus

$$a_0 = 1, \quad ma_m = \sum_{j=1}^m j \alpha_j a_{m-j}, \quad (6.3)$$

and, in terms of the complete exponential Bell polynomial  $Y_m$ ,

$$a_m = \frac{1}{m!} Y_m(1! \alpha_1, 2! \alpha_2, \dots, m! \alpha_m). \quad (6.4)$$

The first coefficients are

$$\begin{aligned} a_0 &= 1, & a_1 &= \frac{1}{18}, & a_2 &= \frac{31}{16200}, \\ a_3 &= \frac{2347}{42865200}, & a_4 &= \frac{1904369}{1311675120000}, & a_5 &= \frac{3309760579}{88561680752160000}. \end{aligned} \quad (6.5)$$

## 6.2 Uniform expansion theorem

**Theorem 6.1** (All-orders prefix expansion). *Fix integers  $r \geq 0$  and  $M \geq 1$ . As  $n \rightarrow \infty$ ,*

$$p_n^{(r)}(x) = \sum_{m=0}^{M-1} (-1)^m a_m \delta_n^{2m} \text{up}^{(r+2m)}(x) + O_{r,M}(\delta_n^{2M}) \quad (6.6)$$

uniformly for  $x \in \mathbb{R}$ .

*Proof.* By equation (2.10),

$$\Phi_n(\xi) = \Phi(\xi) \frac{1}{\Phi(\delta_n \xi)}.$$

Choose  $0 < c < 1$ . On  $|\xi| \leq c/\delta_n$ , the reciprocal is analytic and its Taylor remainder after  $M - 1$  terms is bounded by  $C_M \delta_n^{2M} |\xi|^{2M}$ . Multiplication by  $|\xi|^r |\Phi(\xi)|$  is integrable because  $\Phi$  is Schwartz, so Fourier inversion bounds the low-frequency remainder by  $O_{r,M}(\delta_n^{2M})$ .

For the high-frequency tail, retain any fixed  $K$  factors of  $\Phi_n$ :

$$|\Phi_n(\xi)| \leq C_K |\xi|^{-K}.$$

Choosing  $K > r + 2M + 1$  makes the integral over  $|\xi| > c/\delta_n$  smaller than  $O(\delta_n^{2M})$ . The high-frequency tails of the finitely many  $\Phi(\xi) \xi^{2m}$  terms are superalgebraically small. Finally,

$$\mathcal{F}^{-1}[(2\pi\xi)^{2m} \Phi(\xi)] = (-1)^m \text{up}^{(2m)},$$

which yields equation (6.6) after differentiating  $r$  times.  $\square$

In particular,

$$p_n = \text{up} - \frac{\delta_n^2}{18} \text{up}'' + \frac{31\delta_n^4}{16200} \text{up}^{(4)} - \frac{2347\delta_n^6}{42865200} \text{up}^{(6)} + O(\delta_n^8). \quad (6.7)$$

The leading profile is therefore

$$4^n (p_n^{(r)} - \text{up}^{(r)}) \longrightarrow -\frac{1}{18} \text{up}^{(r+2)} \quad \text{uniformly.} \quad (6.8)$$

The sharp derivative norm of  $\text{up}^{(r+2)}$  is exactly the constant in equation (5.3), explaining why the asymptotic coefficient is already exact at the dyadic extremizer for every admissible  $n$ .

## 6.3 Signed geometric Richardson acceleration

Let  $q = 1/4$  and, for  $M \geq 1$ , define

$$R_{n,M} = \sum_{j=0}^{M-1} w_{M,j} p_{n+j}, \quad (6.9)$$

where

$$\begin{aligned} w_{M,j} &= (-1)^{M-1-j} \frac{q^{\binom{M-j}{2}}}{(q; q)_j (q; q)_{M-1-j}} \\ &= \frac{(-1)^{M-1-j} q^{\binom{M-j}{2}}}{(q; q)_{M-1}} \begin{bmatrix} M-1 \\ j \end{bmatrix}_q. \end{aligned} \quad (6.10)$$

Here  $(q; q)_k = \prod_{\ell=1}^k (1 - q^\ell)$ . These are the Lagrange weights that extrapolate from the geometric nodes  $q^j$ ,  $0 \leq j < M$ , to the origin:

$$\sum_{j=0}^{M-1} w_{M,j} q^{jm} = \delta_{m0} \quad (0 \leq m < M). \quad (6.11)$$

Consequently,

$$R_{n,M}^{(r)} = \text{up}^{(r)} - a_M q^{\binom{M}{2}} \delta_n^{2M} \text{up}^{(r+2M)} + O_{r,M}(\delta_n^{2M+2}). \quad (6.12)$$

The first weight vectors are

$$\begin{aligned} M = 2 : & \quad \left( -\frac{1}{3}, \frac{4}{3} \right), \\ M = 3 : & \quad \left( \frac{1}{45}, -\frac{4}{9}, \frac{64}{45} \right), \\ M = 4 : & \quad \left( -\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835} \right). \end{aligned}$$

Despite alternating signs, the coefficient variation stays uniformly bounded:

$$\sum_{j=0}^{M-1} |w_{M,j}| = \frac{(-q; q)_{M-1}}{(q; q)_{M-1}} < \frac{(-q; q)_\infty}{(q; q)_\infty} = 1.969260353668 \dots \quad (6.13)$$

This supplies a stable signed arbitrary-order method. The next section constructs an entirely positive alternative.

## 7 A positive quadrature hierarchy for the exact tail

### 7.1 Up as a scale mixture of uniform boxes

The function  $\text{up}$  is even, nonnegative, and decreasing on  $[0, 1]$ . Define a measure  $\eta$  on  $[0, 1]$  by

$$\mathrm{d}\eta(r) = -2r \, \text{up}'(r) \, \mathrm{d}r. \quad (7.1)$$

Integration by parts gives  $\eta([0, 1]) = 1$ .

**Theorem 7.1** (Uniform scale mixture). *Let  $R \sim \eta$  and  $U \sim \text{Unif}[-1, 1]$  be independent. Then*

$$X \stackrel{d}{=} RU, \quad (7.2)$$

and

$$\text{up}(x) = \int_0^1 u_r(x) \, \mathrm{d}\eta(r), \quad (7.3)$$

$$\Phi(\xi) = \int_0^1 \text{sinc}(2\pi r \xi) \, \mathrm{d}\eta(r). \quad (7.4)$$

If  $Z = R^2$  and  $\nu$  is its law, then

$$\mathrm{d}\nu(z) = -\text{up}'(\sqrt{z}) \, \mathrm{d}z, \quad 0 < z < 1, \quad (7.5)$$

and

$$m_k := \int_0^1 z^k \, \mathrm{d}\nu(z) = \mathbb{E}R^{2k} = (2k+1)\mathbb{E}X^{2k}. \quad (7.6)$$

*Proof.* For  $x \geq 0$ ,

$$\int_0^1 u_r(x) d\eta(r) = \int_x^1 \frac{1}{2r} (-2r \operatorname{up}'(r)) dr = \operatorname{up}(x) - \operatorname{up}(1) = \operatorname{up}(x).$$

Evenness handles  $x < 0$ , proving the density identity and hence [equation \(7.2\)](#). Fourier transformation gives [equation \(7.4\)](#). The change of variables  $z = r^2$  gives [equation \(7.5\)](#). Finally,

$$\mathbb{E}R^{2k} = \int_0^1 -2r^{2k+1} \operatorname{up}'(r) dr = 2(2k+1) \int_0^1 r^{2k} \operatorname{up}(r) dr = (2k+1) \mathbb{E}X^{2k}.$$

□

The first moments of  $\nu$  are

$$m_0 = 1, \quad m_1 = \frac{1}{3}, \quad m_2 = \frac{19}{135}, \quad m_3 = \frac{583}{8505}. \quad (7.7)$$

All  $m_k$  are rational. Therefore every monic orthogonal polynomial generated from  $\nu$  has rational coefficients, and all Gaussian or Radau nodes are algebraic numbers obtainable by exact arithmetic before root isolation.

## 7.2 Quadrature tail surrogates

Let

$$Q = \sum_{j=1}^J w_j \delta_{z_j}, \quad 0 \leq z_j \leq 1, \quad w_j > 0, \quad \sum_j w_j = 1, \quad (7.8)$$

be a positive quadrature rule for  $\nu$ . In spatial convolution formulas we use the measure convention  $u_0 = \delta_0$ . Let  $Z_Q$  have law  $Q$ , let  $U$  be an independent uniform variable, and define

$$Y_Q = \sqrt{Z_Q} U. \quad (7.9)$$

The associated positive approximant is

$$g_{n,Q} = p_n * \mathcal{L}(\delta_n Y_Q) = \sum_{j=1}^J w_j (p_n * u_{\delta_n \sqrt{z_j}}), \quad (7.10)$$

$$\widehat{g}_{n,Q}(\xi) = \Phi_n(\xi) \sum_{j=1}^J w_j \operatorname{sinc}(2\pi \delta_n \sqrt{z_j} \xi). \quad (7.11)$$

Thus every rule produces a finite positive mixture of explicit piecewise-polynomial box splines.

For a smooth function  $h$ , introduce

$$A_h(z) = \frac{1}{2} \int_{-1}^1 h(\sqrt{z} u) du. \quad (7.12)$$

Repeated integration by parts gives the key transfer identity

$$A_h^{(\ell)}(z) = \frac{1}{2^{2\ell+1} \ell!} \int_{-1}^1 (1-u^2)^\ell h^{(2\ell)}(\sqrt{z} u) du. \quad (7.13)$$

Since

$$\int_{-1}^1 (1-u^2)^\ell du = \frac{2^{2\ell+1} (\ell!)^2}{(2\ell+1)!}, \quad (7.14)$$

we obtain

$$\|A_h^{(\ell)}\|_\infty \leq \frac{\ell!}{(2\ell+1)!} \|h^{(2\ell)}\|_\infty. \quad (7.15)$$

This is the bridge by which matching a moment of the squared scale  $Z$  cancels two spatial derivatives.



### 7.3 General exact quadrature error theorem

Assume  $Q$  is exact for polynomials of degree at most  $\ell - 1$ , and its Peano kernel of order  $\ell$  has one sign. This includes Gaussian, one-endpoint Radau, and Lobatto rules for a positive measure on an interval [11, 12]. Set

$$\varepsilon_{Q,\ell} = Q(z^\ell) - m_\ell, \quad \Gamma_{Q,\ell} = \frac{|\varepsilon_{Q,\ell}|}{(2\ell + 1)!}. \quad (7.16)$$

**Theorem 7.2** (Exact positive quadrature law). *Suppose  $\varepsilon_{Q,\ell} \neq 0$ . For every  $r \geq 0$  and  $n \geq r + 2\ell + 1$ ,*

$$\left\| g_{n,Q}^{(r)} - \text{up}^{(r)} \right\|_\infty = \Gamma_{Q,\ell} 2^{\binom{r+2\ell+1}{2}} 2^{-2\ell n}. \quad (7.17)$$

The extremum occurs at  $x = c_{r+2\ell}$  and has sign  $\text{sgn}(\varepsilon_{Q,\ell})(-1)^r$ .

*Proof.* Put  $f = p_n^{(r)}$  and, for fixed  $x$ , let  $h_x(v) = f(x - \delta_n v)$ . Then

$$g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x) = Q(A_{h_x}) - \nu(A_{h_x}).$$

A one-signed Peano kernel and exactness through degree  $\ell - 1$  imply

$$|Q(A) - \nu(A)| \leq \frac{|Q(z^\ell) - \nu(z^\ell)|}{\ell!} \|A^{(\ell)}\|_\infty.$$

Using equation (7.15) and  $\|h_x^{(2\ell)}\|_\infty = \delta_n^{2\ell} \|p_n^{(r+2\ell)}\|_\infty$  gives

$$|g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x)| \leq \Gamma_{Q,\ell} \delta_n^{2\ell} 2^{\binom{r+2\ell+1}{2}}.$$

At  $x = c_{r+2\ell}$ , every argument  $x - \delta_n \sqrt{z} u$  lies in the constant plateau of  $p_n^{(r+2\ell)}$  from theorem 4.3. Hence  $A_{h_x}^{(\ell)}$  is constant, the Peano inequality is an equality, and the asserted sign follows.  $\square$

Two familiar approximations fit the same theorem. Ordinary truncation is  $Q = \delta_0$ ,  $\ell = 1$ , with  $\Gamma = 1/18$ . The repeated-integration spline is  $Q = \delta_1$ ,  $\ell = 1$ , with  $\Gamma = 1/9$ . Their factor-of-two error difference is therefore a one-line quadrature moment calculation.

### 7.4 Explicit positive accelerated schemes

**Example 7.3** (One-node Gauss: a single variance-matched box). The one-node Gauss rule for  $\nu$  is  $Q = \delta_{1/3}$ . It matches  $m_0, m_1$  and has

$$\varepsilon_{Q,2} = \frac{1}{9} - \frac{19}{135} = -\frac{4}{135}, \quad \Gamma_{Q,2} = \frac{1}{4050}.$$

Thus

$$G_n = p_n * u_{\delta_n/\sqrt{3}} \quad (7.18)$$

is a single positive box spline, and for  $n \geq r + 5$ ,

$$\left\| G_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+5}{2}}}{4050 \cdot 16^n}. \quad (7.19)$$

Its support radius is  $1 - \delta_n + \delta_n/\sqrt{3} < 1$ . Endpoint flatness makes the support deficit superalgebraically small, while the interior error is exactly fourth order.

**Example 7.4** (Two-node Lobatto: exact support and fourth order). The endpoint rule matching  $m_1 = 1/3$  is

$$Q = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1. \quad (7.20)$$

It gives

$$H_n = \frac{2}{3}p_n + \frac{1}{3}(p_n * u_{\delta_n}), \quad (7.21)$$

which is nonnegative, integrates to one, and has exact support  $[-1, 1]$ . Since

$$\varepsilon_{Q,2} = \frac{1}{3} - \frac{19}{135} = \frac{26}{135}, \quad \Gamma_{Q,2} = \frac{13}{8100},$$

$$\left\| H_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{13}{8100} \frac{2^{\binom{r+5}{2}}}{16^n} \quad (n \geq r+5). \quad (7.22)$$

This explicitly realizes the positive, compactly supported, variance-matched  $16^{-n}$  direction proposed but not constructed in the frontier document.

**Example 7.5** (Two-node right Radau: exact support and sixth order). The two-node right-Radau rule, constrained to contain  $z = 1$  and exact through  $z^2$ , is

$$Q = \frac{15}{16}\delta_{13/45} + \frac{1}{16}\delta_1. \quad (7.23)$$

Its first missed moment is

$$\varepsilon_{Q,3} = \frac{704}{42525}, \quad \Gamma_{Q,3} = \frac{44}{13395375}. \quad (7.24)$$

Hence

$$J_n = \frac{15}{16}(p_n * u_{\delta_n \sqrt{13/45}}) + \frac{1}{16}(p_n * u_{\delta_n}) \quad (7.25)$$

is positive with exact support  $[-1, 1]$ , and

$$\left\| J_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{44}{13395375} \frac{2^{\binom{r+7}{2}}}{64^n} \quad (n \geq r+7). \quad (7.26)$$

## 7.5 Arbitrary order with positivity

Standard quadrature immediately produces a hierarchy.

| Rule with $M$ nodes               | Exact degree | $\ell$   | Rate           | Support behavior                        |
|-----------------------------------|--------------|----------|----------------|-----------------------------------------|
| Gauss                             | $2M - 1$     | $2M$     | $2^{-4Mn}$     | Positive; strictly inside $[-1, 1]$     |
| Right Radau (node 1)              | $2M - 2$     | $2M - 1$ | $2^{-(4M-2)n}$ | Positive; exact support $[-1, 1]$       |
| Lobatto (nodes 0, 1, $M \geq 2$ ) | $2M - 3$     | $2M - 2$ | $2^{-(4M-4)n}$ | Positive; exact support, includes $p_n$ |

The constants are not merely asymptotic: once  $n \geq r + 2\ell + 1$ , they are exactly

$$\frac{|Q(z^\ell) - m_\ell|}{(2\ell + 1)!} 2^{\binom{r+2\ell+1}{2}}. \quad (7.27)$$

Thus positive arbitrary-order acceleration reduces to the classical algebra of orthogonal polynomials for the explicitly defined measure  $d\nu(z) = -\text{up}'(\sqrt{z}) dz$ .

## 8 Numerical and exact-arithmetic experiments

### 8.1 Method

The accompanying script uses the exact product samples

$$\Phi_n(k/L), \quad \Phi(k/L),$$

on a periodic box of length  $L = 4$  and applies an inverse FFT with  $2^{17}$  modes. Every function considered is supported in  $[-1, 1]$ , so its periodization has no overlap. Derivatives are obtained by multiplying the transform by  $(2\pi i\xi)^r$ . The infinite product is truncated after 48 factors, beyond double-precision relevance on the active frequency range. Separate routines evaluate [equation \(3.2\)](#) with exact rational arithmetic, generate the  $a_m$ , and generate the Richardson weights.

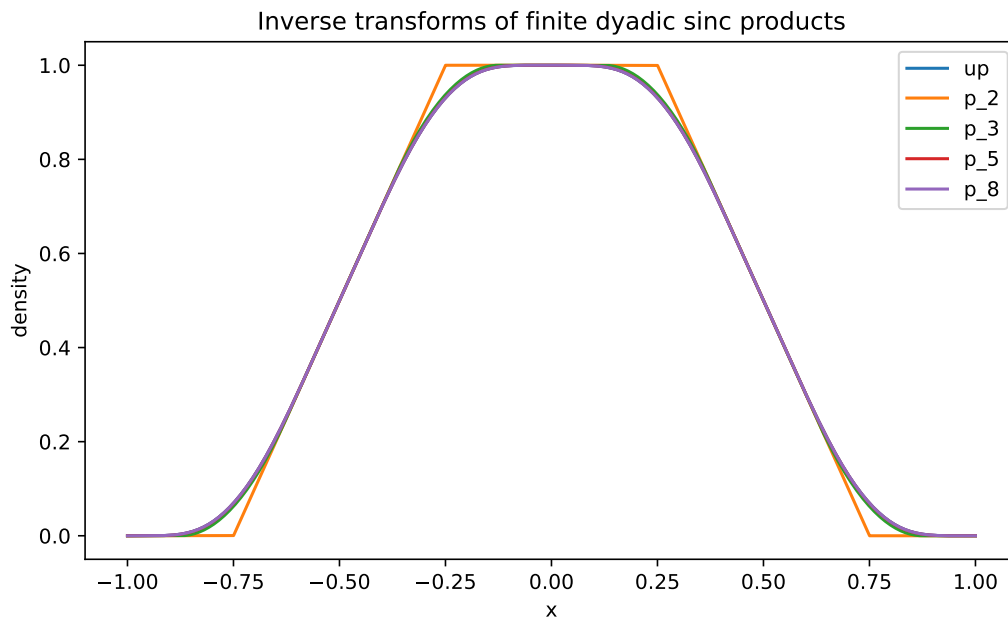


Figure 3: Inverse transforms of the first few finite sinc products. The support expands to  $[-1, 1]$ , the polynomial degree and smoothness increase, and the central plateau shrinks dyadically.

### 8.2 Verification of exact constants

Selected FFT ratios of measured supremum error to the exact formula are shown below.

| Scheme     | $r$ | $n$ | measured error                  | measured/exact |
|------------|-----|-----|---------------------------------|----------------|
| Raw prefix | 0   | 3   | $6.94444444448 \times 10^{-3}$  | 1.000000000006 |
| Raw prefix | 2   | 5   | $5.55555555771 \times 10^{-2}$  | 1.000000000387 |
| Raw prefix | 4   | 7   | 7.11111128754                   | 1.000000024810 |
| Gauss-1    | 0   | 5   | $2.41126543254 \times 10^{-7}$  | 1.000000000183 |
| Lobatto-2  | 0   | 5   | $1.56732253097 \times 10^{-6}$  | 1.000000000069 |
| Radau-2    | 0   | 7   | $1.56639950187 \times 10^{-12}$ | 1.000079775969 |

The raw and fourth-order laws agree essentially to machine precision. The Radau error is already  $1.6 \times 10^{-12}$  at its first admissible  $n$ , so Fourier truncation and roundoff visibly affect the last digits. Full tables are included in `sharp_error_verification.csv` and `positive_acceleration_verification.csv`.

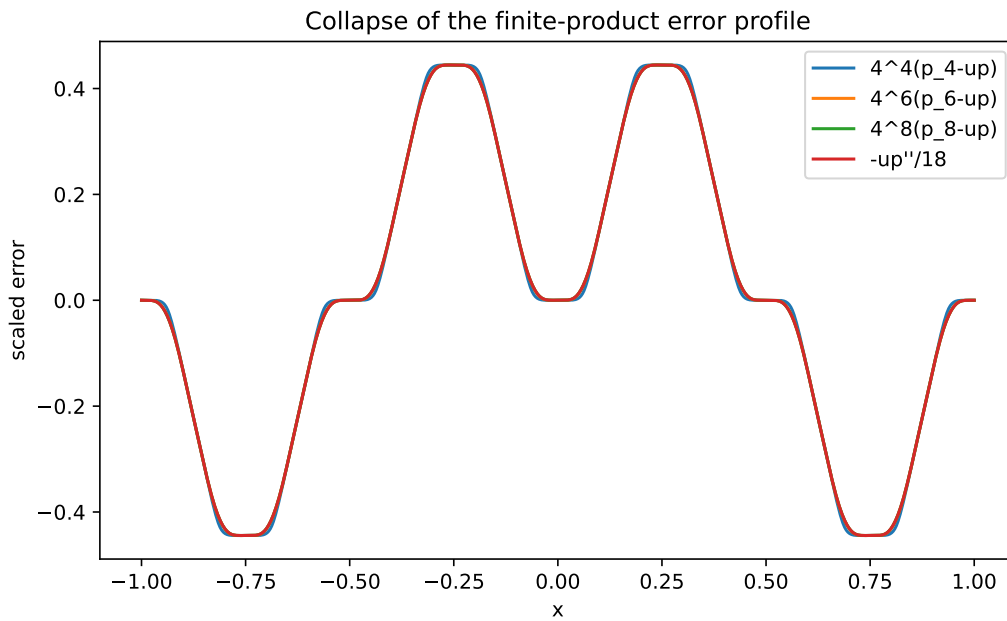


Figure 4: The profiles  $4^n(p_n - \text{up})$  collapse onto  $-\text{up}''/18$ , in agreement with [equation \(6.8\)](#).

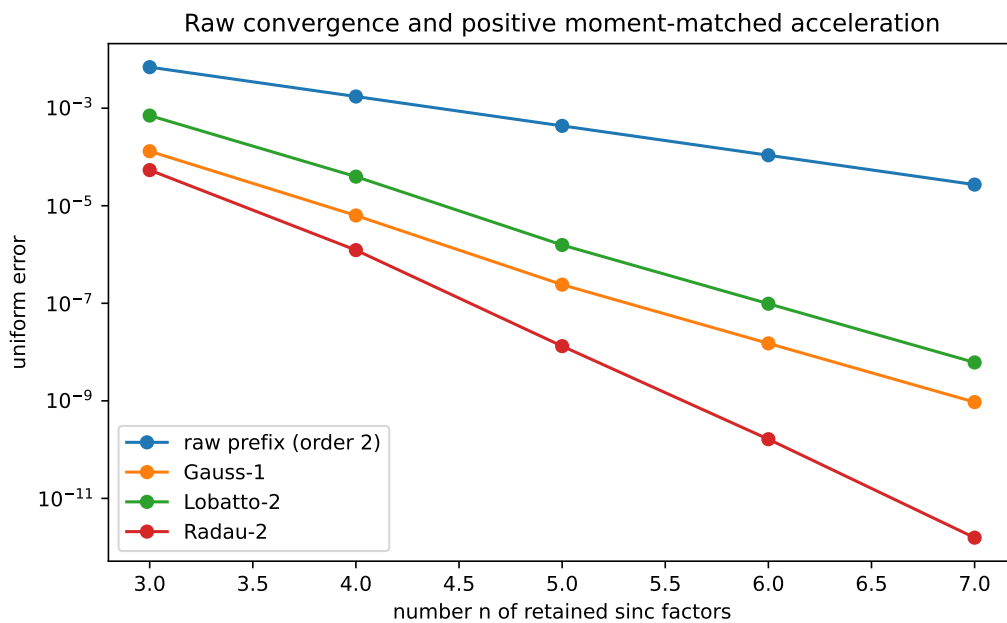


Figure 5: Raw second-order convergence and positive Gauss, Lobatto, and Radau tail acceleration. The highest-order Radau values approach the double-precision floor.

## 9 Algorithms and implementation considerations

### 9.1 Exact cell construction

There are three complementary exact representations.

**Global truncated powers.** Equation (3.2) evaluates a point in  $O(2^n)$  signed rational operations and is ideal for proof and low-order checks. Only knots left of  $x$  need be included.

**Primitive recursion.** If a piecewise-polynomial primitive of  $p_{n-1}$  is stored cellwise, then equation (3.9) evaluates  $p_n$  by two shifted primitive values. This constructs the complete spline with exact coefficients while reusing adjacent-cell data.

**Top-down integration of the Thue–Morse step.** Start from the top derivative values in equations (3.7) and (3.8), integrate cell by cell, and determine constants by continuity and vanishing at the left support endpoint. This uses only additions, multiplications by the mesh, and division by small integers. It is particularly attractive for formal verification.

### 9.2 Positive quadrature construction

The moment sequence  $m_k = (2k+1)\mathbb{E}X^{2k}$  can be generated exactly from the cumulants equation (4.3). A practical  $M$ -node Gauss construction is:

1. compute  $m_0, \dots, m_{2M}$  as rational numbers;
2. perform exact Gram–Schmidt on  $1, z, \dots, z^M$ , or solve the Hankel system for the monic orthogonal polynomial  $P_M$ ;
3. isolate the  $M$  simple roots of  $P_M$  in  $(0, 1)$ ;
4. solve the Vandermonde moment equations for positive weights;
5. insert the nodes and weights in equation (7.11) or equation (7.10).

Right-Radau rules are obtained by orthogonalizing against the modified measure  $(1 - z) d\nu(z)$  and then adjoining  $z = 1$ ; Lobatto rules use  $z(1 - z) d\nu(z)$  and adjoin both endpoints. Exact rational recurrence coefficients make certified algebraic output feasible.

### 9.3 Numerical stability

The global formula equation (3.2) contains large alternating terms and should not be evaluated naively in floating point at high  $n$ . Stable options are:

- exact rational arithmetic followed by one final conversion;
- cellwise Horner evaluation after primitive recursion;
- FFT inversion when values on a dense uniform grid are needed;
- convolution on adaptive polynomial pieces for positive quadrature mixtures.

The Richardson coefficients have uniformly bounded  $\ell^1$  norm by equation (6.13); their cancellation is therefore much milder than a generic high-order extrapolation. Positive quadrature schemes avoid cancellation entirely.

## 10 Further deductions and research directions

### 10.1 Results that appear immediately accessible

**Open problem 10.1** (Lean formalization of the prefix law). *Formalize [theorems 3.1, 4.3 and 5.1](#). The proof ingredients are already close to the repository’s APIs: finite uniform convolutions, Thue–Morse prefix sums, the exact tail decomposition, bounded weak second derivatives, and the formally proved sharp derivative tiling of  $\text{up}$ .*

**Open problem 10.2** (Certified quadrature generator). *Build an exact-arithmetic pipeline that returns the rational orthogonal polynomial, certified algebraic nodes, and positive weights for Gauss and right-Radau rules of  $\nu$ . The output would be a library of positive compactly supported approximants with machine-checkable error constants.*

**Open problem 10.3** (Norms beyond  $L^\infty$ ). *The expansion gives sharp leading constants in every  $L^p$  norm. The exact derivative distribution identities in the primary exposition suggest that some  $L^p$  constants, rearrangement-invariant norms, or signed error distributions may admit closed forms.*

### 10.2 Conjectures

**Conjecture 10.4** ( $q$ -special orthogonal structure). *The monic orthogonal polynomials of  $d\nu(z) = -\text{up}'(\sqrt{z})dz$  have recurrence coefficients admitting closed  $q$ -product or  $q$ -continued-fraction expressions at  $q = 1/4$ . Their rationality is automatic from the rational moments; the conjecture is that the rational numbers have a simple structural form involving  $(q; q)_m$ , Gaussian binomials, or the reciprocal-product coefficients  $a_m$ .*

**Conjecture 10.5** (Lambert edge scaling of quadrature nodes). *The extreme Gauss and Radau nodes have refined endpoint asymptotics controlled by the same lower-Lambert phase that governs the flat endpoint of the Fabius/Rvachev function. The leading zero distribution may be classical, but the edge correction should retain the repository’s log-periodic dyadic phase.*

**Conjecture 10.6** (Positive minimax tail rule). *Among positive  $M$ -component uniform scale mixtures supported in  $[-1, 1]$  and required to include the endpoint scale 1, the right-Radau rule minimizes the first nonzero worst-case  $C^r$  error constant simultaneously for all fixed  $r$ . The exact plateau formula reduces this to an extremal moment problem on  $[0, 1]$ .*

**Conjecture 10.7** (Shape preservation). *For every right-Radau rule of  $\nu$  and every sufficiently large  $n$ , the accelerated density  $g_{n,Q}$  is strictly decreasing on  $(0, 1)$  and has no spurious interior oscillation. Positivity is automatic, but monotonicity of a finite mixture is not; a proof may follow from total positivity of the uniform-box kernel.*

### 10.3 Broader extensions

**Other geometric bases.** For  $0 < q < 1$ , infinite products generated by widths  $(1 - q)q^j$  have the same exact head–tail scaling. The Bell–Bernoulli expansion and positive tail quadrature survive. The dyadic case  $q = 1/2$  is exceptional because subset sums form a uniform grid and the top derivative is exactly Thue–Morse. Determining the sharp  $C^r$  constants for general  $q$  is a natural next problem.

**Multivariate box splines.** Products of directional sinc factors invert to multivariate box splines. A geometric infinite product produces a compactly supported  $C^\infty$  atomic function, and finite prefixes again provide polynomial approximants. Positive quadrature of the tail scale may extend with radial or matrix-valued moments.

**Inverse Fabius propagation.** On compact subintervals where  $\text{up}$  is bounded away from zero, the exact error expansion can be transported through inverse-function calculus to obtain high-order approximations of the inverse Fabius function. Near the flat endpoint, the algebraic prefix expansion must be matched with the Lambert– $W$  asymptotic already developed in the repository.

**Spectral and sampling questions.** The zero multiplicities [equation \(4.2\)](#) suggest dyadic Strang–Fix-type conditions, multiresolution reconstruction formulas, and exact cancellation identities for sampled derivatives. The finite splines could serve as compactly supported preconditioners whose approximation order is enhanced by the positive tail rules.

## 11 Conclusion

The inverse Fourier transforms of finite dyadic sinc products are much more rigid than generic finite convolutions. Binary subset sums force a uniform knot grid; Thue–Morse signs govern the top derivative; and a precise derivative plateau survives every later convolution. Combined with the exact self-similar tail of the Rvachev distribution, this local geometry turns a Taylor bound into an equality and yields the sharp law

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The same mechanism converts one-signed quadrature errors into exact functional errors. The resulting Gauss–Radau–Lobatto hierarchy supplies positive piecewise-polynomial approximants of arbitrary order, with explicit constants and exact support whenever a right endpoint node is included. This closes the repository’s variance-matched  $16^{-n}$  frontier and opens a tractable new interface among dyadic harmonic analysis, Thue–Morse combinatorics, Bell–Bernoulli series, orthogonal polynomials, and certified numerical approximation.

## A Proof of the scale-derivative identity

For completeness, we prove [equation \(7.13\)](#). The case  $\ell = 0$  is the definition. Assume it holds at order  $\ell$ . Differentiating under the integral and using  $\partial_z h^{(2\ell)}(\sqrt{z}u) = u h^{(2\ell+1)}(\sqrt{z}u)/(2\sqrt{z})$  gives

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+2}\ell!\sqrt{z}} \int_{-1}^1 u(1-u^2)^{\ell} h^{(2\ell+1)}(\sqrt{z}u) \, du.$$

Since

$$\frac{d}{du}(1-u^2)^{\ell+1} = -2(\ell+1)u(1-u^2)^{\ell},$$

integration by parts has no boundary term and yields

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+3}(\ell+1)!} \int_{-1}^1 (1-u^2)^{\ell+1} h^{(2\ell+2)}(\sqrt{z}u) \, du.$$

This is the desired induction step. The formula extends continuously to  $z = 0$  by Taylor expansion.

## B Reproducibility manifest

The archive accompanying this report contains:

- `finite_sinc_products_report.tex` – the complete source;
- `finite_sinc_products_report.pdf` – the rendered report;
- `finite_sinc_experiments.py` – documented numerical and exact code;

- `finite_sinc_approximants.pdf` – generated approximation figure;
- `scaled_error_profiles.pdf` – generated error-profile figure;
- `convergence_comparison.pdf` – generated convergence figure;
- `sharp_error_verification.csv` – raw-prefix numerical table;
- `positive_acceleration_verification.csv` – accelerated numerical table;
- `exact_coefficients.txt` – exact series and Richardson coefficients;
- `exact_rational_samples.csv` – exact rational spline samples.

Run

```
python finite_sinc_experiments.py --output-dir . --fft-power 17
```

to regenerate every figure and table.

## References

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## Part IV

# Fourier Images of the Repeated-Integration Approximants

### Abstract

The repeated-integration approximants  $f_n$  to Rvachev's up-function differ from the limit, on the Fourier side, by an exact dilation of a single universal meromorphic transfer function  $A(z) = \text{sinc } z / \Phi(z) = \prod_{m \geq 1} (1 - z^2 / (\pi^2 m^2))^{1 - \nu_2(m)}$ :  $\widehat{f}_n = \Phi \cdot A(2^{-n}t)$ . The valuation-weighted divisor of  $A$  yields digit-sum zero counts, a Thue–Morse sign law, the exact Taylor radius  $4\pi$  with dominant-pole coefficient asymptotics and an arithmetic Darboux hierarchy, the complete finite/limit zero-multiplicity filtration, a sharp  $o(2^n)$  relative-convergence window, forward and inverse conditioning thresholds at  $4\pi 2^n$  and  $\pi 2^n$ , and the impossibility of any globally stable convolutional correction. All-orders expansions hold in every polynomially weighted Fourier  $L^p$  space and every Sobolev space, with explicit leading norm constants; the algebraic Fourier tails have exact mean-square constants and the sharp threshold  $f_n \in H^s$  iff  $s < n + \frac{1}{2}$ . Interpreting the construction as a tail closure and matching more moments of the true self-similar tail produces positive piecewise-polynomial approximants of arbitrary order — atomic, dyadic-atomic, and polynomial-density menus with exact constants — complementing the box-mixture hierarchy of Part III.

## 1 Scope, sources, and relation to the prefix theory

This part is the editorial consolidation of two ninth-wave reports written independently on the same day and on the same subject — the global Fourier geometry of the *repeated-integration* approximants to Rvachev's up-function: *Fourier Images of the Piecewise-Polynomial Approximants to Rvachev's Up-Function* (meromorphic tail transfer, weighted-norm asymptotics, moment-matched acceleration) and *Fourier Images of Recursive Piecewise-Polynomial Approximations to Rvachev's Up Function* (2-adic divisors, critical spectral layers, exact Fourier tails, accelerated closures). Their shared core — the transfer quotient, its cotangent product and divisor, the Taylor radius  $4\pi$  and dominant-pole asymptotics, the finite zero filtration, and the cumulant defects — is stated once below with the stronger of the two treatments; each report's unique layers (respectively: the weighted- $L^p$  master theorem with its norm constants, the arithmetic Darboux hierarchy, and the Gauss/Lobatto/polynomial closure menu; the valuation-weighted canonical product, the digital divisor laws and Thue–Morse sign, the sharp critical window, the two conditioning thresholds, the deconvolution obstruction, the exact mean-square tail constants with the sharp Sobolev threshold, and the dyadic-node closure menu) are kept in full. Every shared constant was verified identical between the sources before merging; the two independently derived zero-order laws, stated in different groupings, agree in all four valuation regimes.

Where Part III ([section 3](#)) studies the ordinary prefix densities  $p_n$  in physical space, the present part studies the twin family  $f_n$  (the prefix with its smallest box doubled) on the Fourier side. The two theories meet in the closure framework of [section 11](#), where the three families of accelerated

tail surrogates — the box mixtures of [section 7](#), and the atomic and polynomial closures below — are compared as quadratures of three different laws.

*Convention.* This part uses radian frequency,

$$\widehat{g}(t) = \int_{\mathbb{R}} g(x) e^{-itx} dx, \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases} \quad (1.1)$$

so that

$$\Phi(t) := \widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \text{sinc}\left(\frac{t}{2^k}\right), \quad \Phi(0) = 1, \quad (1.2)$$

with up the even  $C^\infty$  probability density on  $[-1, 1]$ . The frequency-normalized product used elsewhere in this volume is  $\Phi(2\pi\xi)$ ; in particular the zeros of (1.2) sit at  $2\pi m$ ,  $m \in \mathbb{Z} \setminus \{0\}$ , with order  $1 + \nu_2(m)$ , and  $\Phi(\pi) = 0.5537712758888100953442174163 \dots$  equals the frequency-normalized product at one half.

## 2 The repeated-integration family and its Fourier products

### 2.1 Recursion, box splines, and the head–tail model

With  $f_0 = \frac{1}{2}\mathbf{1}_{(-1,1)}$ , the recursive integration scheme

$$f_n(x) = 2 \int_{-1}^x (f_{n-1}(2t+1) - f_{n-1}(2t-1)) dt = \int_{2x-1}^{2x+1} f_{n-1}(t) dt \quad (2.1)$$

produces compactly supported piecewise-polynomial probability densities converging to up. Writing  $\beta_a = \frac{1}{2a}\mathbf{1}_{[-a,a]}$  for the centered uniform density and unrolling the recursion,

$$f_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \dots * \beta_{2^{-n}} * \beta_{2^{-n}} = p_n * \beta_h, \quad h := 2^{-n}, \quad (2.2)$$

where  $p_n = \beta_{2^{-1}} * \dots * \beta_{2^{-n}}$  is the ordinary prefix of [section 3](#): a univariate box spline of degree  $n$  supported on  $[-1, 1]$ , with the smallest box occurring twice. Probabilistically, if  $X_n$  has density  $f_n$ ,  $X$  has density up, and  $S_n = \sum_{k=1}^n 2^{-k} U_k$  with  $U_k$  independent uniform on  $[-1, 1]$ , then

$$X_n \stackrel{d}{=} S_n + hU, \quad X \stackrel{d}{=} S_n + hX', \quad X' \stackrel{d}{=} X, \quad (2.3)$$

with all variables independent: the finite construction replaces the true self-similar tail  $hX'$  by a single uniform variable  $hU$ . This closure point of view drives the entire part.

### 2.2 Finite products and the master factorization

With  $P_n(t) = \prod_{k=1}^n \text{sinc}(t/2^k) = \widehat{p}_n(t)$ ,

$$F_n(t) := \widehat{f}_n(t) = P_n(t) \text{sinc}(ht), \quad \Phi(t) = P_n(t) \Phi(ht). \quad (2.4)$$

**Definition 2.1** (Tail-transfer quotient). The *tail-transfer function* (or correction quotient) is the even meromorphic function

$$A(z) = \frac{\text{sinc } z}{\Phi(z)}, \quad (2.5)$$

interpreted at common zeros by meromorphic continuation.

**Theorem 2.2** (Master factorization). For every  $n \geq 1$ ,

$$F_n(t) = \Phi(t) A(2^{-n}t), \quad (2.6)$$

as an identity of entire functions after the implicit cancellations, and in the cancellation-free form  $F_n(t) - \Phi(t) = P_n(t)(\text{sinc}(ht) - \Phi(ht))$ .

*Proof.* Divide the two identities in (2.4). □

All stage dependence is thus an exact dilation of one universal meromorphic profile: the finite-stage problem separates into the rapidly decaying global carrier  $\Phi(t)$  and the universal microscopic transfer  $A(2^{-n}t)$ . The identity is stronger than an asymptotic statement — in the renormalized coordinate  $z = 2^{-n}t$  the profile  $\mathcal{R}_n(z) = F_n(2^n z)/\Phi(2^n z) = A(z)$  is independent of  $n$ , so the apparent convergence of the finite products is entirely the critical window moving to infinity, a boundary layer with an exact rather than asymptotic inner solution.

### 3 Exact forms and the canonical valuation product

#### 3.1 Functional equations and renormalization products

The scaling identity  $\Phi(2z) = \operatorname{sinc} z \Phi(z)$  gives three equivalent descriptions of the transfer quotient and a dyadic functional equation:

$$A(z) = \frac{\Phi(2z)}{\Phi(z)^2} = \frac{\cos(z/2)}{\Phi(z/2)}, \quad (3.1)$$

$$\frac{A(z)}{A(z/2)} = \frac{\operatorname{sinc} z}{\operatorname{sinc}^2(z/2)} = \frac{z}{2} \cot \frac{z}{2}, \quad (3.2)$$

the second form of (3.1) by  $\operatorname{sinc} z = \operatorname{sinc}(z/2) \cos(z/2)$  and one cancellation. Iterating (3.2) from  $A(0) = 1$  yields the cotangent renormalization product

$$A(z) = \prod_{j=1}^{\infty} \left( \frac{z}{2^j} \cot \frac{z}{2^j} \right), \quad (3.3)$$

normally convergent on compact pole-free sets since  $u \cot u = 1 - u^2/3 + \mathcal{O}(u^4)$ ; collecting equal cosine scales in  $\operatorname{sinc} z = \prod_{j \geq 1} \cos(z/2^j)$  gives the companion form  $A(z) = \cos(z/2) \prod_{j \geq 3} \cos(z/2^j)^{-(j-2)}$ , which makes the dyadic pole hierarchy visually transparent.

#### 3.2 The valuation-weighted canonical product

The arithmetic structure is clearest in a canonical product obtained by regrouping Euler's  $\operatorname{sinc} z = \prod_{r \geq 1} (1 - z^2/(\pi^2 r^2))$  across the dyadic scales: a fixed factor  $1 - z^2/(\pi^2 m^2)$  occurs in  $\operatorname{sinc}(z/2^k)$  once for every  $k \geq 1$  with  $2^k \mid m$ , i.e.  $\nu_2(m)$  times in  $\Phi$ .

**Theorem 3.1** (Canonical products). *As an entire function,*

$$\Phi(z) = \prod_{m=1}^{\infty} \left( 1 - \frac{z^2}{\pi^2 m^2} \right)^{\nu_2(m)}, \quad (3.4)$$

*and as a meromorphic function,*

$$A(z) = \prod_{m=1}^{\infty} \left( 1 - \frac{z^2}{\pi^2 m^2} \right)^{1-\nu_2(m)}, \quad (3.5)$$

*both locally uniformly in their natural domains (legitimate because  $\sum_m \nu_2(m) m^{-2} = \sum_{j \geq 1} \sum_{2^j \mid m} m^{-2} < \infty$ ).*

### 3.3 The complete divisor and local principal parts

**Theorem 3.2** (Divisor, removable values, and leading jets). *For every nonzero integer  $m$ ,*

$$\text{ord}_{z=\pi m} A = 1 - \nu_2(|m|). \quad (3.6)$$

*Explicitly:  $A$  has a simple zero at every odd multiple of  $\pi$ ; the points  $2\pi q$  with  $q$  odd are removable with the nonzero values*

$$A(2\pi q) = -\frac{1}{\Phi(\pi q)}; \quad (3.7)$$

*and at  $z_0 = \pi 2^r q > 0$  with  $q$  odd and  $r \geq 2$  there is a pole of order  $r - 1$  with leading jet*

$$A(z) = -\frac{z_0^{r-1}}{\Phi(\pi q)} (z - z_0)^{1-r} (1 + \mathcal{O}(z - z_0)). \quad (3.8)$$

*Equivalently, parametrizing the poles as  $z = 4\pi q'$  with  $q' \in \mathbb{Z} \setminus \{0\}$ , the pole order is  $1 + \nu_2(q')$ . In particular the nearest singularities are the simple poles at  $\pm 4\pi$ , with*

$$\text{Res}_{z=4\pi} A = -\frac{4\pi}{\Phi(\pi)} = -22.6923481977102997779\dots, \quad \text{Res}_{z=-4\pi} A = +\frac{4\pi}{\Phi(\pi)}. \quad (3.9)$$

*Proof.* The order formula is immediate from (3.5). For the local jets use  $A(z) = \cos(z/2)/\Phi(z/2)$  and put  $w = z/2, w_0 = z_0/2$ . Iterating the head splitting  $r-1$  times,  $\Phi(w) = (\prod_{k=1}^{r-1} \text{sinc}(w/2^k)) \Phi(w/2^{r-1})$ ; at  $w = w_0$  every displayed factor vanishes linearly, all the associated integer multiples are even except the last, so the product of their sign slopes is  $-1$  and  $\Phi(w) = -((w - w_0)/w_0)^{r-1} \Phi(\pi q)(1 + \mathcal{O}(w - w_0))$ . For  $r \geq 2$ ,  $\cos(z_0/2) = 1$ , giving (3.8); for  $r = 1$ ,  $\cos(\pi q) = -1$  gives (3.7). The residue at  $4\pi$  is the case  $r = 2, q = 1$  of (3.8).  $\square$

The leading-jet formula (3.8) evaluates, at every pole, the top Laurent coefficient in finite terms of  $\Phi$  at odd multiples of  $\pi$  – the first layer of the closed-jet program posed in section 12.

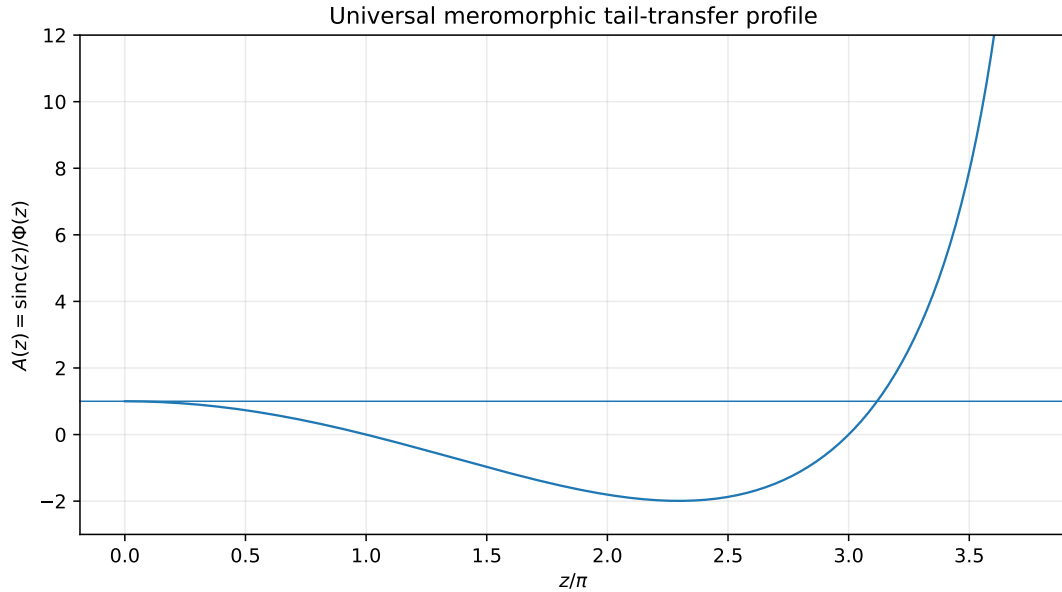


Figure 6: The universal transfer function on the first positive meromorphic cell: simple zeros at  $\pi$  and  $3\pi$ , a removable point at  $2\pi$  with value  $-1/\Phi(\pi)$ , and the first pole at  $4\pi$ .

## 4 Digital divisor laws and the Thue–Morse sign

The signed divisor sequence  $1 - \nu_2(m)$  has three exact arithmetic transforms, and its cumulative parity is a digit sum.

**Theorem 4.1** (Digital divisor laws). *Let  $s_2(M)$  be the binary digit sum. Then*

$$\sum_{m=1}^M \text{ord}_{\pi m} A = \sum_{m=1}^M (1 - \nu_2(m)) = M - \nu_2(M!) = s_2(M), \quad (4.1)$$

*the total pole multiplicity on  $\{\pi, \dots, M\pi\}$  is  $\lceil M/2 \rceil - s_2(M)$ , and the divisor has the Lambert and Dirichlet series*

$$\sum_{m \geq 1} (1 - \nu_2(m)) q^m = \frac{q}{1-q} - \sum_{j \geq 1} \frac{q^{2^j}}{1-q^{2^j}} \quad (|q| < 1), \quad \sum_{m \geq 1} \frac{1 - \nu_2(m)}{m^s} = \zeta(s) \frac{2^s - 2}{2^s - 1} \quad (\Re s > 1). \quad (4.2)$$

*Proof.* Legendre's formula  $\nu_2(M!) = M - s_2(M)$  gives (4.1); the pole count is  $\sum_{m \leq M} \max\{\nu_2(m) - 1, 0\} = \sum_{j \geq 2} \lfloor M/2^j \rfloor = \nu_2(M!) - \lfloor M/2 \rfloor$ . Since  $\nu_2(m) = \sum_{j \geq 1} \mathbf{1}_{2^j | m}$ , summation in  $m$  gives both series, using  $\sum_m \nu_2(m) m^{-s} = \zeta(s)/(2^s - 1)$ .  $\square$

**Corollary 4.2** (Thue–Morse sign law). *For every integer  $M \geq 0$  and every real  $x \in (\pi M, \pi(M+1))$ ,*

$$\text{sgn } A(x) = (-1)^{s_2(M)}. \quad (4.3)$$

*Proof.*  $A > 0$  near 0; crossing  $\pi m$  flips the sign according to the parity of the signed order  $1 - \nu_2(m)$  (poles flip parity exactly as zeros do), and the cumulative parity through  $M$  is  $s_2(M)$  by (4.1).  $\square$

The Thue–Morse sequence thus appears twice in the theory of the same splines, by different mechanisms: as subset-sum parity in the spatial truncated-power coefficients (equation (3.2) for the prefix family), and as a cumulative valuation divisor in the signs of the universal Fourier correction.

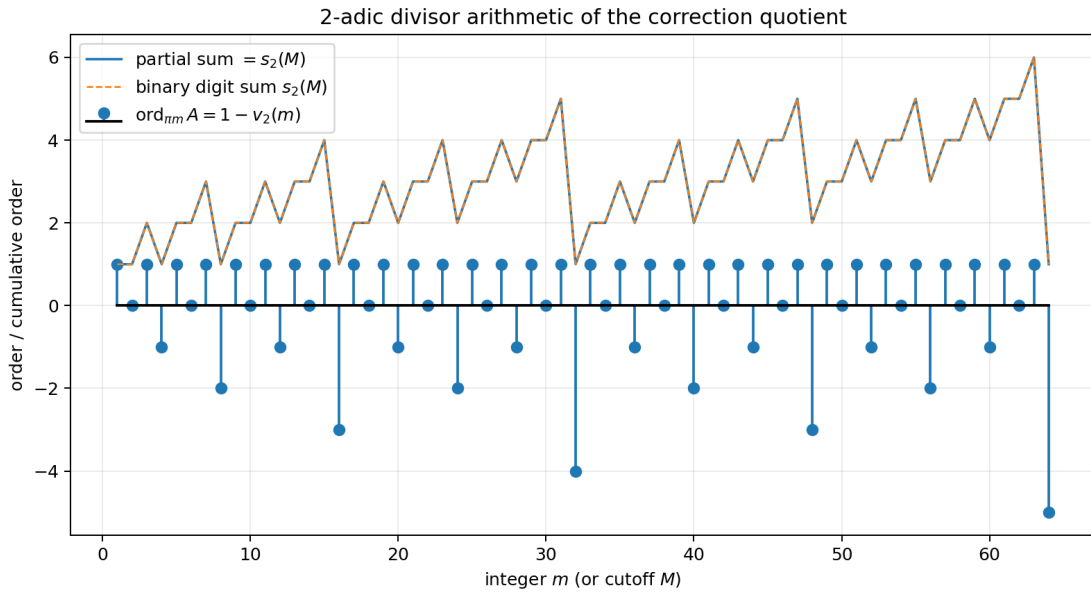


Figure 7: The signed divisor order  $1 - \nu_2(m)$  and its cumulative sum, which coincides exactly with the binary digit sum  $s_2(M)$ .

## 5 Taylor coefficients: Bernoulli series, radius, and dominant poles

### 5.1 The local logarithm and its arithmetic explanation

For  $|z| < \pi$ , the zero-free branch of  $\log A$  at the origin is

$$\log A(z) = \sum_{r=1}^{\infty} \ell_r z^{2r}, \quad \ell_r = -\frac{2^{2r-1}|B_{2r}|}{r(2r)!} \frac{4^r - 2}{4^r - 1}, \quad (5.1)$$

the local series already recorded with the repeated-integration corpus. The canonical product explains its shape arithmetically: logarithmic expansion of (3.5) gives

$$\log A(z) = -\sum_{r=1}^{\infty} \frac{z^{2r}}{r\pi^{2r}} \sum_{m \geq 1} \frac{1 - \nu_2(m)}{m^{2r}} = -\sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} \frac{2^{2r} - 2}{2^{2r} - 1} z^{2r}, \quad (5.2)$$

by the divisor Dirichlet series (4.2); Euler's evaluation of  $\zeta(2r)$  recovers (5.1). The coefficient factor  $(4^r - 2)/(4^r - 1)$  is thus the divisor zeta quotient  $\zeta(2r)^{-1} \sum_m (1 - \nu_2(m))m^{-2r}$ , and simultaneously the exact  $2r$ th cumulant defect ratio of section 9.

Writing  $A(z) = \sum_{r \geq 0} \rho_r z^{2r}$ , differentiation of  $A = \exp \sum \ell_r w^r$  in  $w = z^2$  gives the exact rational recurrence  $r\rho_r = \sum_{k=1}^r k\ell_k \rho_{r-k}$ , whence

$$A(z) = 1 - \frac{z^2}{9} + \frac{2z^4}{2025} + \frac{z^6}{2679075} + \frac{58z^8}{10247461875} + \frac{12622z^{10}}{345944065438125} + \frac{449289436z^{12}}{1933714893977851359375} + \cdots. \quad (5.3)$$

### 5.2 Radius and dominant-pole asymptotics

**Theorem 5.1** (Taylor radius and dominant-pole law). *The Maclaurin series of  $A$  has radius of convergence exactly  $4\pi$ : the zeros at  $\pm\pi, \pm3\pi$  do not obstruct analyticity, the points  $\pm2\pi$  are removable, and the first genuine singularities are the simple poles at  $\pm4\pi$ . Moreover there is a function  $H$  holomorphic in  $|z| < 8\pi$  with*

$$A(z) = \frac{2/\Phi(\pi)}{1 - z^2/(4\pi)^2} + H(z), \quad (5.4)$$

so for every fixed  $\rho < 8\pi$ ,

$$\rho_r = \frac{2}{\Phi(\pi)(4\pi)^{2r}} + \mathcal{O}_\rho(\rho^{-2r}), \quad \lim_{r \rightarrow \infty} \rho_r (4\pi)^{2r} = \frac{2}{\Phi(\pi)} = 3.6115993860280564486832331013 \dots, \quad (5.5)$$

and  $\rho_{r+1}/\rho_r \rightarrow (16\pi^2)^{-1}$ . In particular  $\rho_r > 0$  for all sufficiently large  $r$ .

*Proof.* The paired principal parts at  $\pm4\pi$  determined by (3.9) sum to the rational model in (5.4); subtraction removes both first poles, the next poles are the double poles at  $\pm8\pi$ , and Cauchy's estimate on  $|z| = \rho < 8\pi$  bounds the remainder coefficients.  $\square$

Exponentiating the convergent logarithmic series is justified only for  $|z| < \pi$ , but the resulting formal recurrence computes the Taylor series of the meromorphic continuation, convergent throughout  $|z| < 4\pi$  — the ordinary series reaches four times farther than the logarithm.

The dominant-pole approximation is accurate remarkably early: exact rational computation gives

| $r$                  | 3       | 5       | 10       | 20           | 30                 | 50                           |
|----------------------|---------|---------|----------|--------------|--------------------|------------------------------|
| $\rho_r (4\pi)^{2r}$ | 0.40698 | 3.58279 | 3.611538 | 3.6115993859 | 3.6115993860280563 | 3.61159938602805644868323101 |

with  $\rho_r > 0$  verified for  $2 \leq r \leq 50$ .

**Proposition 5.2** (Arithmetic Darboux hierarchy). *For  $q \geq 1$  put  $R_q = 4\pi q$  and  $d_q = 1 + \nu_2(q)$ , and write the principal part of  $A$  at  $R_q$  as  $\sum_{\ell=1}^{d_q} c_{q,\ell}(z - R_q)^{-\ell}$ . Then for each  $Q \geq 1$  and  $R_Q < \rho < R_{Q+1}$ ,*

$$\rho_r = \sum_{q=1}^Q \sum_{\ell=1}^{d_q} 2(-1)^\ell c_{q,\ell} R_q^{-2r-\ell} \binom{2r+\ell-1}{\ell-1} + \mathcal{O}_{Q,\rho}(\rho^{-2r}) : \quad (5.6)$$

*each positive integer  $q$  contributes the exponential scale  $(4\pi q)^{-2r}$  times a polynomial in  $r$  of degree  $\nu_2(q)$ , so the 2-adic hierarchy of pole orders becomes an arithmetic hierarchy of coefficient corrections*

$$\rho_r \sim \frac{2}{\Phi(\pi)(4\pi)^{2r}} + \frac{\alpha_1 r + \alpha_0}{(8\pi)^{2r}} + \frac{\beta_0}{(12\pi)^{2r}} + \frac{\gamma_2 r^2 + \gamma_1 r + \gamma_0}{(16\pi)^{2r}} + \dots, \quad (5.7)$$

*with the top constant of each block determined by the leading jet (3.8).*

*Proof.* Evenness pairs the principal part at  $-R_q$  with signs  $(-1)^\ell$ ; the coefficient of  $z^{2r}$  in a paired principal part is  $2(-1)^\ell c_{q,\ell} R_q^{-2r-\ell} \binom{2r+\ell-1}{\ell-1}$ . Subtract all paired principal parts with  $q \leq Q$  and apply Cauchy's estimate in  $|z| < R_{Q+1}$ .  $\square$

**Conjecture 5.3** (Global coefficient positivity). *Apart from  $\rho_1 = -1/9$ , every Taylor coefficient of  $A$  is positive.*

The dominant-pole law proves this for all large  $r$ , and exact arithmetic verifies it through  $r = 50$ , but all nonconstant coefficients of  $\log A$  are negative, so exponentiation produces alternating combinatorial sums and the finite gap is genuinely open. Candidate routes: an explicit positive representation of  $A(z) - 1 + z^2/9$  in  $z^2$ ; a majorization of the negative logarithmic cumulants after exponentiation; a pole-subtracted Mittag-Leffler expansion with positivity-preserving remainder; or a sign-regular interpretation of the tail comparison.

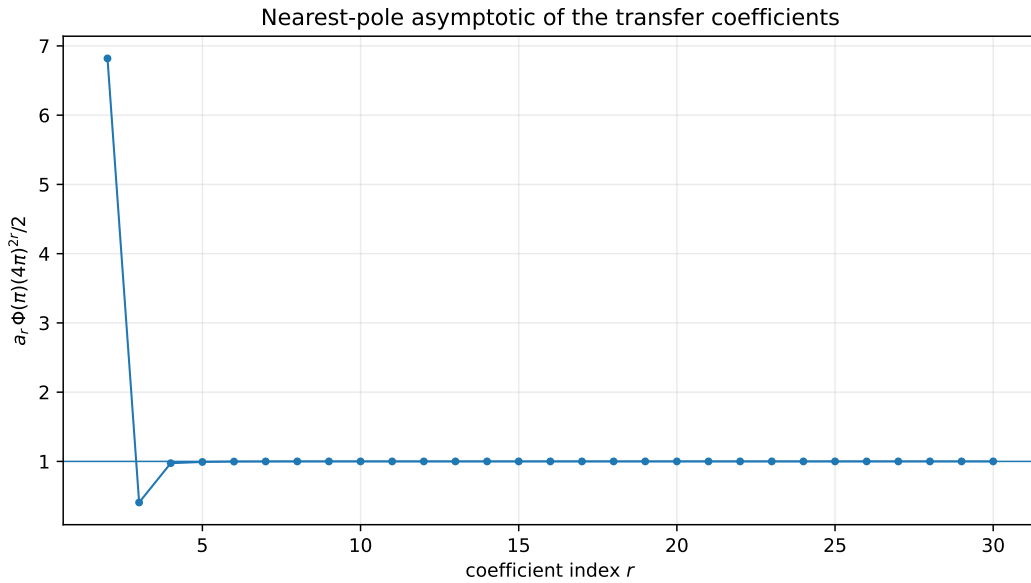


Figure 8: Normalized exact rational coefficients  $\rho_r \Phi(\pi)(4\pi)^{2r}/2$  against the nearest-pole prediction 1; the asymptotic is already highly accurate from  $r = 4$  onward. The nearly linear semilogarithmic error profile is consistent with the double poles at  $\pm 8\pi$ , whose contribution carries a linear polynomial factor in  $r$  per equation (5.7).



## 6 The finite zero filtration and its divisor realization

Both  $F_n$  and  $\Phi$  vanish exactly on  $2\pi\mathbb{Z} \setminus \{0\}$ , but with different multiplicities — a distinction invisible in magnitude plots and precisely responsible for the zeros and poles of the quotient.

**Theorem 6.1** (Exact finite multiplicities). *Let  $m \in \mathbb{Z} \setminus \{0\}$  and  $v = \nu_2(|m|)$ . Then  $\text{ord}_{2\pi m} \Phi = v + 1$ , while*

$$\text{ord}_{2\pi m} F_n = \min(n, v+1) + \mathbf{1}_{v \geq n-1} = \min(v+1, n-1) + 2 \cdot \mathbf{1}_{v \geq n-1} = \begin{cases} v+1, & v \leq n-2, \\ n+1, & v \geq n-1. \end{cases} \quad (6.1)$$

(The two groupings, from the factorizations  $F_n = P_n \cdot \text{sinc}(t/2^n)$  and  $F_n = (\prod_{k < n} \text{sinc}(t/2^k)) \text{sinc}^2(t/2^n)$  respectively, were derived independently in the two sources and agree in all regimes.) Consequently the defect  $\Delta_n(m) = \text{ord}_{2\pi m} F_n - \text{ord}_{2\pi m} \Phi$  is

$$\Delta_n(m) = \begin{cases} 0, & v \leq n-2 \quad (\text{exact jet resolution}), \\ 1, & v = n-1 \quad (\text{over-resolved by one order}), \\ 0, & v = n \quad (\text{exact again}), \\ n-v, & v \geq n+1 \quad (\text{unresolved high dyadic jet}). \end{cases} \quad (6.2)$$

*Proof.*  $\text{sinc}(t/2^k)$  vanishes at  $t = 2\pi m$  iff  $2^{k-1} \mid m$ , i.e.  $k \leq v+1$ ; among  $k = 1, \dots, n$  this contributes  $\min(n, v+1)$ , and the duplicated smallest box one more exactly when  $v \geq n-1$ .  $\square$

**Corollary 6.2** (Divisor-level realization). *For every nonzero integer  $M$ ,  $\text{ord}_{t=2^n\pi M} (F_n/\Phi) = 1 - \nu_2(|M|)$ : at the critical scaling  $t = 2^n z$  the three exceptional layers of (6.2) are exactly the divisor of  $A$  — the over-resolved layer  $v = n-1$  is the simple zero of  $A$  at odd multiples of  $\pi$ , the re-agreement layer  $v = n$  is the removable point at  $2\pi \cdot \text{odd}$ , and each unresolved layer  $v \geq n+1$  is the pole of order  $v-n$ .*

## 7 The critical spectral boundary layer

### 7.1 The sharp relative-convergence window

Absolute convergence  $F_n \rightarrow \Phi$  is strong on the real axis because both sides decay; relative convergence has an exact scale.

**Theorem 7.1** (Sharp window criterion). *Let  $T_n > 0$ . With the quotient understood meromorphically,*

$$\sup_{|t| \leq T_n} \left| \frac{F_n(t)}{\Phi(t)} - 1 \right| \rightarrow 0 \quad \Longleftrightarrow \quad \frac{T_n}{2^n} \rightarrow 0, \quad (7.1)$$

and in the convergent regime

$$\sup_{|t| \leq T_n} \left| \frac{F_n(t)}{\Phi(t)} - 1 \right| = \frac{T_n^2}{9 \cdot 4^n} (1 + \mathcal{O}((T_n/2^n)^2)). \quad (7.2)$$

For fixed  $0 < c < 4\pi$  the critical supremum  $\sup_{|t| \leq c2^n} |F_n/\Phi - 1| = \max_{|z| \leq c} |A(z) - 1|$  is independent of  $n$  and nonzero; more generally, truncating the Taylor series of  $A$  after  $J-1$  terms leaves a relative error  $\mathcal{O}(4^{-J\epsilon n})$  on the subcritical band  $|t| \leq 2^{(1-\epsilon)n}$ .

*Proof.* By theorem 2.2,  $F_n/\Phi = A(t/2^n)$ . If  $T_n/2^n \rightarrow 0$ , the local expansion  $A(z) = 1 - z^2/9 + \mathcal{O}(z^4)$  gives both convergence and (7.2). Otherwise a subsequence has  $T_n/2^n \geq \epsilon > 0$ , and since  $A$  is nonconstant analytic,  $\max_{|z| \leq \epsilon} |A(z) - 1| > 0$  (infinite if the window reaches a pole). The critical identity is a change of variables.  $\square$

Hence  $2^n$  is not merely a convenient scale but the exact boundary between uniform relative convergence and a nontrivial limiting profile.



## 7.2 Two conditioning thresholds and the deconvolution obstruction

Bandlimited to  $|t| \leq c 2^n$ , the exact relation between the finite and limiting transforms is the Fourier multiplier identity  $\Pi f_n = A(2^{-n}D) \Pi$  up with symbol  $A(2^{-n}t)$ . Forward application is governed by the first pole, inversion by the first zero.

**Theorem 7.2** (Forward and inverse thresholds). *Let  $M_+(c) = \sup_{|z| \leq c} |A(z)|$  and  $M_-(c) = \sup_{|z| \leq c} |A(z)|^{-1}$ . Then  $M_+(c) < \infty$  iff  $c < 4\pi$ , and  $M_-(c) < \infty$  iff  $c < \pi$ , with the blowup laws*

$$M_+(c) \sim \frac{4\pi}{\Phi(\pi)(4\pi - c)} \quad (c \uparrow 4\pi), \quad M_-(c) \sim \frac{\pi \Phi(\pi)}{\pi - c} \quad (c \uparrow \pi), \quad (7.3)$$

the second from the local factorization  $A(z) = \frac{\pi - z}{\pi \Phi(\pi)}(1 + \mathcal{O}(\pi - z))$  near the first zero (using  $\Phi(\pi/2) = \frac{\pi}{2}\Phi(\pi)$ ).

**Theorem 7.3** (No global stable convolutional correction). *There is no compactly supported distribution, finite measure, or  $L^1$  kernel  $K_n$  with  $f_n = \text{up} * K_n$ , and none  $L_n$  with  $\text{up} = f_n * L_n$ .*

*Proof.* The first identity forces  $\widehat{K_n}(t) = A(2^{-n}t)$  everywhere by meromorphic continuation; the transform of a compactly supported distribution is entire and that of a finite measure is bounded on  $\mathbb{R}$ , while  $A(2^{-n}t)$  has a real pole at  $4\pi 2^n$ . The reverse identity requires the symbol  $1/A(2^{-n}t)$ , with a real pole already at  $\pi 2^n$ .  $\square$

The obstruction is stronger than ill-conditioning: the global translation-invariant correction kernel does not exist in any of the usual stable classes. Away from the divisor one still has the exact shell transfer

$$\log |F_n(2^n z)| - \log |\Phi(2^n z)| = \log |A(z)|, \quad (7.4)$$

so every asymptotic decay law for the limiting product on the sequence  $t = 2^n z$  transfers to the finite products with an exact  $O(1)$  profile correction — a direct bridge from the Fourier-decay corpus (which studies the envelope and transfer-operator behaviour of  $\Phi$ ) to the finite piecewise-polynomial approximants, to be supplemented at divisor points by the multiplicity filtration (6.1).

## 8 All-orders norm asymptotics

### 8.1 The weighted- $L^p$ master theorem

The local Taylor series of  $A$  promotes to a global norm theorem in every polynomially weighted Fourier space, because the head product supplies arbitrarily many powers of algebraic decay: for every fixed  $N$  and  $n \geq N$ ,  $|P_n(t)| \leq \prod_{k=1}^N \min(1, 2^k/|t|) \leq C_N(1 + |t|)^{-N}$ . For  $s \geq 0$ ,  $1 \leq p \leq \infty$ , write  $\|g\|_{L_s^p} = \|(1 + |t|)^s g\|_{L^p(\mathbb{R})}$ .

**Theorem 8.1** (Weighted Fourier expansion). *Let  $J \geq 1$ ,  $s \geq 0$ ,  $1 \leq p \leq \infty$ . As  $n \rightarrow \infty$ ,*

$$4^{Jn} \left[ F_n(t) - \Phi(t) \sum_{j=0}^{J-1} \rho_j 4^{-jn} t^{2j} \right] \longrightarrow \rho_J t^{2J} \Phi(t) \quad \text{in } L_s^p(\mathbb{R}), \quad (8.1)$$

and hence  $\left\| F_n - \Phi \sum_{j < J} \rho_j 4^{-jn} t^{2j} \right\|_{L_s^p} = |\rho_J| 4^{-Jn} \|t^{2J} \Phi\|_{L_s^p} + o(4^{-Jn})$ .

*Proof.* Let  $D_J(z) = \text{sinc } z - \Phi(z) \sum_{j < J} \rho_j z^{2j}$ , an entire function with  $D_J(z) = \rho_J z^{2J} + \mathcal{O}(z^{2J+2})$ , and let  $H_J(z) = D_J(z)/z^{2J}$  (value  $\rho_J$  at 0), continuous and bounded on  $\mathbb{R}$  since  $|D_J(z)| \leq C_J |z|^{2J}$  for  $|z| \geq 1$ . The two identities (2.4) give exactly

$$4^{Jn} \left[ F_n(t) - \Phi(t) \sum_{j < J} \rho_j 4^{-jn} t^{2j} \right] = t^{2J} P_n(t) H_J(2^{-n}t).$$

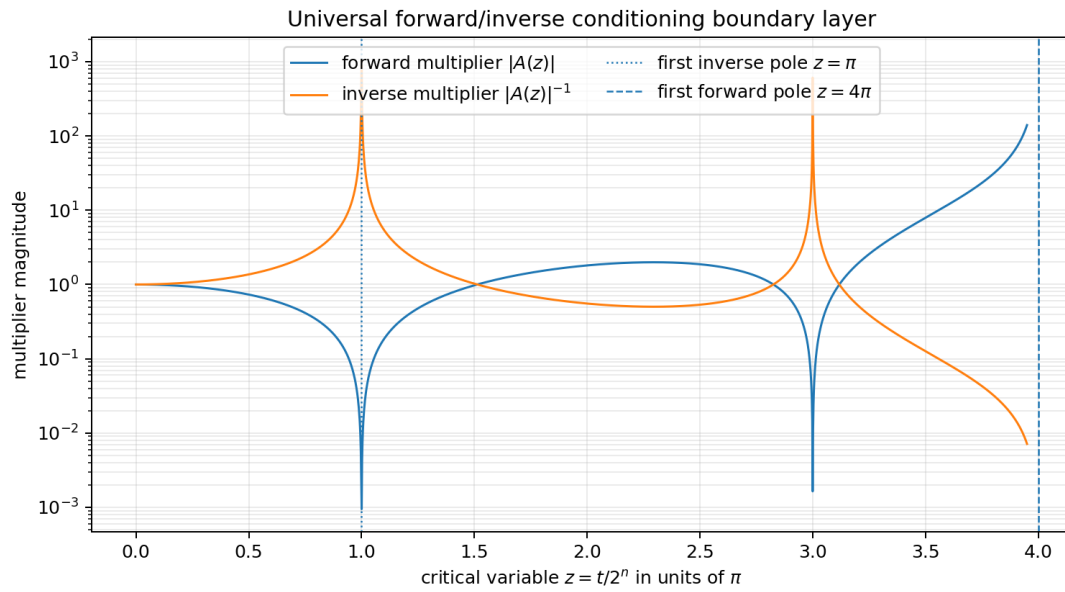


Figure 9: Forward and inverse multiplier magnitudes on the critical scale  $z = t/2^n$ : inverse instability begins at the first zero  $z = \pi$ , forward instability at the first pole  $z = 4\pi$ , with further inverse peaks at the odd multiples of  $\pi$ .

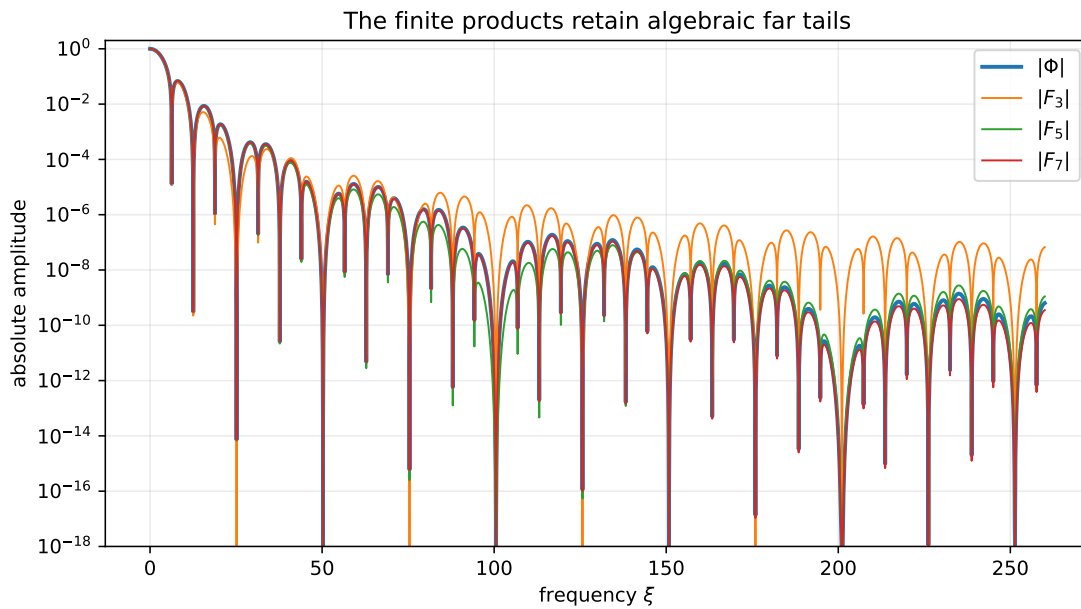


Figure 10: Absolute finite and limiting spectra on a logarithmic scale: each fixed finite product has only algebraic far-field decay (section 10), while  $\Phi$  is rapidly decreasing; increasing  $n$  pushes the discrepancy outward to the boundary layer  $|t| \asymp 2^n$ .

Pointwise,  $P_n \rightarrow \Phi$  and  $H_J(2^{-n}t) \rightarrow \rho_J$ ; choosing  $N > s + 2J + 1/p$  in the head-decay bound gives an  $n$ -independent  $L^p$  majorant after multiplication by  $(1 + |t|)^s$ , and dominated convergence concludes.  $\square$

**Corollary 8.2** (Universal leading profile and norm constants).  $4^n(F_n - \Phi) \rightarrow -\frac{t^2}{9}\Phi$  in every  $L_s^p$ , and  $4^n \|F_n - \Phi\|_{L^p} \rightarrow \frac{1}{9} \|t^2\Phi\|_{L^p}$ . The unweighted constants, by lobe-wise quadrature, are

$$C_\infty = \frac{1}{9} \sup_t t^2 |\Phi(t)| = 0.650851567393992 \dots \quad (\text{maximizer } t_* = 3.750573332377785 \dots), \quad (8.2)$$

$C_1 = \frac{1}{9} \int t^2 |\Phi| dt = 11.0486184433431 \dots$  and  $C_2 = \frac{1}{9} (\int t^4 |\Phi|^2 dt)^{1/2} = 2.00390775107366 \dots$  (reproducible numerical values, not certified enclosures); the measured scaled norms at  $n = 8$  agree with all three limits to five significant digits.

## 8.2 Sobolev expansions, energies, and monotonicity

Since  $\widehat{\text{up}^{(2j)}} = (-1)^j t^{2j} \Phi$ , the  $p = 2$  case transfers to physical space: for every fixed real  $s \geq 0$  and  $J \geq 1$ , with  $a_j = (-1)^j \rho_j$ ,

$$4^{Jn} \left[ f_n - \sum_{j=0}^{J-1} a_j 4^{-jn} \text{up}^{(2j)} \right] \longrightarrow a_J \text{up}^{(2J)} \quad \text{in } H^s(\mathbb{R}) \quad (8.3)$$

(also in every  $C^m$  via  $p = \infty$ ) – the known physical-space expansion  $f_n = \text{up} + \frac{4^{-n}}{9} \text{up}'' + \frac{2 \cdot 16^{-n}}{2025} \text{up}^{(4)} - \frac{64^{-n}}{2679075} \text{up}^{(6)} + \dots$  is simultaneously an asymptotic expansion in every fixed Sobolev topology. On the energy level, with the convolution coefficients  $d_q = \sum_{j=0}^q \rho_j \rho_{q-j}$  and  $c_q = \sum_{j=1}^{q-1} \rho_j \rho_{q-j}$ , expanding  $|F_n|^2 = |\Phi|^2 A(ht)^2$  gives, for every fixed  $s$  and  $M$ ,

$$\|f_n\|_{\dot{H}^s}^2 = \sum_{q=0}^{M-1} d_q 4^{-qn} \|\text{up}\|_{\dot{H}^{s+q}}^2 + \mathcal{O}_{s,M}(4^{-Mn}) = \|\text{up}\|_{\dot{H}^s}^2 - \frac{2 \cdot 4^{-n}}{9} \|\text{up}\|_{\dot{H}^{s+1}}^2 + \frac{29 \cdot 16^{-n}}{2025} \|\text{up}\|_{\dot{H}^{s+2}}^2 + \dots \quad (8.4)$$

$$\|f_n - \text{up}\|_{\dot{H}^s}^2 = \sum_{q=2}^{M-1} c_q 4^{-qn} \|\text{up}\|_{\dot{H}^{s+q}}^2 + \mathcal{O}_{s,M}(4^{-Mn}) = \frac{16^{-n}}{81} \|\text{up}\|_{\dot{H}^{s+2}}^2 - \frac{4 \cdot 64^{-n}}{18225} \|\text{up}\|_{\dot{H}^{s+3}}^2 + \dots \quad (8.5)$$

Two consequences: the error-norm ratio  $\|f_{n+1} - \text{up}\|_{\dot{H}^s} / \|f_n - \text{up}\|_{\dot{H}^s} \rightarrow \frac{1}{4}$ , and  $\|f_n\|_{\dot{H}^s}$  is eventually strictly increasing to  $\|\text{up}\|_{\dot{H}^s}$  (from the negative  $4^{-n}$  term in (8.4)).

## 9 Cumulant defects and finite moment polynomials

The transfer function is the quotient of the characteristic function of a uniform tail by that of the true self-similar tail, so cumulants diagonalize the comparison: comparing  $\log A(ht)$  with the cumulant series of (2.3) gives, for every  $r \geq 1$ ,

$$\kappa_{2r}(X) = \frac{2^{2r} B_{2r}}{2r(4^r - 1)}, \quad \kappa_{2r}(X_n) - \kappa_{2r}(X) = \frac{2^{2r} B_{2r}}{2r} \cdot \frac{4^r - 2}{4^r - 1} \cdot 4^{-rn}, \quad (9.1)$$

with all odd cumulants zero: the  $r$ th term of the Bernoulli series (5.1) is the defect of the  $2r$ th cumulant, and the leading  $4^{-n}$  rate is exactly the variance mismatch  $\text{Var}(X_n) - \text{Var}(X) = \frac{2}{9} 4^{-n}$  between the uniform tail (variance  $\frac{1}{3}$ ) and the true tail (variance  $\frac{1}{9}$ ). Bell-polynomial conversion gives finite-stage moment polynomials in  $q = 4^{-n}$  of degree at most the half-order,

$$\mathbb{E}X_n^2 = \frac{1}{9} + \frac{2}{9}q, \quad \mathbb{E}X_n^4 = \frac{19}{675} + \frac{4}{27}q + \frac{16}{675}q^2, \quad \mathbb{E}X_n^6 = \frac{583}{59535} + \frac{38}{405}q + \frac{16}{405}q^2 - \frac{16}{59535}q^3, \quad (9.2)$$

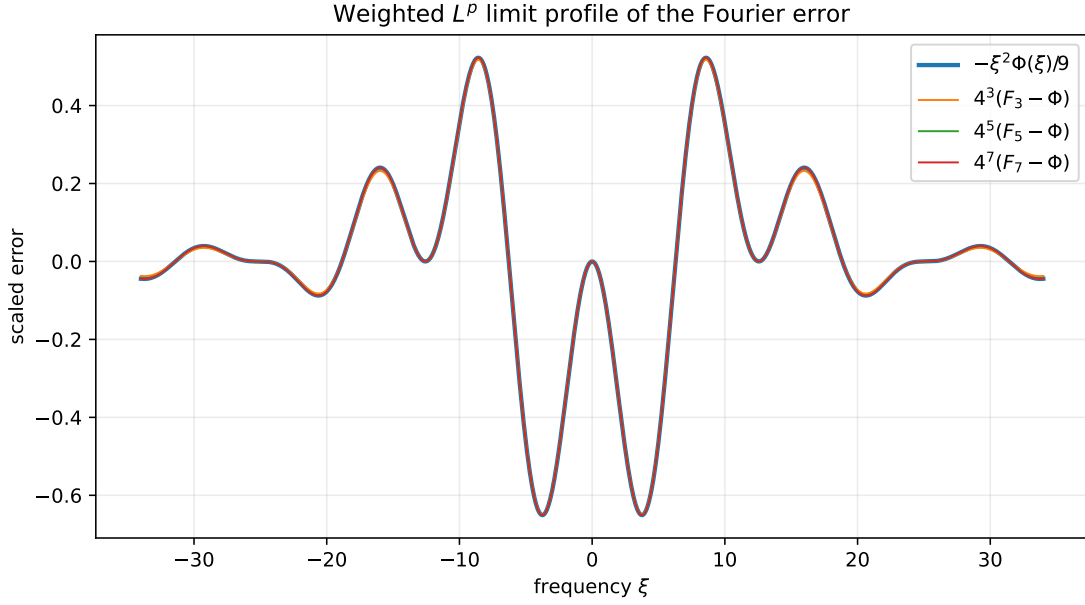


Figure 11: The scaled errors  $4^n(F_n - \Phi)$  converge globally to  $-t^2\Phi/9$ , not only near the origin; [theorem 8.1](#) makes the convergence valid in every polynomially weighted  $L^p$ .

against the limiting moments  $\mathbb{E}X^2 = \frac{1}{9}$ ,  $\mathbb{E}X^4 = \frac{19}{675}$ ,  $\mathbb{E}X^6 = \frac{583}{59535}$ ,  $\mathbb{E}X^8 = \frac{132809}{32531625}$  (cumulants  $\frac{1}{9}$ ,  $-\frac{2}{225}$ ,  $\frac{16}{3969}$ ,  $-\frac{16}{3825}$ ) that determine the accelerated rules of [section 11](#).

## 10 Exact algebraic Fourier tails and the Sobolev threshold

For fixed  $n$  the finite spline has only finite smoothness, visible in the exact algebraic factorization

$$F_n(t) = 2^{n(n+3)/2} t^{-(n+1)} \left( \prod_{k=1}^{n-1} \sin(t/2^k) \right) \sin^2(t/2^n). \quad (10.1)$$

The oscillatory numerator prevents a pointwise constant, but its mean square is exactly computable through a dyadic trigonometric integral.

**Theorem 10.1** (Exact mean-square tail). *Let  $I_n = \frac{1}{\pi} \int_0^\pi \sin^4 u \prod_{j=1}^{n-1} \sin^2(2^j u) du$ . Then*

$$I_n = \frac{2^{n+2} + (-1)^{n+1}}{3 \cdot 2^{2n+1}}, \quad I_1 = \frac{3}{8}, I_2 = \frac{5}{32}, I_3 = \frac{11}{128}, I_4 = \frac{21}{512}, I_5 = \frac{43}{2048}, \dots, \quad (10.2)$$

and consequently

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T t^{2n+2} |F_n(t)|^2 dt = 2^{n(n+3)} I_n, \quad \int_{|t| > T} |F_n|^2 dt = \frac{2^{n(n+3)+1} I_n}{2n+1} T^{-2n-1} + \mathcal{O}_n(T^{-2n-2}), \quad (10.3)$$

with the ideal low-pass error  $\|f_n - \Pi_T f_n\|_{\dot{H}^s}^2 = \frac{2^{n(n+3)} I_n}{\pi(2n+1-2s)} T^{-(2n+1-2s)} + \mathcal{O}_n(T^{-(2n+2-2s)})$  for  $s < n + \frac{1}{2}$ , and the sharp threshold

$$f_n \in H^s(\mathbb{R}) \iff s < n + \frac{1}{2}. \quad (10.4)$$

*Proof.* Let  $Q_n(u) = \prod_{j=1}^{n-1} \sin^2(2^j u)$ , with constant Fourier coefficient  $2^{1-n}$  and  $\cos 4u$ -coefficient  $c_n$ . Since  $\sin^4 u = (3 - 4 \cos 2u + \cos 4u)/8$  and all frequencies of  $Q_n$  are multiples of 4,  $I_n = \frac{3}{8} 2^{1-n} + \frac{c_n}{16}$ .

The highest-frequency coefficient of  $Q_n$  is  $(-1)^{n-1}2^{3-2n}$  at frequency  $2^{n+1} - 4$ ; multiplying by  $\sin^2(2^n u) = (1 - \cos(2^{n+1}u))/2$  gives the recurrence  $c_{n+1} = \frac{c_n}{2} - \frac{(-1)^{n-1}2^{3-2n}}{4}$  with  $c_2 = -\frac{1}{2}$ , solved by  $c_n = -\frac{2^{2-n}}{3} - \frac{(-1)^n 2^{3-2n}}{3}$ ; substitution gives (10.2). The tail laws follow by substituting  $u = t/2^n$  in (10.1), averaging the bounded periodic numerator against  $t^{-\alpha}$  by integration by parts, and applying Plancherel; positivity of  $I_n$  makes the weighted integral diverge at  $s = n + \frac{1}{2}$ , proving sharpness.  $\square$

The threshold is the Fourier counterpart of the spline's final piecewise-constant derivative stage; (10.3) adds the leading constants, not just the exponents.

## 11 Positive moment-matched closures

### 11.1 The closure framework and the rate theorem

The construction (2.3) replaces the exact scaled tail law by a tractable one — a *closure*. For a symmetric probability measure  $\nu$  on  $[-1, 1]$  with characteristic function  $\psi_\nu(z) = \int e^{-izx} d\nu(x)$ , define

$$g_{n,\nu} = p_n * (D_h \nu), \quad \widehat{g_{n,\nu}}(t) = P_n(t) \psi_\nu(2^{-n}t), \quad (11.1)$$

where  $D_h$  scales by  $h = 2^{-n}$ . The original spline is  $\nu = \text{Unif}[-1, 1]$ ; every finitely supported  $\nu$  gives a positive finite sum of shifted head splines  $g_{n,\nu} = \sum_j w_j p_n(\cdot - 2^{-n}t_j)$ , a piecewise-polynomial density of degree  $n - 1$ ; a  $\nu$  with piecewise-polynomial density of degree  $d$  gives degree at most  $n + d$ ; and the support is exactly  $[-1, 1]$  iff  $\text{supp } \nu$  reaches both endpoints (since  $\text{supp } p_n = [-1 + h, 1 - h]$ ).

**Theorem 11.1** (Moment matching determines the spectral order). *If  $\int x^k d\nu = \mathbb{E}X^k$  for  $0 \leq k \leq 2M - 1$  and  $c_M = \frac{(-1)^M}{(2M)!} (\int x^{2M} d\nu - \mathbb{E}X^{2M}) \neq 0$ , then for every  $s \geq 0$ ,  $1 \leq p \leq \infty$ ,*

$$4^{Mn} (\widehat{g_{n,\nu}} - \Phi) \longrightarrow c_M t^{2M} \Phi(t) \quad \text{in } L_s^p(\mathbb{R}), \quad (11.2)$$

and correspondingly in every  $C^m$  and  $\dot{H}^s$  in physical space.

*Proof.*  $\psi_\nu(z) - \Phi(z) = c_M z^{2M} + \mathcal{O}(z^{2M+2})$  by symmetry and Taylor expansion at 0; the exact difference is  $P_n(t)(\psi_\nu(ht) - \Phi(ht))$ , and the bounded-remainder plus head-decay argument of [theorem 8.1](#) applies verbatim.  $\square$

**Theorem 11.2** (Arbitrary-order positive support-filling closures). *For every  $r \geq 0$  there is a symmetric, finitely supported probability measure  $\nu_r$  on  $[-1, 1]$  with positive mass at both endpoints matching the up moments through degree  $2r + 1$ ; consequently  $\|g_{n,\nu_r} - \text{up}\|_{C^m} = \mathcal{O}_{m,r}(4^{-(r+1)n})$  and likewise in every  $\dot{H}^s$ , with  $g_{n,\nu_r}$  nonnegative, of total mass one, and of support exactly  $[-1, 1]$ .*

*Proof.* Push the up law forward under  $x \mapsto x^2$  to a measure on  $[0, 1]$ ; a positive Gauss–Radau rule containing the endpoint 1 and exact through degree  $r$  exists by standard quadrature theory; lifting each node  $y_j$  to the symmetric pair  $\pm\sqrt{y_j}$  with half weights (a zero node to 0) gives  $\nu_r$ . The rates follow from [theorem 11.1](#).  $\square$

### 11.2 The explicit menu

Solving the moment equations with the rational moments of [section 9](#) yields two complementary explicit families: rules with minimal atom counts (irrational nodes allowed), and rules with dyadic rational nodes. Both sources' overlapping entries agree exactly; the variance-matched three-point rule below was derived independently in both.

| closure          | tail measure $\nu$                                                                                                                                               | $M$ | $c_M$                    | st                                                                                                                                                                                                                                                                                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| original uniform | $\text{Unif}[-1, 1]$                                                                                                                                             | 1   | $-1/9$                   | ex<br>su<br>po<br>th<br>sp<br>$f_n$<br>it<br>se<br>na<br>su<br>po<br>on<br>re<br>bo<br>ex<br>su<br>po<br>th<br>sh<br>na<br>su<br>po<br>th<br>sh<br>ex<br>su<br>po<br>fi<br>dy<br>sh<br>ex<br>su<br>po<br>fi<br>sh<br>ex<br>su<br>po<br>se<br>dy<br>sh<br>ex<br>su<br>po<br>on<br>$C$<br>de<br>si<br>of<br>de<br>gr<br>$n$<br>6 |
| variance box     | $\text{Unif}[-1/\sqrt{3}, 1/\sqrt{3}]$                                                                                                                           | 2   | $-1/4050$                |                                                                                                                                                                                                                                                                                                                                |
| Lobatto-3        | $\frac{8}{9}\delta_0 + \frac{1}{18}(\delta_{-1} + \delta_1)$                                                                                                     | 2   | $7/2025$                 |                                                                                                                                                                                                                                                                                                                                |
| Gaussian-3       | $\frac{32}{57}\delta_0 + \frac{25}{114}(\delta_{-a} + \delta_a), a = \sqrt{19/75}$                                                                               | 3   | $1238/334884375$         |                                                                                                                                                                                                                                                                                                                                |
| dyadic-5         | weights $\frac{376}{675}, \frac{896}{2025}, \frac{1}{2025}$ at $0, \pm\frac{1}{2}, \pm 1$ (pair totals)                                                          | 3   | $71/21432600$            |                                                                                                                                                                                                                                                                                                                                |
| Lobatto-5        | $\frac{78976}{153675}\delta_0 + \frac{2470629}{10262075}(\delta_{-b} + \delta_b) + \frac{619}{270450}(\delta_{-1} + \delta_1), b = \sqrt{683/3087}$              | 4   | $1006207/24604155961875$ |                                                                                                                                                                                                                                                                                                                                |
| dyadic-7         | pair totals $\frac{10411}{2679075}, \frac{1003648}{2679075}, \frac{581632}{2679075}$ at $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}$ , center $\frac{120376}{297675}$ | 4   | —                        |                                                                                                                                                                                                                                                                                                                                |
| polynomial $q_3$ | density $\frac{35}{32}(1 - x^2)^3$                                                                                                                               | 2   | $2/22275$                |                                                                                                                                                                                                                                                                                                                                |

The convergence scales are  $4^{-Mn}$ : rate  $16^{-n}$  for the three  $M = 2$  rules,  $64^{-n}$  for Gaussian-3 and dyadic-5,  $256^{-n}$  for Lobatto-5 and dyadic-7. Rules including  $\pm 1$  preserve the exact support;

Gaussian rules maximize order per atom; the polynomial density achieves  $16^{-n}$  with a single continuous piecewise-polynomial representation and the smallest leading constant of the  $M = 2$  entries ( $C_{q3,\infty} = 0.0507\dots$  against  $C_{L3,\infty} = 1.952\dots$  and  $C_{VB,\infty} = 0.1394\dots$ , computed against  $\sup |t|^4|\Phi| = 564.7796\dots$  at  $t = 16.3025\dots$ ). A fixed dyadic ladder is not guaranteed to retain positivity indefinitely, so [theorem 11.2](#) is the robust existence result while the dyadic menu gives convenient low-order formulas. Cancellation makes direct double-precision verification of the  $64^{-n}$  and  $256^{-n}$  rules unreliable; the verification packages evaluate the scaled point errors at 120 decimal digits and reproduce the predicted leading profiles (Gaussian–3 at  $n = 11$ : 3.21182 against the limit 3.21168; Lobatto–5 at  $n = 14$ : 873.5187 against 873.5103).

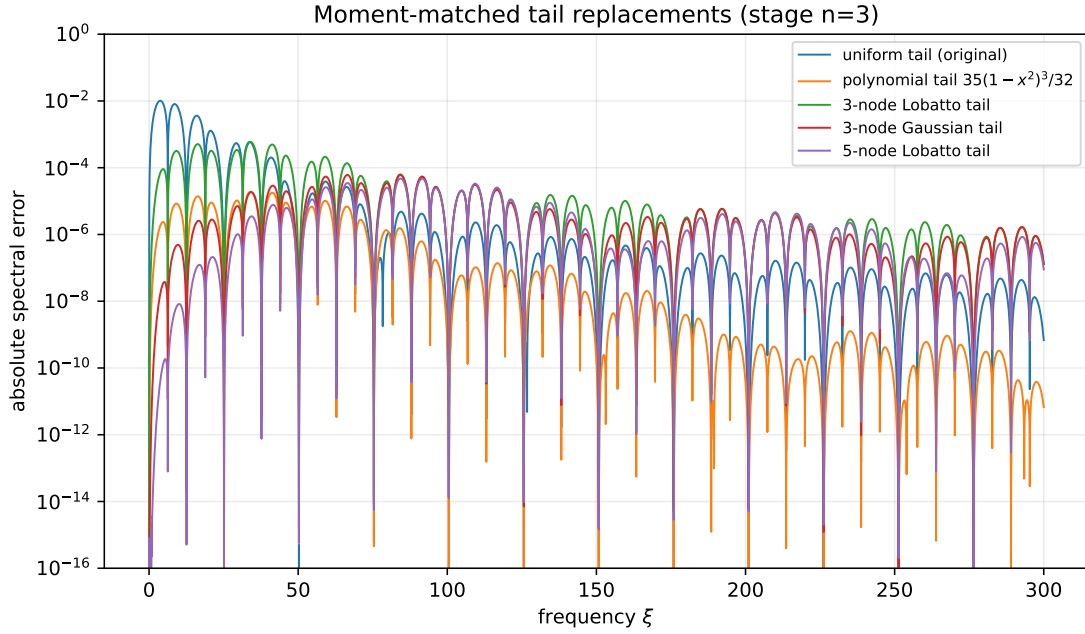


Figure 12: Absolute Fourier errors of the closure menu at stage  $n = 3$ ; the faster rates are visible over most of the frequency range, with leading profiles attaining maxima on progressively more distant lobes.

### 11.3 Three closure families and the prefix quadratures

Part III and this part together produce three structurally different positive acceleration families over the *same* head spline  $p_n$ :

1. *box mixtures* ([equation \(7.10\)](#)): quadrature of the squared-scale-mixture law ( $X \stackrel{d}{=} RU$  with  $Z = R^2 \sim \nu$ ) – each node contributes a box  $p_n * u_{\delta\sqrt{z_j}}$ , giving mixtures of box splines of full degree;
2. *atomic closures* (this part): quadrature of the up law itself (equivalently, Gauss–Radau on the pushforward under  $x \mapsto x^2$  lifted symmetrically) – each node contributes a translate  $p_n(\cdot - ht_j)$ , giving sums of shifted head splines of degree  $n - 1$ ;
3. *polynomial closures*: a continuous tail density, giving one smooth piecewise-polynomial density of slightly higher degree.

The variance-matched members of the first two families have the same leading constant  $|c_2| = 1/4050$  – the box mixture  $p_n * u_{2^{-n}/\sqrt{3}}$  and the variance box above are the same approximant seen from the two quadrature variables – while the endpoint-preserving members differ genuinely (the box family’s



Lobatto rule keeps  $p_n$  itself as a component; the atomic family’s endpoint atoms shift the head to  $\pm 1$ ). Moment order controls the global scale  $4^{-Mn}$  in every family; what distinguishes them is support exactness, smoothness, atom count, and the divisor of the critical transfer  $B_\nu = \psi_\nu/\Phi$ , which for generic rules has poles on the full dyadic zero lattice — the original uniform tail is special in sharing many zeros with  $\Phi$ , which is what produced the removable/pole hierarchy of [theorem 3.2](#). Moment order and critical zero matching are independent design objectives ([section 12](#)).

## 12 Directions, conjectures, and formalization targets

1. *Positivity* ([theorem 5.3](#)) and the pole-resolved transseries ([5.7](#)): compute the full Laurent jets at  $4\pi q$  from finite products of  $\Phi$ -values at odd multiples of  $\pi$  (the leading jets are [\(3.8\)](#)), reorganize into an optimally truncated arithmetic transseries, and decide eventual monotonicity of the normalized ratio.
2. *Hybrid tail design*: for prescribed  $M, K$ , find positive  $\nu$  matching moments through  $2M - 1$  while  $\psi_\nu$  shares prescribed zero jets with  $\Phi$  at  $2\pi, \dots, 2\pi K$  — a finite-dimensional positive trigonometric moment problem (semidefinite search, then exact certificates); minimize atoms or polynomial degree. Rational or Padé spectral multipliers encoding the first poles offer an alternative postprocessing route around the kernel obstruction of [theorem 7.3](#), with evenness, Thue–Morse sign intervals, and no spurious real poles below a chosen band as natural constraints.
3. *Optimal closures*: minimize the number of atoms at fixed moment order with endpoint masses; minimize the first unmatched moment; determine the largest order retaining positivity on a fixed dyadic node ladder; minimize the forward multiplier norm on a target band (quadrature meets minimax filter design).
4. *Modified local recursions*: derive finite vector-valued refinement systems whose  $n$ th iterate is exactly a Gaussian, Lobatto, or polynomial-tail closure, preserving the one-line locality of [\(2.1\)](#) at higher order.
5. *Base- $q$  analogues*: for  $\Phi_q(z) = \prod_{k \geq 1} \text{sinc}(z/q^k)$  and  $A_q = \text{sinc } z/\Phi_q$ , the Euler regrouping suggests  $\text{ord}_{\pi m} A_q = 1 - \nu_q(m)$  with divisor Dirichlet series  $\zeta(s)(q^s - 2)/(q^s - 1)$ ; determine which spectral and sign phenomena survive and which are genuinely binary (for  $q = 2$  the cumulative divisor collapses to the digit sum and the functional equation to the cotangent product).
6. *Two-parameter asymptotics*: allow the Fourier weight  $s_n$  to grow with  $n$  and locate the transition between the fixed-weight theorem and the exact boundary layer; on the physical side this quantifies how much spectral regularity a finite spline possesses at a given stage, refining [\(10.4\)](#).
7. *Transfer-operator bridge*: insert the shell identity [\(7.4\)](#) into the Fourier-decay corpus to obtain a joint envelope/profile theorem for finite products, with exceptional arithmetic points from [\(6.1\)](#) and uniformity near the first zero and pole.
8. *Lean targets*: the finite multiplicity formula [\(6.1\)](#); the divisor count and digit-sum identity [\(4.1\)](#); the Taylor radius and first residues; the  $I_n$  recurrence and closed form [\(10.2\)](#); the cumulant defect [\(9.1\)](#); exact moment verification, positivity, and support of the explicit closures; and the shift-sum identity behind [\(11.1\)](#). The weighted- $L^p$  theorem is modular: a bounded remainder, a finite-product decay lemma, and dominated convergence.

## 13 Verification and reproducibility

The two absorbed packages provide independent verification pipelines, retained verbatim under `assets/Rvachev_Piecewise_Approximation_Fourier_Images/` (exact SymPy coefficients



through order 30, 120-digit mpmath evaluation of  $\Phi(\pi)$ , the residue, and the cancellation-sensitive accelerated errors, lobe-wise maximization, composite Gauss–Legendre norms on 700 lobes, symbolic verification of every quadrature rule; figures and CSV audit artifacts) and `assets/rvachev_fourier_frontier_report/` (exact coefficients through order 50, functional-identity checks, divisor partial sums against  $s_2$ , finite multiplicity tables, 100-digit quadrature of  $I_n$  against the closed form, rational verification of the dyadic closures; figures, coefficient CSV, and text summary). Both scripts regenerate every figure and table from scratch (`fourier_piecewise_experiments.py`, `rvachev_fourier_experiments.py`). All shared constants agree between the two packages; the numerical work is ancillary throughout — every theorem above has a complete proof independent of the computations.

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